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WORKING GROUP ON MULTISPECIES ASSESSMENT METHODS (WGSAM; outputs from 2025 meeting)

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i Executive summary

The Working Group on Multispecies Assessment Methods (WGSAM) aims to advance the operational use of knowledge on predator-prey interactions for advice on fisheries and ecosystem management. Key runs are multispecies and ecosystem models reviewed against high quality criteria. They are developed to contribute to ICES advice.

This report presents the definition of a key run and an update to the SMS model key run for the Baltic Sea. The updated model extends input time-series to 2024 by extending the data for herring, sprat and cod. The key run provides updated estimates of cod predation mortality for the Baltic Sea sprat and central Baltic Sea herring stocks. Low cod stock size leads to low predation on both herring and sprat. Forecasts of cod abundance are used for parts of 2023 and 2024 due to the most recent Eastern Baltic Sea cod assessment not being accepted. Stomach samples are important for determining predator-prey interactions and alternative stomach sampling designs for the Baltic Sea are evaluated. This provides a framework for how to balance sampling effort with the provision of useful data.

Continuous updates to the Baltic Sea key run require non-forecasted data on cod abundance. Due to the low stock size for cod, inclusion of other sources of predation, such as seals or birds, should be considered.

ii Expert group information

Expert group name	Working Group on Multispecies Assessment Methods (WGSAM)
Expert group cycle	Multiannual
Year cycle started	2025
Reporting year in cycle	1/3
Chairs	Nis Sand Jacobsen, Denmark
	Michael Thomson, United Kingdom
Meeting venue and dates	6–10 October 2025, Copenhagen, Denmark (26 participants)

1 WGSAM Key Run Definition

This section serves to describe WGSAM's approach to so called 'key runs'. A 'key run' is a peer reviewed model run developed by WGSAM that serves to inform with a specific advice product for stocks or ecosystems. This is different from model reviews, which WGSAM historically has also done, where ecosystem or multispecies models are reviewed in the group. Key runs are currently and historically part of the terms of reference (ToRs) for WGSAM. The last decade the Baltic Sea SMS model and the North Sea SMS model have been through a key run process every three years, where the main advice product has been natural mortality estimates for the stock assessment groups HAWG, WGNSSK and WBGFAS.

The criteria for reviewing key runs largely follow the decision table provided by WGSAM (ICES, 2019, 2021), where the model run must go through 6 steps

1. Appropriateness of the model for the problem (problem identification)
2. Assumptions (scientific basis, computational infrastructure; adequacy of conceptual model)
3. Input data quality
4. Comparison with observations
5. Uncertainty/sensitivity analysis
6. Peer review (WGSAM's role, but consider previous reviews from model construction)

The 6 steps can be understood in detail from the description on the WGSAM github[1].

A key run is only valid for model configuration, input data and software as evaluated during the key run review process and only to address the specific problem reviewed in the group. Models cannot be considered to have key run status when any of these aspects change. If the model is applied to another problem, the configuration of the model changes, new input data or alternative software is used the resulting model will not have key run status unless reviewed again by WGSAM.

WGSAM provide key run updates every 3 years for the North and Baltic Sea, which provides an indicative time frame for when updates will need to be considered.

[1] WGSAM Key Run Review Criteria. 2025. ICES Working Group on Multispecies Assessment Methods (WGSAM). https://ices-eg.github.io/wg_WGSAM/ReviewCriteria.html

2 Baltic Sea SMS key run review

2.1 Is the model appropriate for the problem?

The SMS model will be used to provide natural mortality estimates by age and year as input to single species assessments of Eastern Baltic herring (Subdivisions 25–29 and 32 (excluding Gulf of Riga)) and Baltic sprat (Subdivisions 22–32). Natural mortality estimates are only used as input for the historic part and no forecast is needed. M estimates by age and quarter are a direct output of the model. However, an assumption is needed for residual mortalities M1 while the predation mortalities M2 are estimated (M = M1+M2). The model provides estimates for the years 1974 to 2024.

The eastern Baltic cod occurs in the SD24–32 with largest abundances in SD25–29, and the western Baltic cod is found in the SD22–24. The distribution of the eastern Baltic stock overlaps entirely with the distribution of the central Baltic herring stock (SD25–29,32), while only to a large extent with the distribution of Baltic sprat which occurs throughout the SD22–32. The Baltic SMS key run includes the eastern Baltic cod but not the western Baltic stock, hence what referred to cod predation in the model is in fact limited to predation by the eastern Baltic stock.

Cod is the only predator in the model, feeding on herring, sprat, and an 'other food' component. Cod is a constant predator in the SMS model due to age reading problems with the stock. For some years, only length-based input such as catch at length have been provided for the assessment of the cod stock. Such input cannot be handled in SMS. Therefore, numbers at age derived from the Eastern Baltic cod assessment are used as known input without error since the 2019 key run. Predation by other likely important predators to herring and sprat (e.g., seal and seabirds) is not considered in in the current key run. In recent years cod abundance has been at a historical low, and other predators such as seal and cormorants have been increasing. The model might therefore not be accounting for a large part of the predation part of natural mortality on the two prey species.

Overall, the model is appropriate to provide information on natural mortalities as input for the assessments of central Baltic herring and Baltic sprat. However, results depend to a large extent on the input from the Eastern Baltic cod assessment. The assessment for cod went from a category 1 assessment with absolute age-structured estimates to a category 3 assessment with relative estimates only indicative of trends in 2024 (ICES, 2024b), because of lack of trust in the data and model. It was therefore chosen to use the outputs of the 2023 cod assessment to inform cod abundance in SMS for the period 1974–2022 and abundances coming from the cod forecast in 2023–2024. Cod abundance in 2023–2024 is therefore very uncertain and dependent on the forecast assumptions, notably regarding recruitment and fishing mortality. In addition, cod inputs are assumed to be known without error, SMS uncertainty estimates around M are likely underestimated and cannot be considered informative.

2.2 Is the scientific basis of the model sound?

The SMS model is an established and reviewed model that has previously been applied in the Baltic Sea to provide inputs for assessments of commercially exploited stocks (Baltic sprat and herring). It is also applied for the North Sea food web (ICES, 2014, 2018, 2021, 2024a) to provide input natural mortalities for the assessments of many commercially exploited stocks in the North Sea (e.g., North Sea cod and herring). The Baltic SMS model has been reviewed by ICES on several occasions (ICES, 2012a, 2012b, 2019, 2023a).

2.3 Is the input data quality and parameterization sufficient for the problem?

Model setup

For herring and sprat, the data (1974–2024), in quarterly time steps, used as input to SMS consists of: catch at age, proportion landed, mean weight at age in the catch and the stock, proportion mature, proportion mortality before spawning, natural mortality at age, survey effort and survey catch at age (Lewy and Vinther, 2004). For cod, the data used as input to SMS consists of: cod numbers and mean weight per 2-cm length class from the ICES WGBFAS, Stock-Synthesis 3 (SS3) assessment from 1974–2024 (with data through 2022 and projection years 2023 and 2024) and Latvian cod stomach data from 1974–2014. SMS is fitted to the Cod stomach data and the catch at age and survey indices for herring and sprat. SMS estimates values for F, SSB, recruitment and M2 (i.e., cod predation mortality) for herring and sprat. In total, 300+ parameter values are estimated by the model.

Data quality

SMS uses the same data as used for input to the single species assessments of herring and sprat in the Baltic Sea (catch at age, mean weights, proportion mature, survey indices). These data have been benchmarked and therefore no further review on these data has been carried out. However, while single species assessments start with age 1 as recruits, SMS starts with age 0. Given that predation mortalities are important for age 0, this is understandable. But e.g., the mean weight at age is highly uncertain for the 0 group. WGBFAS assumes that mean weight at age in the sea is the same as mean weight at age in the catch. In general, the assumption that mean fish weight in the catch equals the mean fish weight in the stock is a major assumption that can have an influence on model results. SMS uses smoothed quarterly values as input for the age 0 (Q3 & Q4) and age 1 (Q1 & Q2). Even though there is a large uncertainty around the input values for weight at age in the sea, especially for the youngest ages.

In addition to the data that is used for the single species assessments, SMS relies on cod stomach data. A new stomach data set for cod became available in recent years. Due to the better quality of the new stomach contents data, this data set was applied as default in the key run of 2019 (ICES, 2019) and continued in 2022 and the present key run. The stomachs have been collected mainly by the Latvian institute and provide detailed information based on individual stomachs. The dataset provides information for the period 1974–2014, stomach data from more recent years is not available. Overall, data from 64 000 stomachs were used as input for SMS. Stomach data are reported per individual cod with predator size classes in mm or cm. Although in general the eastern Baltic is well covered, sampling focuses around the Latvian coast and there are some gaps in space, especially in the 1990s. The main distribution areas for cod are the SD 25, 26 and 28 and stomachs from these areas are used for SMS both in this as in previous key runs. The three subdivisions have not simultaneously been sampled in all years.

The compilation procedure followed to parameterise stomach contents in the current and 2022 key run were performed via the 'FishStomachs' R package[2]. This introduced some differences in terms of relative stomach contents (ICES, 2023a, Annex 2) compared to the beta version of FishStomachs used for the 2019 key run, although these are minimal.

Consumption rates are calculated based on the cylinder gastric evacuation rate model from Andersen and Beyer (2005a, b). In order to consider recent changes in cod consumption rate, the

[2] FishStomachs R package. 2025. Morten Vinther. https://github.com/MortenVinther/FishStomachs

relationship between average quarterly consumption rate and total length (a priori parametrized as $C=aL^b$ with C the average quarterly consumption rate and L total length) was estimated separately for three different periods (1974–1989, 1990–1999, 2000–2014).

Overall, the data quality is considered decent input to the model. That the model uses key input from the single species assessments can be seen as a strength because these data were already scrutinized during a full ICES benchmark process. It is a positive development that the stomach data compilation process has been documented in an R package, this has facilitated the reproducibility of the stomach data compilation process for 2022–2025 and in the future and avoids discrepancies in stomach data used in subsequent key runs.

Assumptions and parameterization

The parameterization of the diet selection sub-model is based on several assumptions. First, there is only one vulnerability parameter per interaction for the full model time-series. The assumption of constant vulnerability may be violated if e.g. the spatial predator-prey overlap changes. Given an assumed constant overlap, the implied Holling type II functional feeding response as used in SMS is well known to lead to instability when prey items become low in abundance and makes them vulnerable to extinction in the model. However, this is mainly an issue for forecasts when trying to make predictions outside the range of observations.

Another important assumption is a time constant biomass pool of "Other Food" that for the eastern Baltic cod includes a considerable proportion of benthic preys. If the availability of important Other Food prey items changes over time, this can lead to biased predictions of relative stomach contents and therefore predation mortalities. As shown in Figures 2.7 and 2.8 in the section 2.4 below, this could be an issue for this key run. The current SMS key run is based on a log-normal size selection of the predators (see Annex 3).

Overall, the parameterization and assumptions are consistent with recent scientific knowledge. However, some of the assumptions regarding constant vulnerabilities and constant "Other Food" availability may be relaxed in future key runs and further investigations are needed to fully utilize the extensive time-series of stomach data (also in relation to spatial patterns) to optimise the parameterization of the diet selection sub-model.

WGSAM notes that with eastern Baltic cod being at a historically low level, and with the cod stock assessment future in question, identifying and including other predators or sources of mortality for herring and sprat should be prioritized for future Baltic key runs.

2.4 Does model output compare well with observations?

The SMS key run was able to estimate total catch over time sufficiently well for herring and sprat (Figure 2.1), and the main features of the time-series are well covered by the model. For herring, the catch-at-age residual plots by quarter do not show any discernible pattern (Figure 2.2). For sprat however, there are clear clusters of positive and negative residuals, except for the oldest age classes all show more positive residuals in the earlier part of the time-series, especially for Q3 (Figure 2.3). Catch-at-age residuals for sprat are also larger overall than for herring.

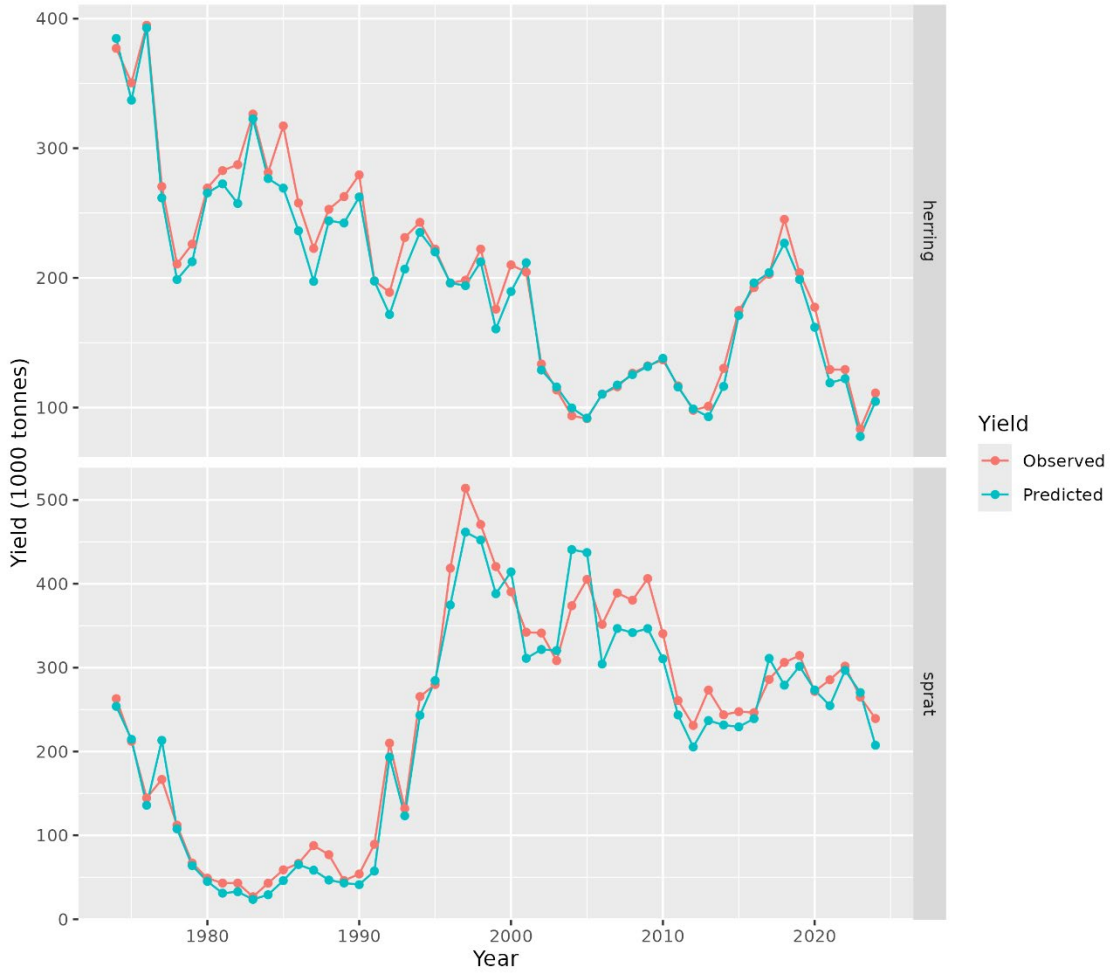


Figure 2.1. Comparison between predicted and observed catch time-series.

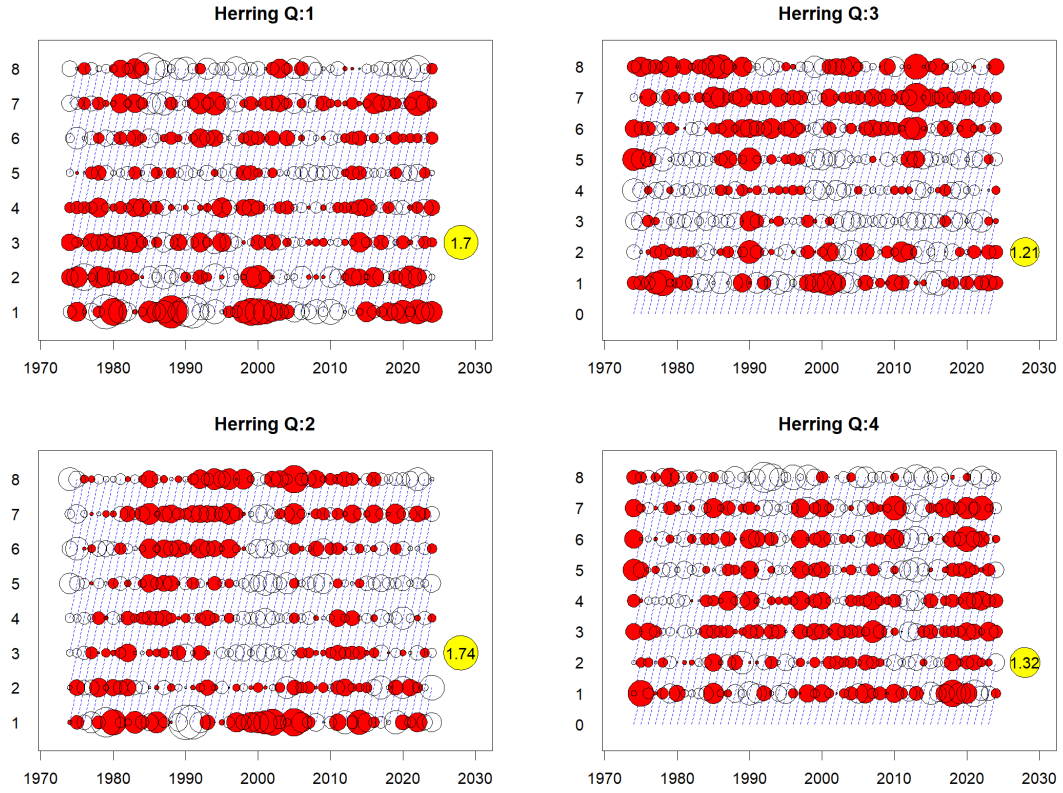


Figure 2.2. Herring catch observations Residuals, log(Survey observed CPUE) - log(expected CPUE). Red is positive, white is negative. Q:9 means annual data. The yellow circle illustrates the size of the largest value as a reference.

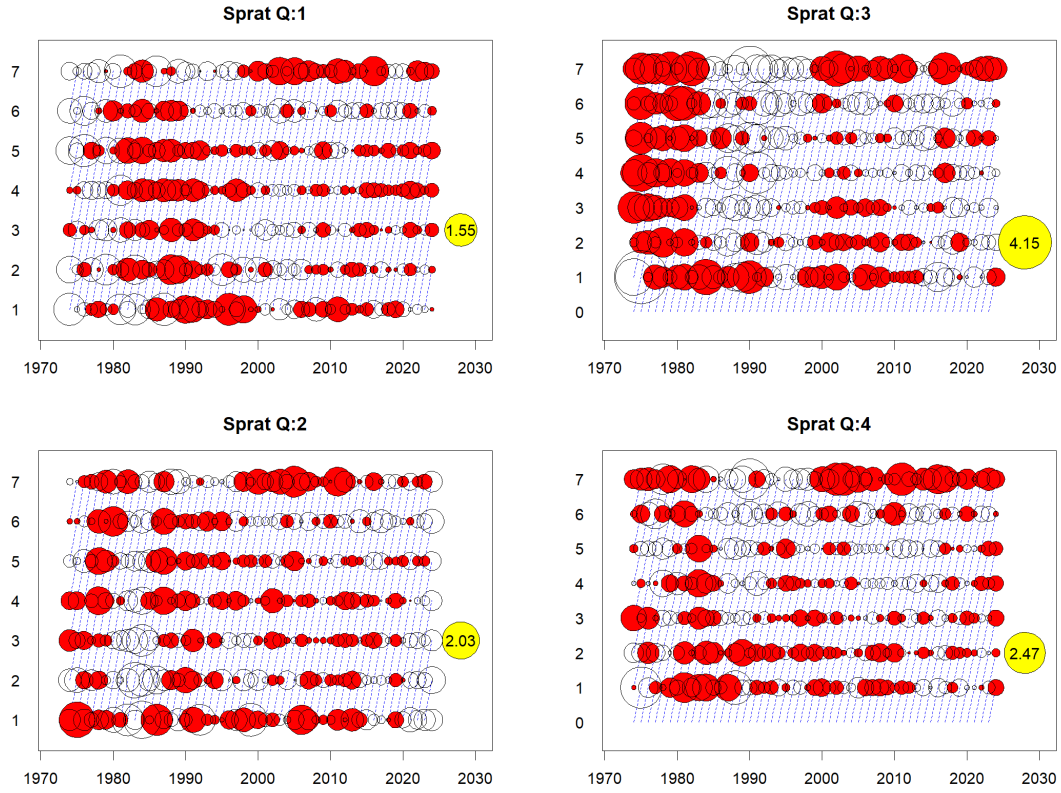


Figure 2.3. Sprat catch observations Residuals, log(Survey observed CPUE) - log(expected CPUE). Red is positive, white is negative. Q:9 means annual data. The yellow circle illustrates the size of the largest value as a reference.

Survey time-series have been updated since the last benchmark of herring and sprat (ICES, 2023c) and in this updated key run accordingly. Some of the survey time-series have been shortened in length (i.e., herring acoustic, sprat recruitment index) which resulted in the perception of an improved fitting of some of the indices such as the sprat recruitment index (i.e., age 1 sprat in the LAT RUS acoustic survey) compared to the 2022 key run. The fit to survey data continues to capture the overall trends but with a better fit to sprat data (Figure 2.5) than to herring data (Figure 2.4). There are clear "year effects" (consistent under- or overestimation in a given year) in the residuals (Figure 2.6). Overall, no sign of overfitting to a particular data source is apparent.

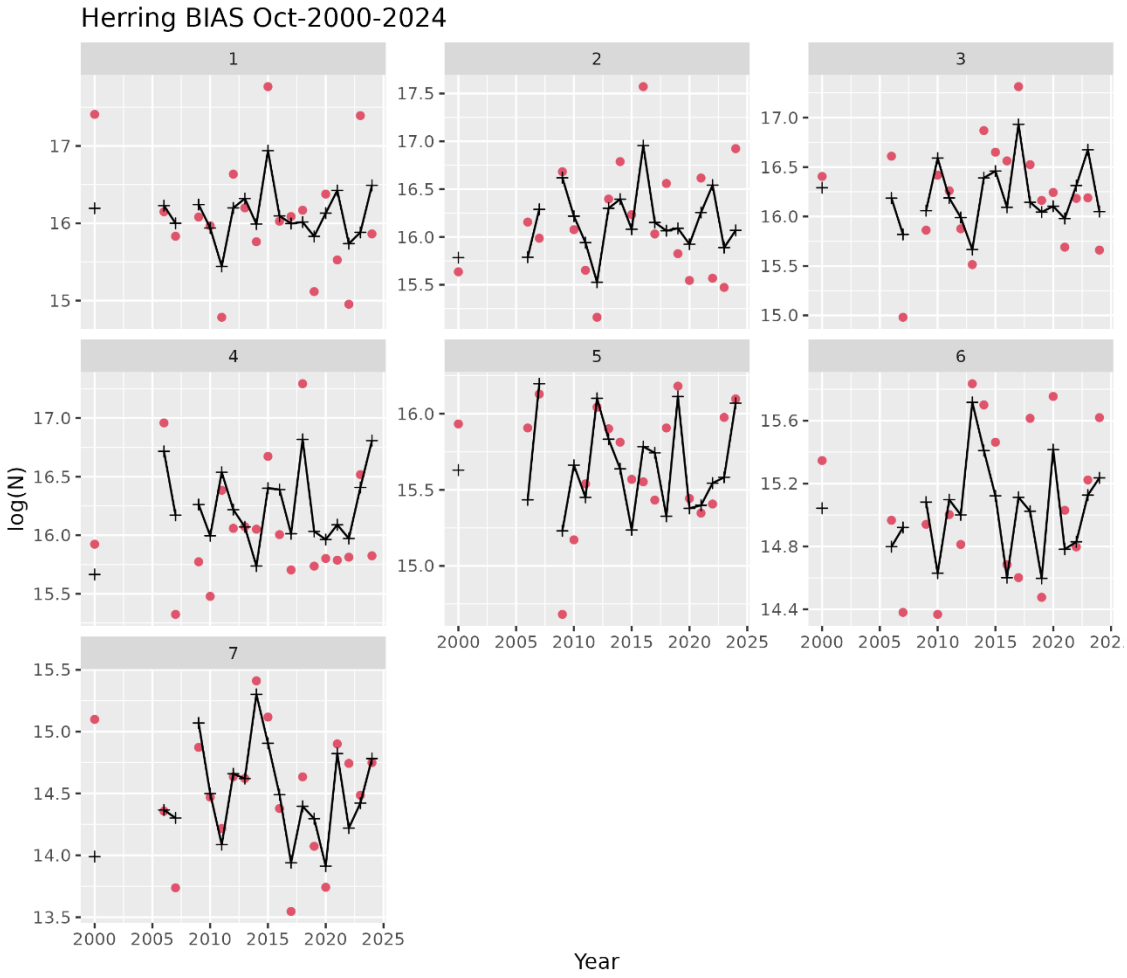
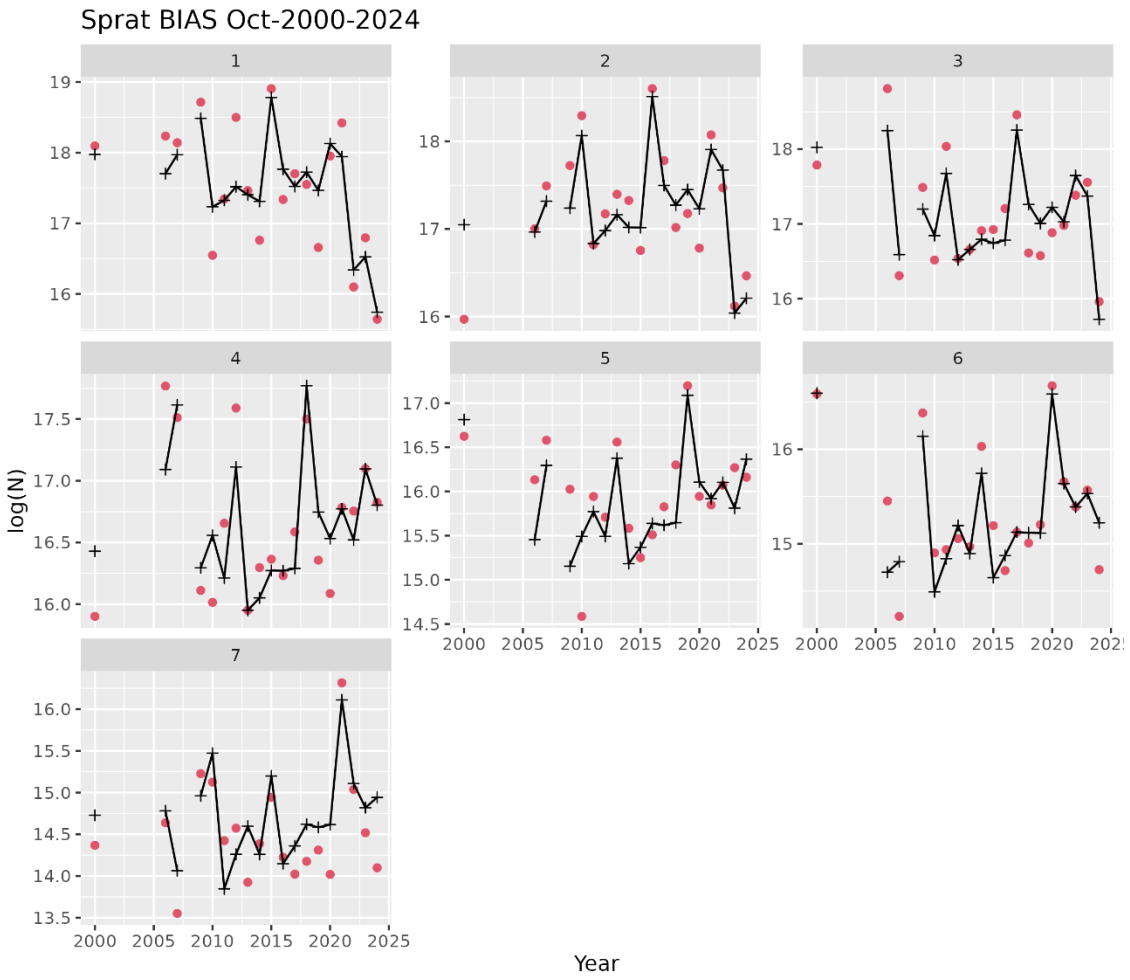


Figure 2.4. Herring surveys observation residuals (log scale). The red dots are the observed values, the lines and crosses are the predictions.



Sprat surveys observation residuals (log scale). The red dots are the observed values, the lines with crosses are the predictions.

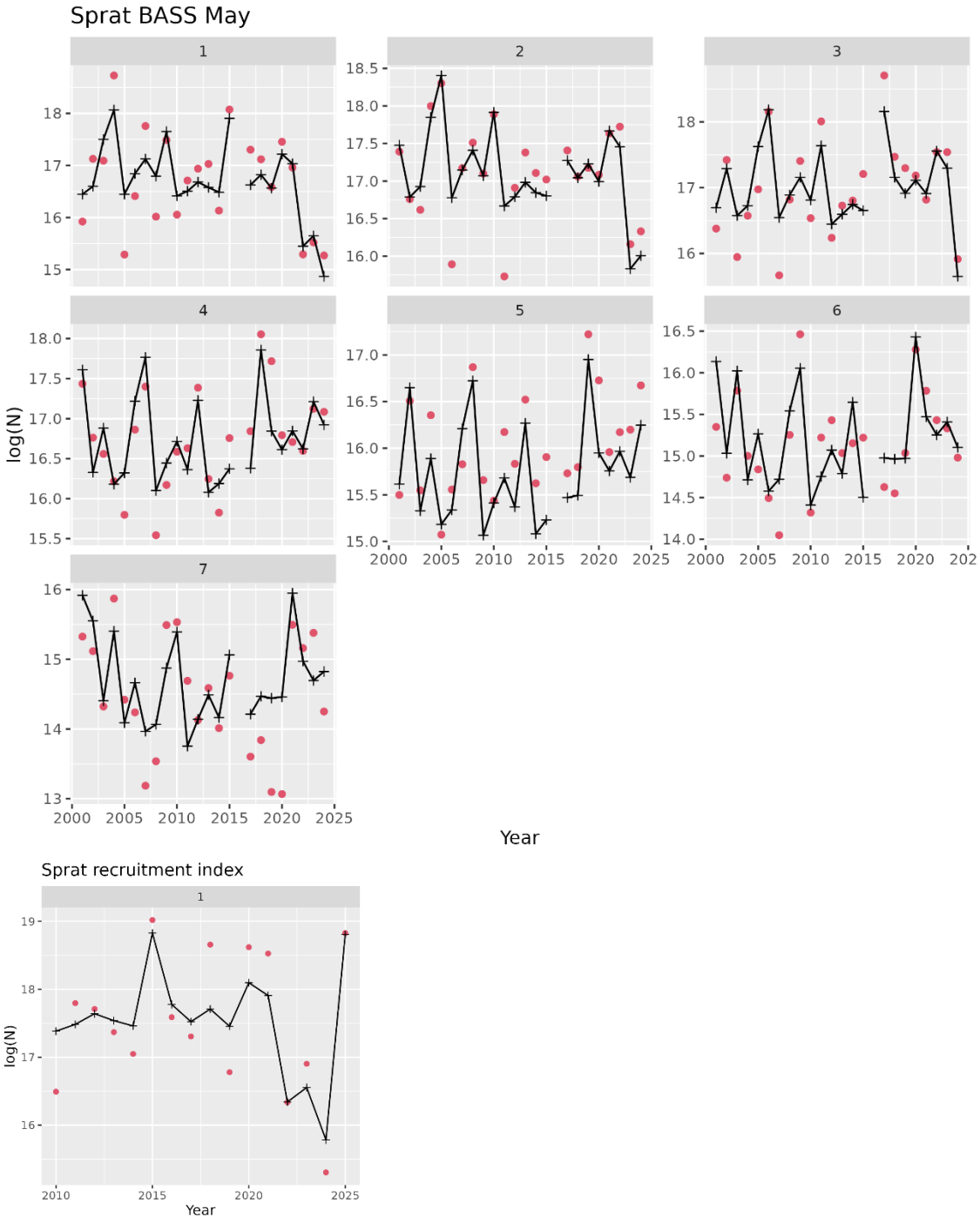


Figure 2.5. Sprat surveys observation residuals (log scale). The red dots are the observed values, the lines with crosses are the predictions.

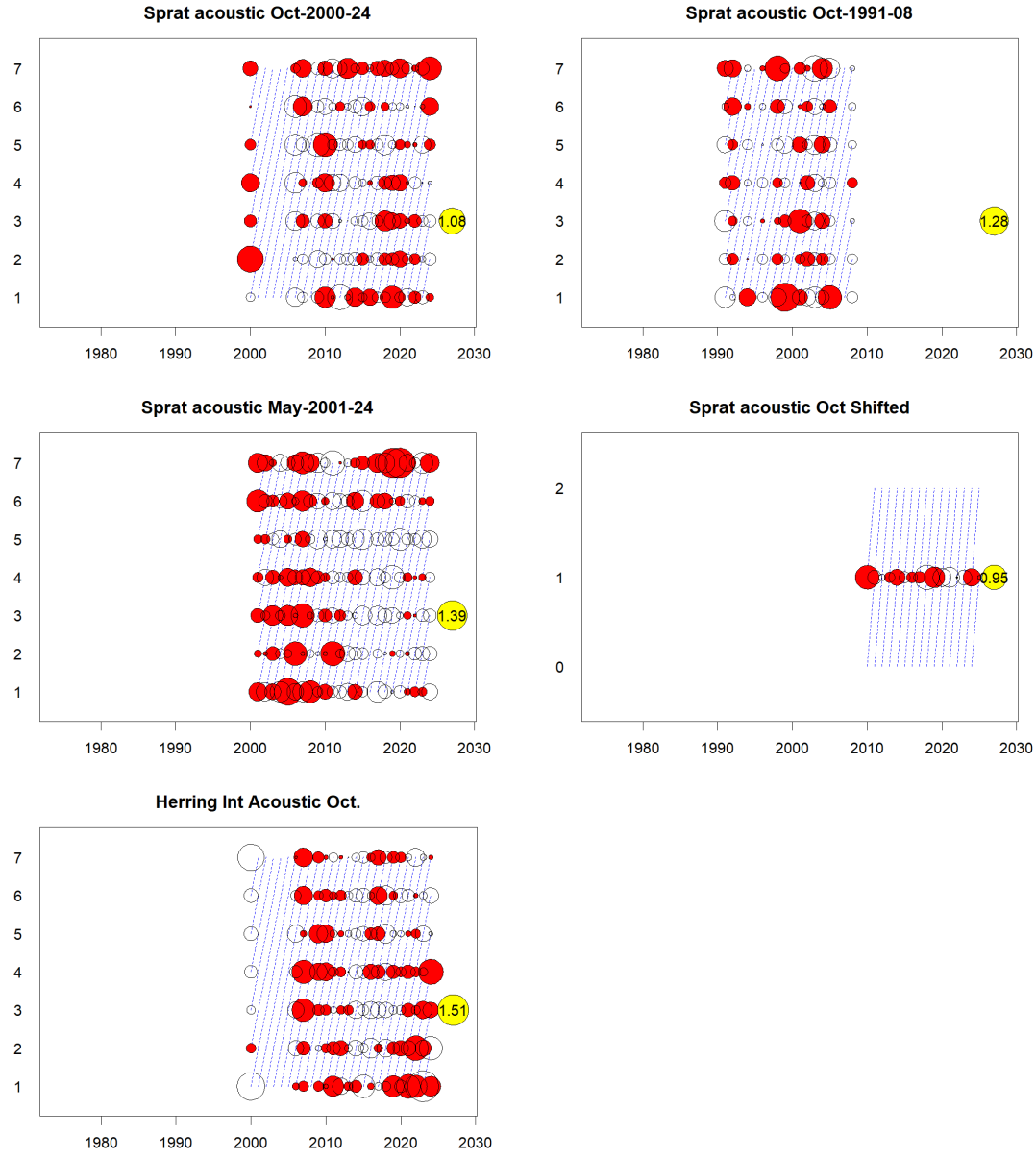


Figure 2.6. Survey observations residuals, log(Survey observed CPUE) - log(expected CPUE). Red is positive, white is negative. The yellow circle illustrates the largest value for reference.

The fit to stomach data reveals residual patterns, where there is a consistent over- or underestimation (cluster of residuals switching from positive to negative and vice-versa) of other food, sprat and herring in certain time periods (Figure 2.7). The box plots of residuals also show a clear switch towards positive residuals in the 1990s for cod eating herring, and a clear declining trend for cod eating other (Figure 2.8). This suggests possible (non-constant) trends in processes such as prey vulnerabilities or availability of other food, which are so far not captured by the model.



Figure 2.7. Cod stomach observation residuals (raw = (observed proportion) - (expected proportion)). For residual plots, white is positive, green is negative.

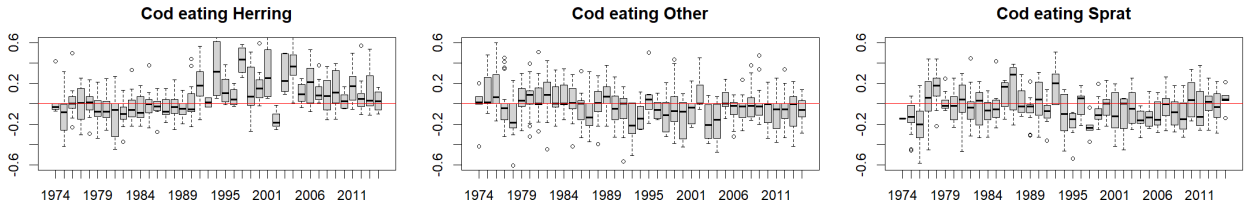
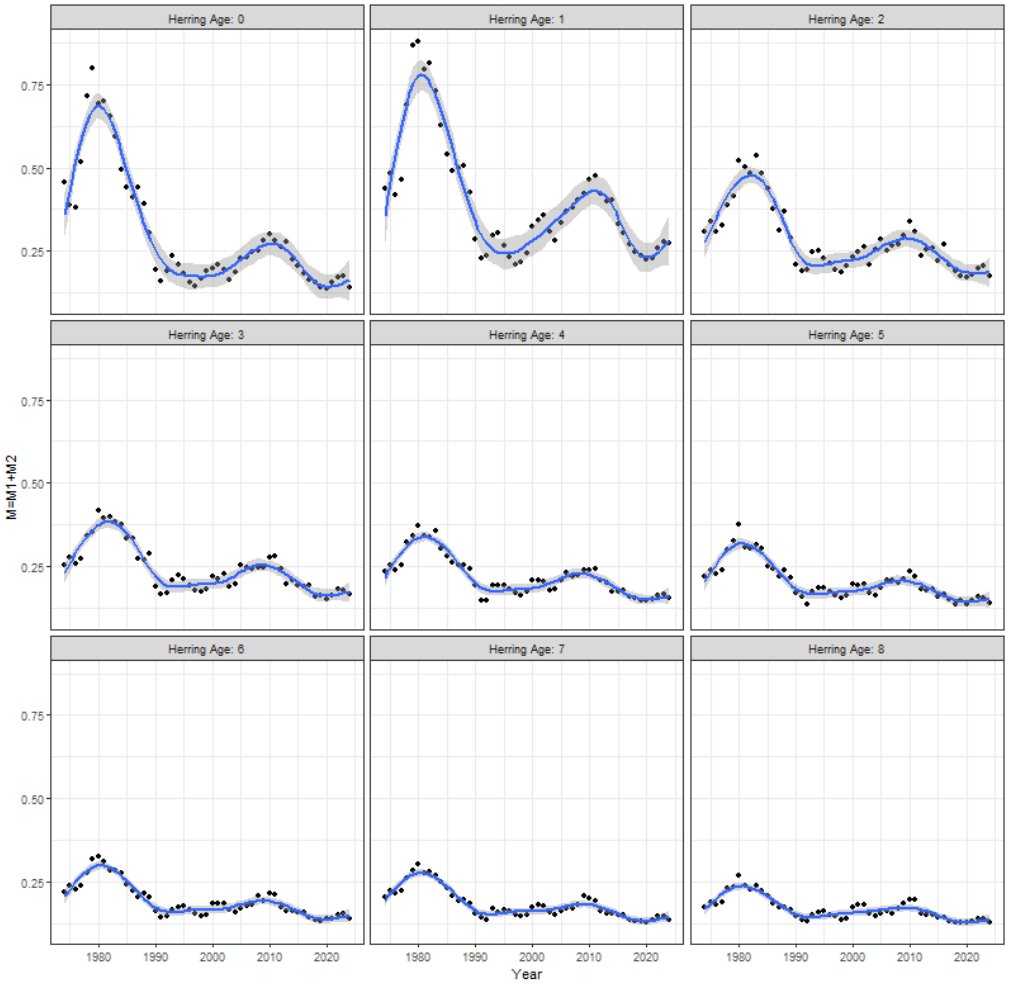


Figure 2.8. Box plots of cod stomach observation residuals by year.

2.5 Uncertainty

The uncertainty of any one selected output variable in SMS is estimated using the Hessian matrix and the Delta method approximation (see Annex 3). However, we note that these do not include possible error in the input variables and are thus likely an underestimation of the uncertainties in the model estimates. For example, the stock numbers for cod (a so-called 'other predator' within the model) are assumed to be known without errors while considerable uncertainty is associated to the stock assessment estimates. The uncertainty estimates of the model are for this reason considered uninformative and are not presented in detail in the stock annex (see Annex 3) or used further.

Predation mortality (M2) decreases in general by prey size (age) as expected (Figure 2.9); however for sprat (and for herring to a lesser extent) M (M2 + constant M1) is estimated to be highest for age 1, as was the case in the last key run. This pattern might be explained by a lower spatial overlap between the very small preys and cod or may be an effect of the rather uncertain mean weight at age applied for the 0-group. WGBFAS assumes that mean weight at age in the sea is the same as mean weight at age in the catch. This assumption could be violated for the age 0 individuals, as the representation of age 0 individuals in the catches and surveys are likely the largest individuals in their cohorts. The age 0 individuals are likely underrepresented in the catch and survey data. While the M values needed for the single species assessments start with age 1 and M values for age 0 are less critical, shifts in the cod diet portion of age 0 herring and sprat could lead to changes in M estimates for the other prey age classes. It is therefore recommended to test the robustness of the model outcomes to the values chosen for the mean weights of the age 0-group.



Annual natural mortalities (M=M1+M2) by species and age. Black dots are the sum of quarterly M1 and M2; the blue line is a gam spline estimate.

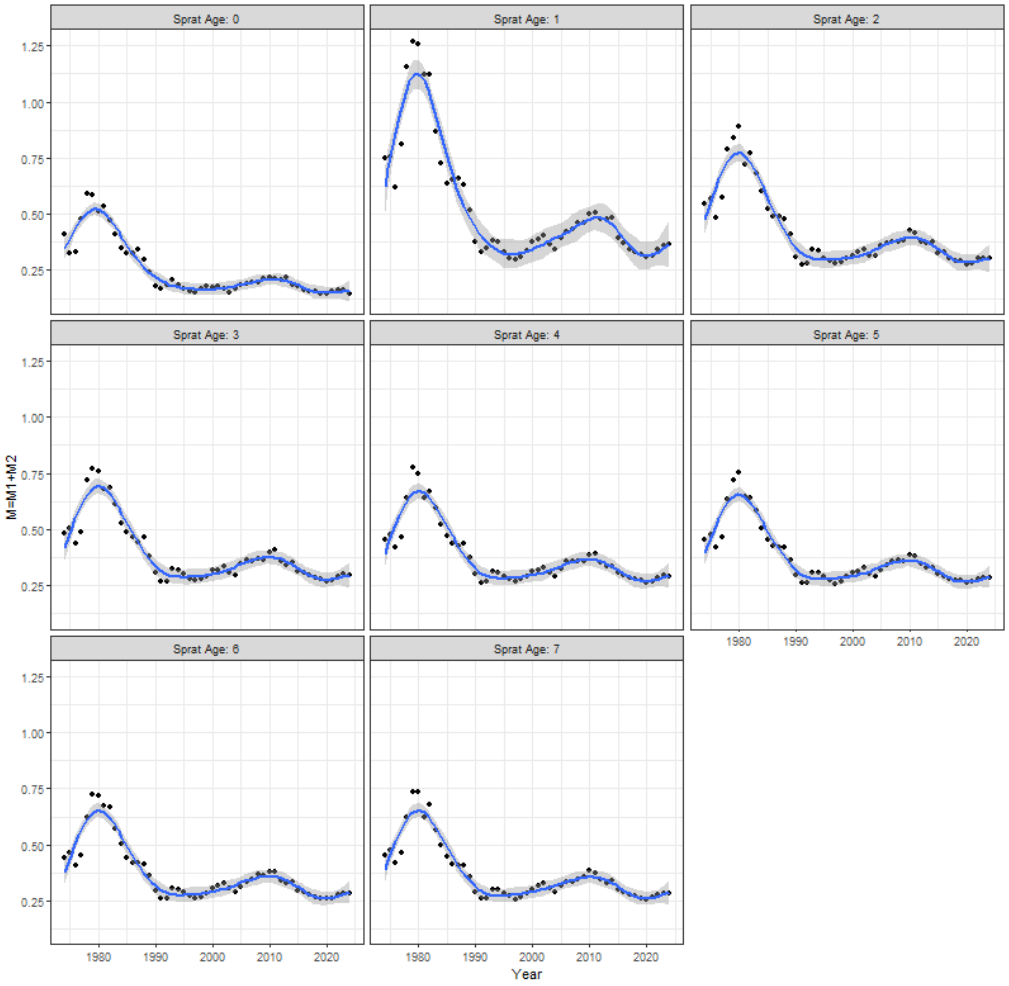


Figure 2.9. Annual natural mortalities (M=M1+M2) by species and age. Black dots are the sum of quarterly M1 and M2; the blue line is a gam spline estimate.

Some of the sensitivities of the model were quantified by running the following protocol:

1. Retrospective analysis (successive 5-year peel of all input data);
2. Robustness test of results towards a change in the residual natural mortality (M1).

In addition, the effects of small model changes that had to be made, such as a new compilation method of the cod stomach contents, a correction of the food ration calculations, and a change in the fishing parameter configuration were tested on the model outcomes, effects of these changes on the model outcomes are described in section 2.6. In 2019, several other robustness tests were conducted (overview in Table 2.1). These were not repeated for the current model input data and are therefore not formally part of this review.

The retrospective analysis results in small shifts of F, recruitment and SSB (Figure 2.10), yet consistent estimates of successive M2 values across clupeid age classes were obtained as additional years of assessment data were added to the key run configuration (Figure 2.11). We therefore conclude that these point estimates might be regarded as robust to the variability in assessment data of herring and sprat and not strongly dependent on one specific year of data.

Table 2.1. Changes of parameters values tested on the predictions of SMS in 2019, 2022 and 2025. The tests conducted in 2019 were not repeated for the current model input data and are therefore not formally part of the review.

Test	Reference	Tests performed	Executed	Remarks
Retrospective analysis	Annex 3, section 5.2.1	Peeling off 5 years of data	2025	
Sensitivity to M1 value	Annex 3, section 5.2.7	Double of original value for sprat and herring	2025	
Retrospective analysis	Annex 3, section 5.2.1	Peeling off 5 years of data	2022	
Change in M1 value	Annex 3, section 5.2.7	Half of original value for sprat and herring	2022	
Change in aggregation period stomach data	Annex 3, section 5.2.4	5 or 10 year aggregation period	2019	
Change in consumption rates	Annex 3, section 5.2.6	Value at half of original value	2019	
Overlap index other food	Annex 3, section 5.2.5	With and without index	2019	
Change in stomach data	Annex 3, section 5.2.2	New and old stomach data	2019	New data used since 2019
Change in shape size selection function	Annex 3, section 5.2.3	Uniform, constraint uniform, Log-normal	2019	Log-normal size selection was chosen and used in 2022
Comparison with Baltic Sea Gadget model	WGSAM 2019		2019	



Figure 2.10. Retrospective analysis for herring and sprat. Summary output.

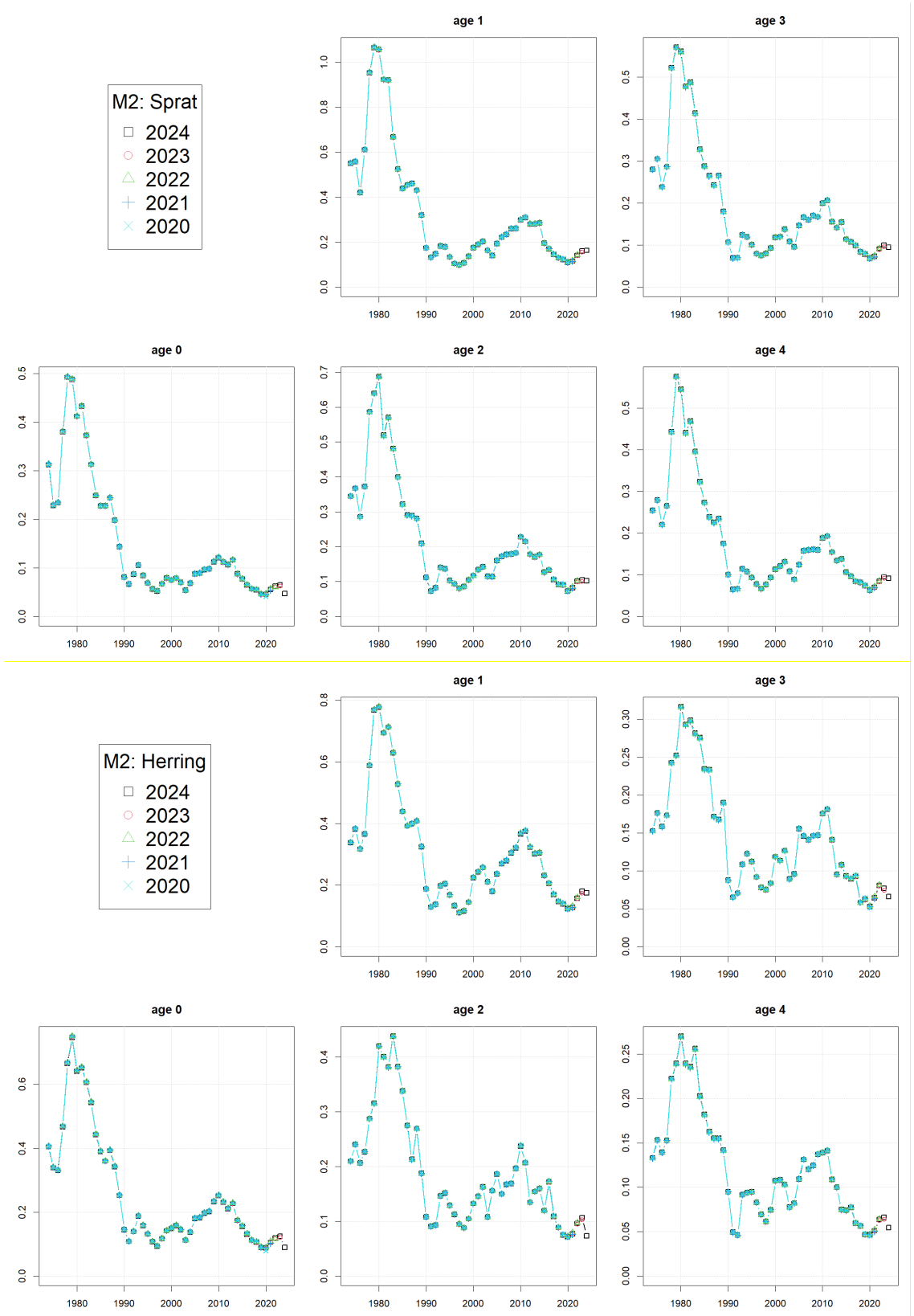


Figure 2.11. Retrospective analysis for herring and sprat, M2 at age.

Residual mortalities M1 are assumed to be constant and independent of age class in the model parameterization. M1 values for Herring and Sprat are respectively set to 0.1 and 0.2 in this as in the previous key run; there is no empirical evidence available to support a choice in this value. To test the effect of M1, a simulation was performed with a doubling of the value for M1 for sprat and herring. The results show a better overall fit mainly to stomach contents and catches (Table 2.2) and lower M2 estimates for both clupeids across all age classes (Figure 2.13). Furthermore, the higher M1 values resulted in higher recruitment, a lower F and a higher SSB estimate for both herring and sprat (Figure 2.12).

Table 2.2. Unweighted objective function contributions for the baseline and sensitivity (M1*2) runs.

	Catch	CPUE	SSB/R	Stomach	SUM
Baseline key run					
Cod	0	0	0	-331.1	-331
Herring	596.9	62.8	40.1	0	700
Sprat	1092.3	198.4	36.1	0	1327
SUM	1689.1	261.2	76.2	-331.1	1696
Sensitivity (M1*2) run					
Cod	0	0	0	-399.8	-400
Herring	591.8	83.3	11.4	0	687
Sprat	1078.4	190.5	36.5	0	1307
SUM	1670.2	273.8	47.8	-399.8	1592

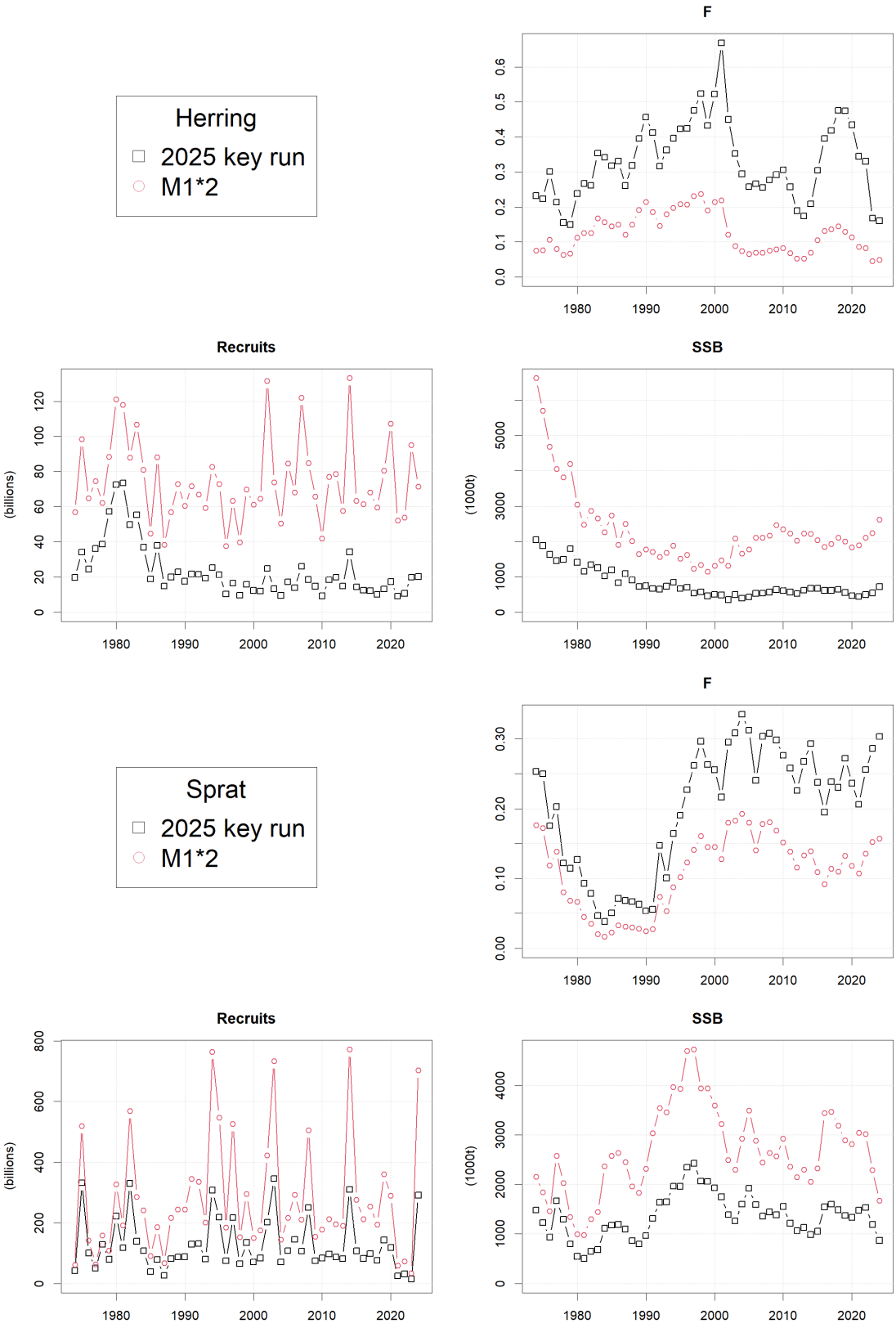


Figure 2.12. Effect of increased M1 values on F, Recruitment and SSB for herring and sprat.

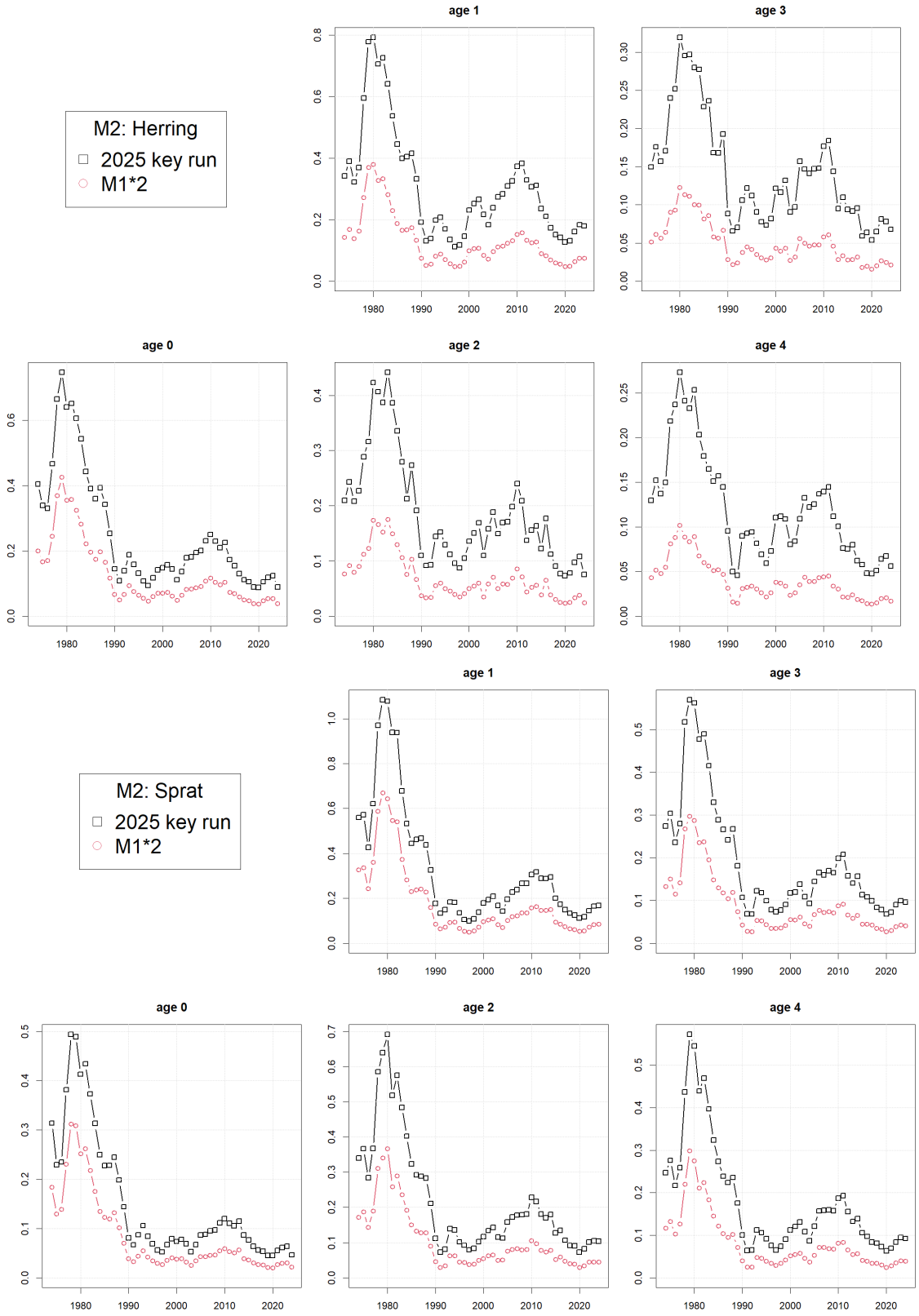


Figure 2.13. Effect of increased M1 values on age-specific M2 estimates for herring and sprat.

The 2019 review states that the predator-prey size selection has quite a strong effect on old M2 estimates (ICES, 2019). In general, the diet partitioning of cod between herring, sprat and the other food component in the model is uncertain. The residuals of the diet data fitting show a significant deviation in the model fit for recent years (Figures 2.7 and 2.8), which is an indication that the model is perhaps not capable of catching changes in cod diet, such as a shift from benthic to fish or from herring to sprat. Additional analysis of the stomach data may allow for a better understanding of potential changes in the diet of cod and the changes in the model setup that may be needed to simulate these dynamics.

One of the main uncertainties in the model is probably the cod stock sizes. Due to the issues with the age readings in Baltic cod, estimates for weight and catch at age that SMS needs for its calculations are currently not available. Therefore, cod is currently considered as an 'external predator' in SMS. In practice this means that the cod numbers are kept fixed and assumed to be known without error. In reality, the estimates of the cod numbers are likely to be quite uncertain, especially with the current problems with the assessment of the cod stock in the Baltic (ICES, 2022). The effect of different values for the cod abundance on the model estimates should be tested in the future.

2.6 Previous peer review

The SMS methodology has been reviewed in ICES (ICES, 2012a, 2012b, 2019, 2023a).

No changes in the model were made in the 2025 key run compared to the 2022 key run, except that cod abundances now are forecasted values (2023–2024). The two key runs show highly consistent results for the summary output of recruitment, SSB and mean F (Figure 2.14). For herring, estimates of F, SSB and recruitment are very similar. After 2016, the two key runs deviate slightly, with the 2022 key run estimating slightly higher recruitment compared to the 2025 key run. For sprat, the estimates of SSB and recruitment are similar for the entire time-series between the two key runs, while for F, the 2025 key run estimated slightly lower F across 2005 and onwards..

In terms of M2 (Figure 2.15), the two key runs show similar results with some discrepancies for certain age groups and years in herring. For herring, the 2022 key run estimates slightly higher M2 for age 1 herring after 2017.

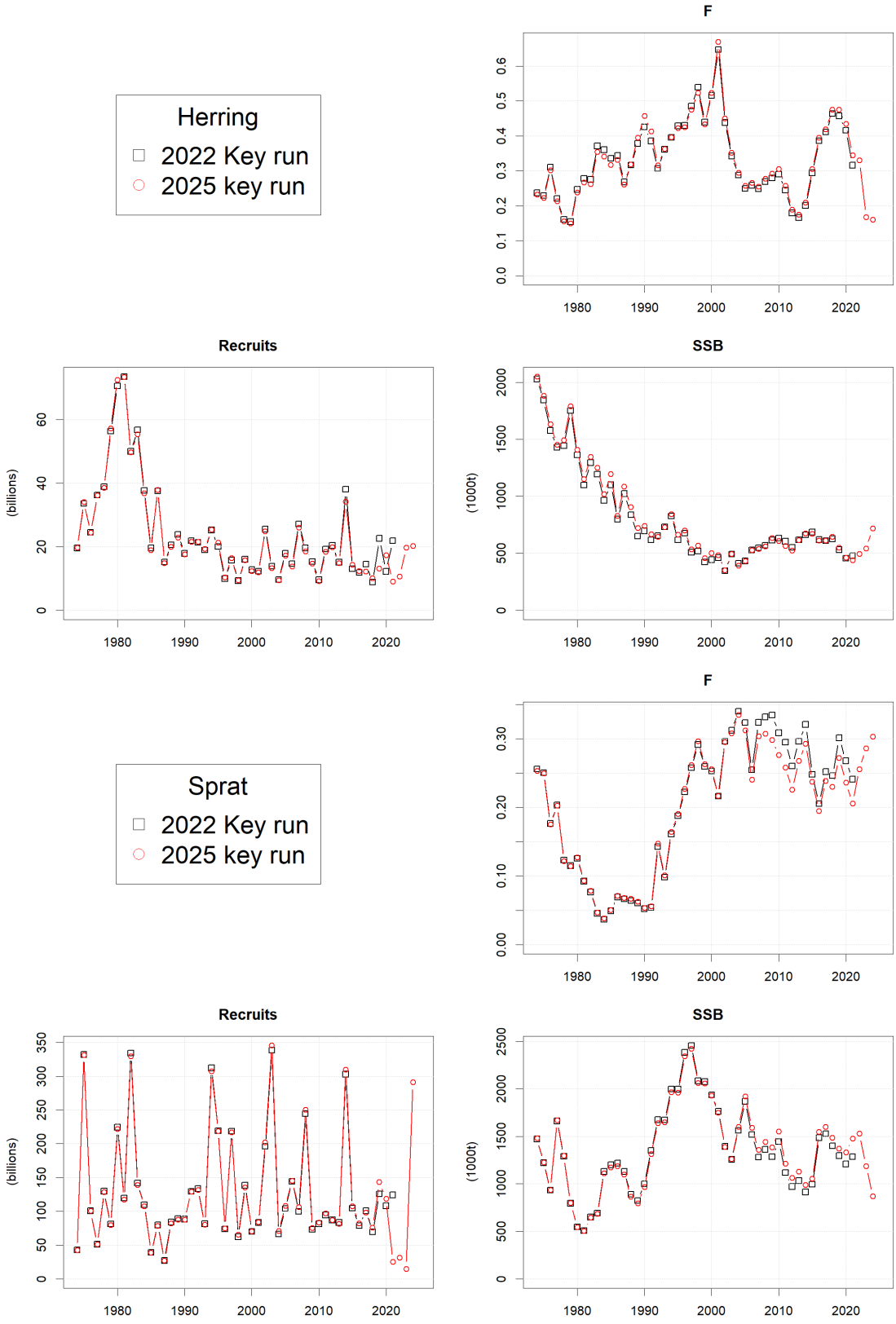


Figure 2.14. Comparison of fishing mortality (F), recruitment (Recruits) and spawning stock biomass (SSB) estimates between the 2019 (black squares) and the 2022 (red circles) Baltic Sea key runs for herring (above) and sprat (below).

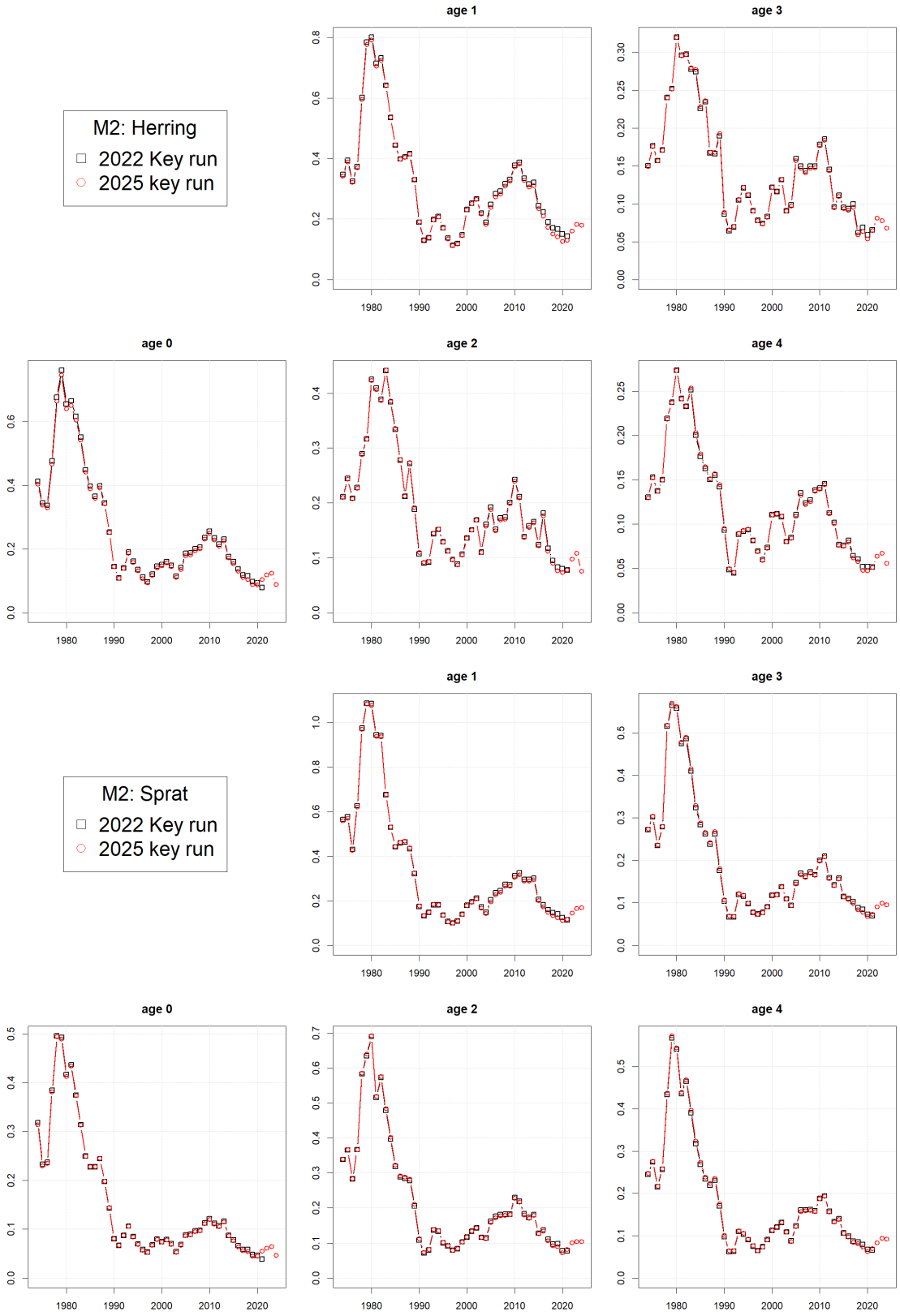


Figure 2.15. Comparison of natural mortality (M2) estimates between the 2019 (black squares) and the 2022 (red circles) Baltic Sea key runs for herring (above) and sprat (below).

2.7 Review recommendations

WGSAM accepts the model output from SMS as key run with the settings given in the Stock Annex (Annex 3 of this report).

Key run summary sheet

AREA	BALTIC SEA
Model name	SMS
Type of model	Age–length structured statistical estimation model
Run year	2025
Predatory species	Assessed species: Herring , Sprat
Prey species	Herring, Sprat
Time range	1974–2024
Time step	Quarterly
Area structure	Eastern Baltic Sea, ICES sub-divisions 25-29, excl Gulf of Riga
Stomach data	Cod: 1974-2014
Purpose of key run	Making historic data on natural mortality available and multispecies dynamics
Model changes since last key run	All time-series updated.Cod abundance in the model is the forecasted values from the 2023 assessment. Survey time-series updated since benchmark for sprat and herring. .
Output available at	Sharepoint/data/Eastern Baltic SMS key run and https://github.com/ices-eg/wg_WGSAM
Further details in	Report of the Working Group on Multispecies Assessment Methods 2025

WGSAM considers the key run as currently best possible run with SMS to provide natural mortality estimates. WGSAM recommends to use these values as input to single species stock assessments. Whether the raw or smoothed natural mortality values should be used in the herring and sprat assessment models is up to the stock assessors and is usually a decision that is reviewed and agreed upon during Benchmarks. The full time-series should be used and not only an update for the years after the last key run in 2012.

However, there are also clear limitations with the approach and results have been shown to be sensitive to e.g., consumption rates and treatment of "Other Food" as well as the size selectivity of cod (ICES, 2019). In addition, the results are likely to depend to a large extent on the outcome of the ICES Eastern Baltic cod assessment. The Eastern Baltic Cod assessment was not updated for 2024 due to issues with recent data. WGSAM used the last accepted assessment from WGBFAS (ICES, 2023b) with the associated forecasted values for cod abundance. The group recognizes that this is a hard limitation in the model and emphasizes that the current key run cannot be performed in the future without estimates of cod abundance.

Any bias in the cod stock assessment directly influence the predation mortality estimates. Assumptions around other food and constant vulnerabilities and uncertainties in the body sizes and abundance of age 0 prey individuals may also bias the natural mortality estimates to some extent.

For further work on the Baltic Sea SMS key run, WGSAM recommends the following:

1. Additional analyses of the cod diet to get a better process understanding of what is driving the systematic changes in relative stomach contents. Inclusion of the newer stomach data from after 2014.
2. A split of "Other Food" in parts where the time dynamic can be considered (e.g., flounder and *Saduria entomon*) directly or via a proxy and a part that still needs to be assumed constant in time may be beneficial. The availability of the different food groups for cod in the Baltic Sea have changed strongly over time (e.g. Haase *et al*., 2020).
   In 2019, the WGSAM reviewers made these two recommendations, on which no work has been done since then. The overestimations of the model in the amount of herring eaten by cod and the underestimations of other food and sprat (all in recent years, Figure 2.8), indicate that it is still a good idea to consider these recommendations. The recent changes in the condition of Baltic cod indicate that there has been likely a change in the food availability for cod. Although it could be difficult to implement these changes they could be quite important for reliable estimates of the model.
3. The inclusion of spatial dynamics in the food consumption of cod (either directly or via overlap coefficients) may improve the fit to data sources.
   In 2019, the WGSAM reviewers made this recommendation, which has not been considered so far. As long as the spatial overlap between prey and predator is constant through time, incorporating an overlap coefficient should not have a strong effect on the model predictions as this is currently addressed via the vulnerability parameter. However, if there is a temporal trend in the spatial overlap between predator and the prey species, and/or if the prey age classes are unevenly distributed through space, an overlap coefficient (per prey age class) could make a difference for the model predictions.
4. 4. A run with age 1 as recruits could be tried because input for the 0 group is highly uncertain.
   In 2019, the WGSAM reviewers made this recommendation, which has not been considered so in the present key run. The idea is that the weight at age data for the age 0 classes of the prey species are so uncertain that the model cannot estimate the consumption correctly based on the data that is currently used. However, this may be a difficult change for the model as age 0 herring and sprat form an important part of the cod diet. It is relevant to highlight the uncertainty and potential issues with the quality of the estimates for the age 0 prey classes before they could be used as input in a single species stock assessment. Discussion with the stocks experts from the assessment working group on possible alternative data sources on the weight-at-age of age 0 clupeids would be beneficial.
5. 5. Account for the uncertainty in cod number at age in the model.
   In 2019, the WGSAM reviewers made this recommendation, which has not been considered so far. Due to the current problems regarding the age-reading of Baltic cod, an age-length-based model is used for the stock assessment. Cod number-at-age in the stock are retrieved from the stock assessment and used to represent the cod stock in the SMS key run without uncertainty. In 2025 this issue is compounded by the key run using the forecasted cod data for 2024 and parts of 2023.
6. Provide a multispecies F_{MSY} estimation that can be used as a comparison with the F_{MSY} values from the stock-by-stock (SS) frameworks currently used for advice.
   The current model version of SMS of the Baltic Sea is not suitable for predictions of multispecies F_{MSY} values. Unfortunately, due to the non-dynamic nature of cod in the 2025 key run, it is not possible to predict how fisheries mortality may affect the other species in the food web through indirect effects. It is advisable to calculate this metric in the future if possible. The consideration of optimal yield trade-offs amongst the commer-

cially exploited species in the complex is an important facet in improving fisheries management advice and an ecosystem-based approach to fisheries management.

7. Consider inclusion of other predators than cod, such as seals or birds.
8. The value of M1 changes the estimate of M2 and F. A likelihood profile analysis could reveal which values of M1 for sprat and herring could lead to the best fit to data and additionally test of the sensitivity of the current assumptions.

3 Evaluating alternative stomach sampling designs

Prior to the meeting, the Baltic International Fish Survey Working Group (WGBIFS) expressed that the current sampling design of cod stomachs is too excessive. To assess the impact of reduced stomach sampling for standardized indices stomach content data, we conducted two model-based simulation experiments, using spatiotemporal GLMs with Gaussian Markov Random Fields (Anderson *et al.*, 2024; Lindmark *et al.*, 2025).

The first experiment is similar to the approach in Anderson *et al.* (2025), and can be described as follows:

1. Assume that all data collected between 1993–2023 was the status quo scenario.
2. Fit spatiotemporal models to individual-level relative prey weight (ratio of prey weight to predator weight) to three key prey groups (herring, sprat and *Saduria entomon*.
3. Estimate area-expanded indices of relative prey weight for these prey groups.
4. Define four alternative sampling strategies. For each scenario, randomly subsample the data, assuming that sampling strategy was in place in the entire time-series.
5. Replicate previous step 30 times.
6. For each of the 30 samples, and for each sampling strategy, refit the same spatiotemporal model and estimate the index.
7. Evaluate how much the coefficient of variation of the annual indices ($\mathrm{CV}_t = \sqrt{\exp(\mathrm{SE}_t^2) - 1}$), where SE_t is the standard error of the model-based annual index in year t, increase relative to the status quo index on average over the 30 year time-series, for each prey group, and how the median absolute annual relative error (where annual relative error is $(\hat{\theta}_t - \theta_t)/\theta_t$, where $\hat{\theta}_t$ is the index from the reduced sampling and θ_t is the index from the full data set.

The alternative sampling strategies considered where: one random haul per ICES rectangle, one fish per cm class (S1), one random haul per ICES rectangle and strata, one fish per cm class (S2), all hauls, two fish from each 5-cm class (S3), current method, but sampling every other year (S4).

In the second analysis, we wanted to assess how well declines in a predation index could be retrieved given different sample sizes and magnitudes of decline over a 10-year time period. The approach for the simulation-based power analysis was as follows:

1. Fit a simplified version of the model for each prey separately to the last 10 years of data (2014–2023)
2. Generate spatial data sets by sampling 100 random trawl stations, with 10 predators at each (with lengths assigned randomly), for 10 years
3. Simulate observations to above data given parameters estimated at 1). Replace the linear year slope with experimental values ranging from -0.05 to 0.
4. Repeat 30 times.

Results from the simulation experiments indicated that sampling according to scenario S3, i.e., the current sampling scenario but only 2 fish per 5 cm length class, would reduce the number of stomachs collected by nearly half, but only lead to only a small (~5%) increase in uncertainty and a minor increase in the median relative error (Figure 3.1). The power analysis indicated that the true year slope generally had to be smaller than 0.02 (slightly lower for *S. entomon*), for it to be retrieved in more than 90% of iterations even in the full data set (1000 annual samples); (Figure 3.2). For *S. entomon* and herring in particular, there was a tendency for reduced power to detect negative trends in more than 90% of iterations if the fraction subsampled was lower than 0.6.

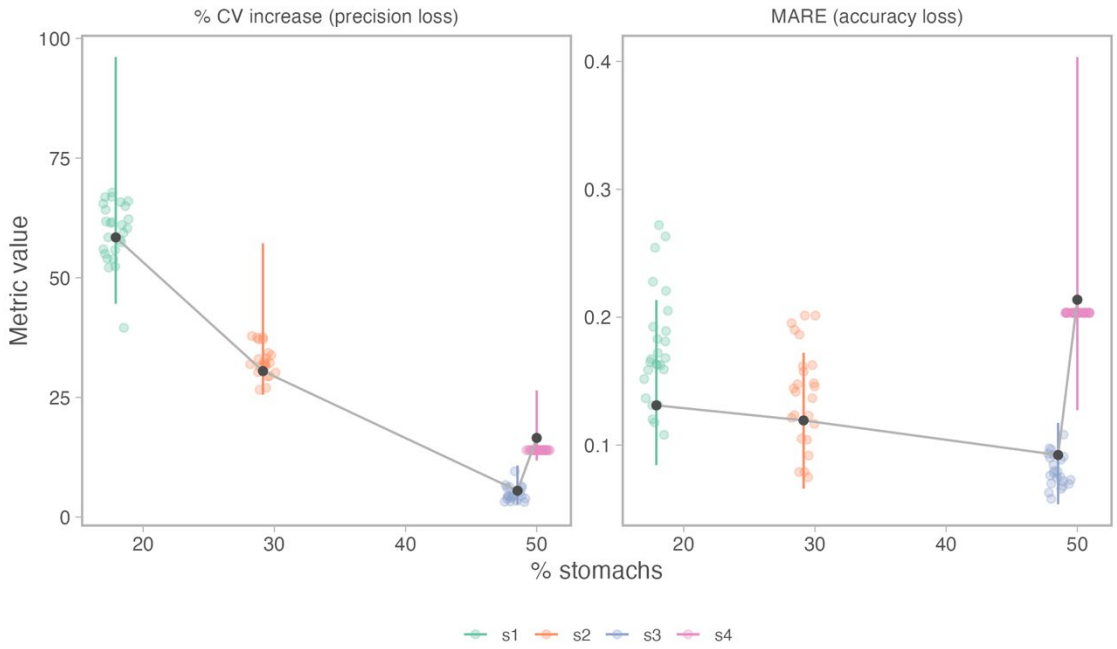


Figure 3.1. Effects of reduced sampling on index metrics. Relationship between % increase in CV (left) and the percentage of stomachs retained in each subsampling scenario (indicated by colour), and the median absolute relative error (right) and the percentage of stomachs retained. Each point represents an index value from a specific iteration and prey species.

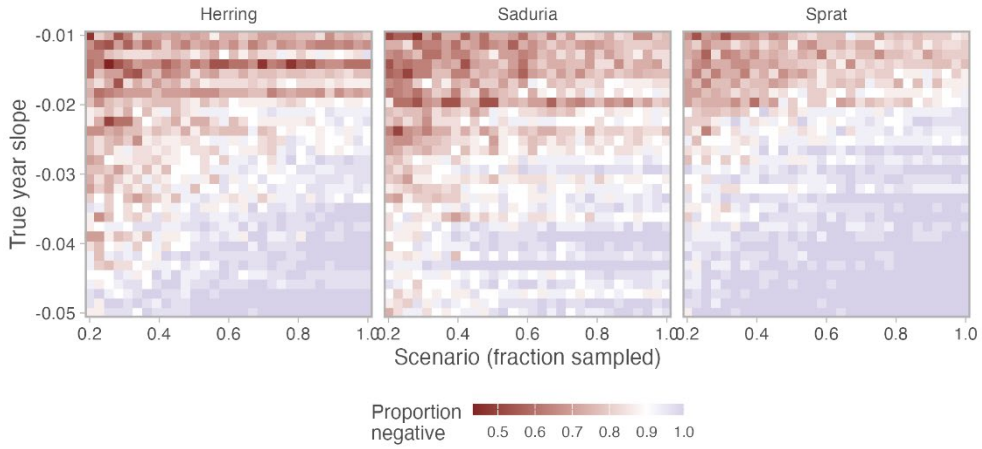


Figure 3.2. Ability of the spatiotemporal model to correctly identify a negative trend in data given different true slopes (y-axis) and fraction of stomachs retained (x-axis), where 1000 stomachs were simulated each year.

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Annex 1: List of participants

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Annex 2: WGSAM resolution

The **Working Group on Multispecies Assessment Methods** (WGSAM), chaired by Nis Sand Jacobsen, Denmark; and Michael Thomson, UK; will work on ToRs and generate deliverables as listed in the Table below.

	Meeting dates	Venue	Reporting details	Comments (change in Chair, etc.)
Year 2025	6–10 October	ICES HQ, Copenhagen, Denmark	Reports on keyrun reviews to be provided after each review is complete	
Year 2026	5–9 October	Leiden, Netherlands	Reports on any keyrun reviews that are completed	
Year 2027	October		Final report by 15 November to SCICOM	

ToR descriptors

ToR	Description	Background	Science Plan Codes	Duration	Expected Deliverables
a	Regional updates: Review further progress and deliver key updates on multispecies modelling and ecosystem data analysis contributing to modeling throughout the ICES region.	This ToR acts to increase the speed of communication of new results across the ICES area.	5.1; 5.2; 6.1	3 years	Report on further progress and key updates. Review and collaborate with appropriate EGs to revise sections on "species interactions" in the Fisheries Overviews.
b	Key-runs: Parametrisation of multispecies and ecosystem key-run models for different ICES regions. This includes standard update (limited to inclusion of recent data), extensive update (incl. new data and processes), and new key-runs.	Key-runs are models checked against high quality criteria, which are developed to contribute to a variety of operational objectives as part of the ICES advice, i.e. provide information on natural mortality for inclusion in single species assessments, estimates of multispecies reference points, large operating ecosystem models for MSE, etc.	5.1; 5.2; 6.1	3 years	Report on output of multispecies models including stock biomass and numbers and natural mortalities for use by single species assessment groups and external users.
c	Skill assessment: Establish benchmarking methods to assess the performance of multispecies models in-	This work is aimed at assessing the performance of models intended for strategic or tactical management advice. Evaluation will	5.1; 6.1; 6.3	3 years	Report on technical requirements for cross-models standardisation and comparison. Manuscript(s) on performance assessment

	tended for operational advice.	require work towards standardisation for cross- model comparison. This ToR will also deal with evaluation of methods for fitting to data, model calibration, diagnostics, and data treatment in the context of multispecies modelling.		of wide array of multispecies models based on a large simulation study.
d	Multi-model advice: Evaluate methods for generating advice by comparing and/or combining multiple models.	This work is aimed at addressing structural uncertainty in advice by reviewing methods for multi-model inference and multispecies applications	5.1; 6.1; 6.3	3 years	Report on methods for comparing models and for constructing model ensembles. Update and refine software implementing ensemble models. Report on case examples from both simulation testing and real studies.
e	MSE: Evaluate methods and applications for multispecies and ecosystem advice, including evaluation of management procedures under the uncertainties of environmental change.	This ToR uses multispecies and ecosystem approaches to understand the resistance and resilience of ecosystems to human activities under the uncertainties of environmental change. Through the use of simulations, management strategies and harvest control rules can be evaluated for robustness to uncertainties related to species interactions, changing climate, and ocean uses.	2.5; 5.2; 6.1	3 years	Review methods to evaluate populations and ecosystem resilience. Review of methods for management strategy evaluation which incorporate the effects and uncertainties of climate change.
f	Reference points: Define, calculate and adjust multi-species reference points considering interactions between species.	This ToR will look into biological reference points in a multispecies context. This includes multispecies reference points and single-species reference points applied in multispecies approaches, as well as ex-	1.4; 5.1; 5.3; 6.3	3 years	Report on methods for multispecies effects on reference points.

	ploring the feasibility of single-species reference points when accounting for species interactions.	

Summary of the Work Plan

Year 1	All ToRs
Year 2	All ToRs, update keyrun North Sea
Year 3	All ToRs

Supporting information

Priority	The current activities of this Group will lead ICES into issues related to the ecosystem effects of fisheries under multiple sources of uncertainties incl. climate change. The activities will provide information (e.g., natural mortality estimates, performance of indicators, multispecies reference points) and tools (e.g., multi-model ensembles, keyrun models) valuable for the implementation of an integrated advice and the application of a precautionary approach in several North Atlantic ecosystems. Consequently, these activities are considered to have a high priority.
Resource requirements	The research programmes which provide the main input to this group are already underway, and resources are already committed. The additional resource required to undertake additional activities in the framework of this group is negligible.
Participants	The Group is normally attended by some 20–25 members and guests. Expertise in ecosystem dynamics, trophic interactions, modelling and fish stock assessment from across the whole ICES region.
Secretariat facilities	Standard EG support.
Financial	No financial implications.
Linkages to ACOM and groups under ACOM	ACOM, FRSG and assessment Expert Groups.
Linkages to other committees or groups	WGMIXFISH, WGBIFS, IBTSWG, WGECO, all EAMSG groups, WKCLIMAD.
Linkages to other organizations	None

Annex 3: Stock Annex for the ICES Eastern Baltic Sea SMS configuration

Working Group	Working Group on Multispecies Assessment Methods (WGSAM)
Date	October 2025 (WGSAM 2025 meeting)
Predatory species	Cod
Prey species	Assessed species: Herring, Sprat
Stock Assessor	Morten Vinther

Summary

The key run uses the SMS model (Lewy and Vinther, 2004) which is a stock assessment model including biological interactions estimated from a parameterised size-dependent food selection function. The model is formulated and fitted to observations of total catches, survey cpue and stomach contents for the Eastern Baltic Sea (ICES Sub-divisions 25-32, excluding the Gulf of Riga). Parameters are estimated by maximum likelihood and the variance/covariance matrix is obtained from the Hessian matrix.

In the present SMS analysis, cod is a predator, and herring and sprat are preys. The population dynamics of cod were estimated outside the model by ICES WGBFAS, whereas key runs before 2019 estimated cod stock size and cod cannibalism within the SMS.

Substantial changes of input data were part of the 2019 key run, but the 2022 and 2005 key runs are mainly an addition of stock assessment data of the last three years and a small correction of the food ration calculation. The 2025 estimated predation mortalities (M2) are consistent with the M2 values from the previous key run in 2022.

2025 key run

This key run includes an update of the input data (2022-2024) supplied by WGBFAS. The Eastern Baltic herring and sprat stocks were through ICES benchmarks in 2023 and the SMS data were updated accordingly. The cod assessment was benchmarked in 2019, but the assessment was downgraded in 2024 from a Category 1 assessment to a Category 3 assessment indicative of trends because of limited input data and inconsistencies in model estimates. SMS makes used of the ICES assessment from 2023 as this was last assessment approved by ICES. Cod diet data and food ration and were left unchanged since the 2022 key run.

Despite some changes in the input data for the two prey species due to their benchmark in 2023, the 2025 estimated predation mortalities (M2) are consistent with the M2 values from the previous key run in 2022, with only very small changes for the historic M2.

2022 key run

The 2022 key run includes revisions and updates of the input data (1974-2021). New assessment input data (e.g. catch at age numbers, mean weight and survey indices) , 2019-2021, from WGBFAS were included. Cod diet data were recalculated from the same stomach data observations (1974-2014) as used in the 2019 key run. Cod food rations were also recalculated, to correct for seasonal variation. The model configuration was changed slightly for fishing mortality. Overall these changes are not considered substantial, and the estimated key run results are close to the results from the 2019 key run.

2019 key run

A key run for the Eastern Baltic Sea SMS model, including data for the period 1974–2018 was produced at the 2019 WGSAM. This key run replaced the 2012 key run. The 2019 key run included revisions and updates of the input data. Major modifications were that cod is treated as an external predator and the use of newly available data on cod stomach contents sampled mainly by the Latvian Institute.

SMS was updated with the most recent data from WGBFAS 2019, i.e. data for Herring in subdivisions 25–29 and 32, excluding the Gulf of Riga (central Baltic Sea) and for Sprat in subdivisions 22–32.

Due to age reading problems for cod in the eastern Baltic, ICES has since 2019 applied an age-length based analytical assessment with the Stock Synthesis model (SS3). Natural mortality of cod is estimated within the SS3 model. Without input data by ages, and with estimated high and time variable natural mortality, SMS is no longer able to estimate cod stock numbers and predation mortality estimates on cod due to cannibalism. Instead, cod is now considered as an "other predator" where stock number and size distribution are assumed to be known without errors. Population numbers and size distributions were extracted from the SS3 output.

Consumption (food ration) of cod was revised to reflect the most recent knowledge of evacuation rates and temporal trends in cod consumption rates.

Diet data for cod were substantially extended by including the stomach content data from the EU Stomach Tender. This addition of data did not change predation rates on herring and sprat substantially, but increased the weighting of the stomach data in the model likelihood, indicating a higher quality of stomach data compared to the previously used data.

Model description

The SMS model (Lewy and Vinther, 2004) is a stock assessment model including biological interactions estimated from a parameterised size-dependent food selection function. The model is formulated and fitted to observations of total catches, survey cpue and stomach contents for the main stocks in the Baltic Sea. Parameters are estimated by maximum likelihood and the variance/covariance matrix is obtained from the Hessian matrix.

The following predator and prey stocks are available:

- External predator: cod;
- Prey: herring and sprat

The population dynamics of herring and sprat are estimated within the model.

A detailed description of the model can be found in Appendix 1.

Input data

The description of input data is divided into four main sections:

Analytical assessment stocks: Stocks for which analytical age-based assessments are done by ICES or can be done from data available from ICES. Data input are similar to those applied by ICES "single-species" assessments used for TAC advice, with some additional data.

External predator stocks: Stocks for which stock numbers are assumed known and given as input to SMS.

Diet and ration data: Diet data and food ration data for all predators (analytical stocks and external predators) derived from observed stomach contents data.

Additional data: Miscellaneous data.

Analytical assessment stocks

This group of stocks includes:

1) Herring;
2) Sprat;

"Single-species" input data, by default given by quarterly time steps, include

- Catch-at-age in numbers (SMS input file canum.in);
- Proportion of the catch-at-age landed, assumed 100% (file proportion_landed.in);
- Mean weight-at-age in the catch (file weca.in);
- Mean weight-at-age in the stock (file west.in);
- Proportion mature-at-age (file propmat.in);
- Proportion of M and F before spawning (file proportion_M_and_F_before_spawning.in);
- M, single-species natural mortality-at-age (file natmor.in);
- Survey catch-at-age and effort (file fleet_catch.in).

SMS uses quarterly time steps, so input catch data should preferably also be given by quarter which is also the case for the Baltic Sea.

Table 2.1.1. Overview of "dynamic" fish stocks used in SMS and their basis from ICES single-species advice.

Species	SMS		ICES Assessment			
	Species code	Max age	Stock area	First year	Age range (data)	time step
Herring	HER	8+	SD 25–29 and 32, excluding the Gulf of Riga (central Baltic Sea)	1974	1–8+	Year
Sprat	SPR	7	SD 22-32	1974	1–8+	Year

Discarding is considered to be negligible for both stocks.

Herring

Catch numbers at age

Data for 2021-2024 were copied from Excel sheets stored at the ICES WGBFAS SharePoint. ICES WGBFAS provided quarterly catch-at-age number and mean weights for herring for the period 2002-2018. The full data series are not presented in the WGBFAS report, but were kindly made available by Tomas Gröhsler. Catch at age for the period before 2002 data were copied from the 2012 SMS key run.

Mean weight at age

WGBFAS assumes that mean weight at age in the sea is the same as mean weight at age in the catch. This assumption is fairly unbiased for older fish even though fisheries may be concentrated in areas (southern part of the EB) with the largest individuals. Mean weight at age in the catch for the youngest fish is higher than the mean weight in the sea as these size classes are not fully selected in the fishery. The mean weight at age as used by WGBFAS (Figure 0-1) shows a clear temporal trend with a decreasing mean weight in the period 1974-2000.

The quarterly mean weight at age data from WGBFAS (2002-2021) combined with the 2012 key run data for the period 1974-2001 are presented in Figure 0-2 for the youngest ages 0 and 1. It is clearly seen that the mean weights for age 1 in quarter 2 do poorly link to quarter 1 and not at all to quarter 3.

It is assumed that the mean weight in the sea are the same as the observed mean weight in the catch. However, when calculation the mean weight at age in the sea, the observed mean weight at age in the catch for age 1 in quarter 2 was discarded and substituted by the average of the observed mean weight at age in quarter 1 and 3 (Figure 0-3). As the observed mean weights for the ages 0-1 are highly variable, the smoothed values, ages 0 and 1, were finally used as mean weight in the sea.

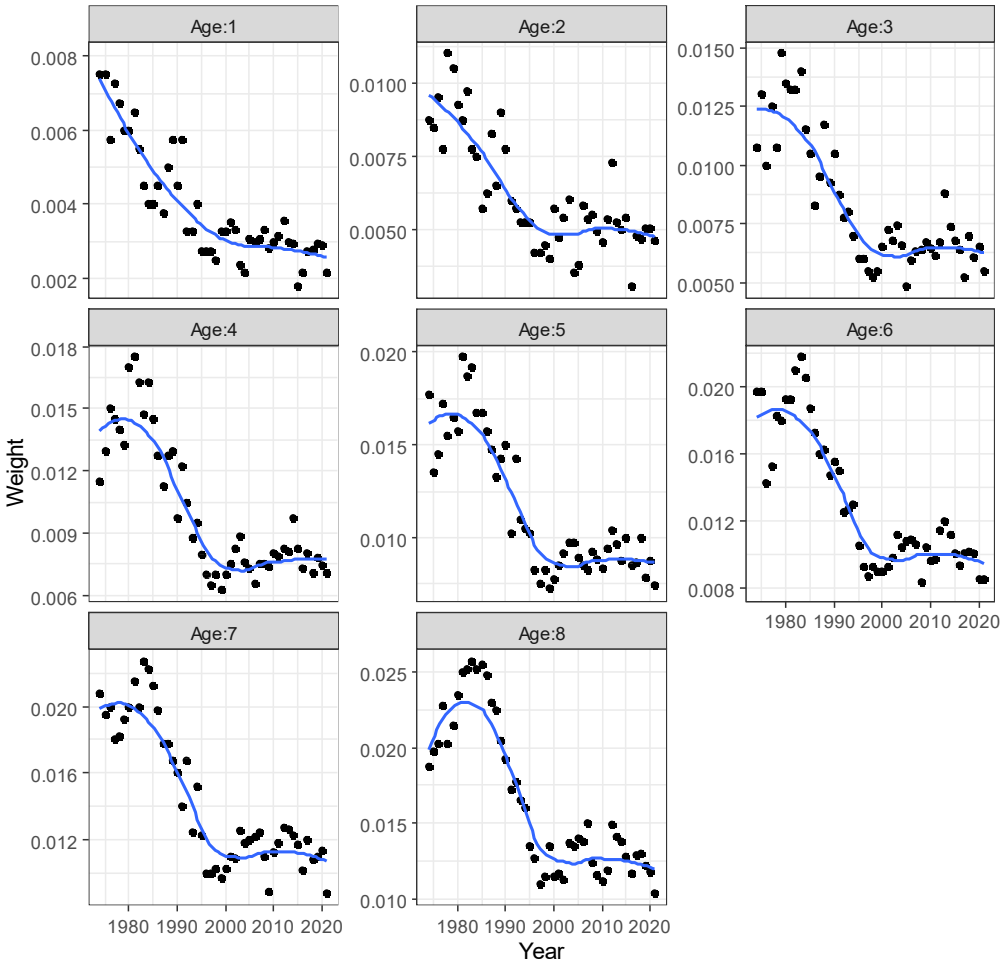


Figure 0-1. Herring mean weight at age in the catch (and in the sea) as used by WGBFAS. Dots show data points and the blue line is a loess smoother.

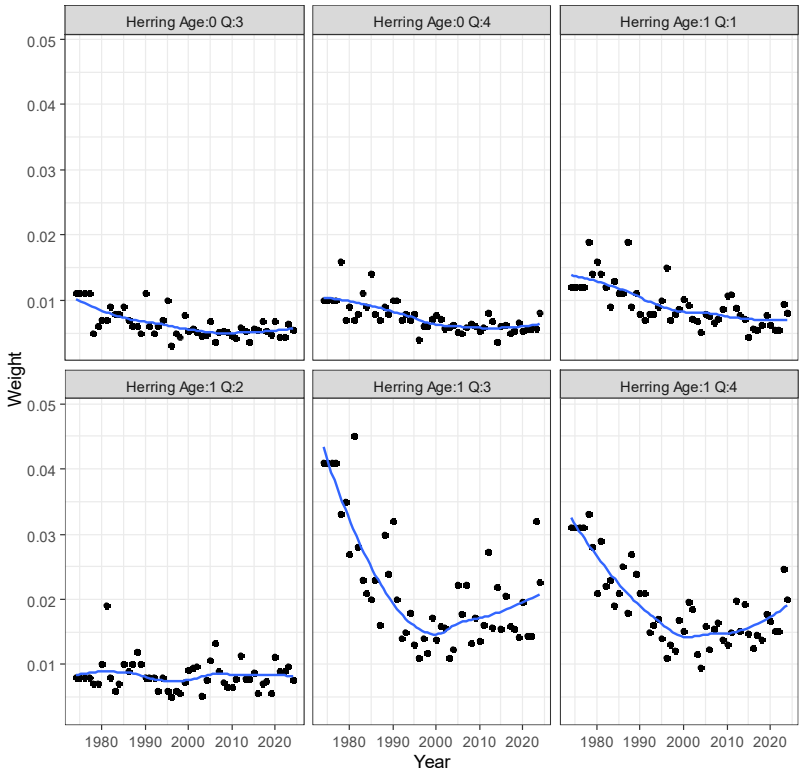


Figure 0-2. Quarterly herring mean weight at ages 0 and 1 in the catch as provided by WGBFAS. Dots show data points and the blue line is a loess smoother.

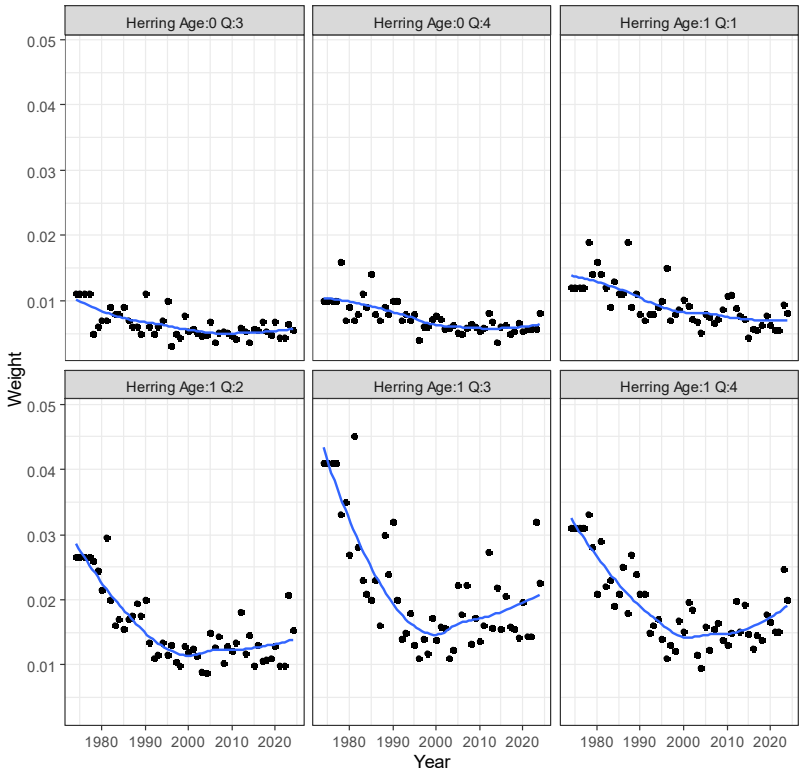


Figure 0-3. Quarterly herring mean weight at ages 0 and 1 in the sea as used by SMS. Dots show data points and the blue line is a loess smoother.

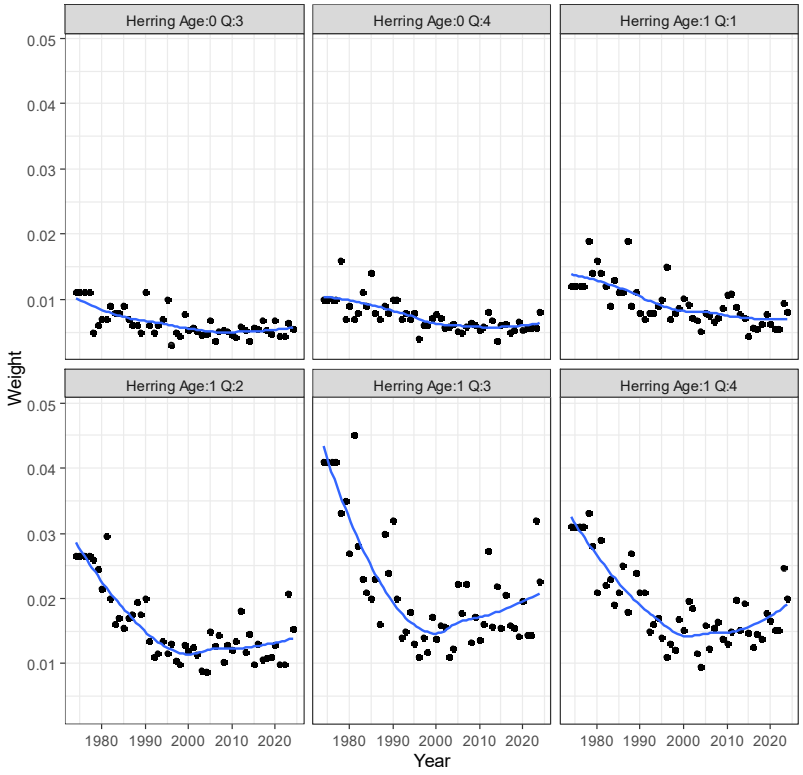
Other biological data

Proportion mature and M (used for "single species" SMS) at age data are copied from single-species data. WGSAM 2019 decided to use M1 at 0.025 per quarter for all ages. The 2012 key run applied 0.05, but for consistency with herring in the North Sea this was changed to 0.025 in 2019 and this is also the value used in the 2022 key run.

Survey data

Survey data are copied from the ICES single-species assessment.

SMS name	Years	Ages	alfa and beta	Source
Herring International Acoustic Survey October	2000-2024	1–7	0.0-0.3 (Q3)	WGBFAS, 2025



Sprat

Catch at age

Data for 2021-204 were copied from Excel sheets stored at the ICES WGBFAS SharePoint. Quarterly catch-at-age number and mean weights for sprat, 1998-2018, were provided by ICES WGBFAS. The full data series are not presented in the WGBFAS report, but were kindly provided by Tomas Gröhsler. Older quarterly catch at age data were copied from the 2012 SMS key run.

Mean weight at age

WGBFAS assumes that mean weight at age in the sea is the same as mean weight at age in the catch. This assumption is probably unbiased for older fish even though fisheries may be concentrated in areas (south-western part of the EB) with the largest

individuals. Mean weight at age in the catch for the youngest fish is probably higher than the mean weight in the sea as these size classes are not fully selected in the fishery. The mean weights at age as used by WGBFAS (Figure 0-4) show a clear temporal trend with a peak in mean weight around 1987 followed by a decrease until around 2003.

The quarterly mean weight at age in the catch from WGBFAS (1998-2018) combined with the 2012 key run data for the period 1974-1978 are presented in Figure 0-5 for the youngest ages 0 and 1. It is clearly seen that the mean weights for age 0 and age 1 in quarter 1 and 2 are highly variable from one year to the next. The same can be said about age 1 in quarter 3 and 4, but these quarters follow better the overall trend presented for the WGBFAS data (Figure 0-4). Due to the high (observation) variation in catch mean weights for ages 0-1, the smoothed values were used as mean weight at age in the sea (Figure 0-5) by SMS.

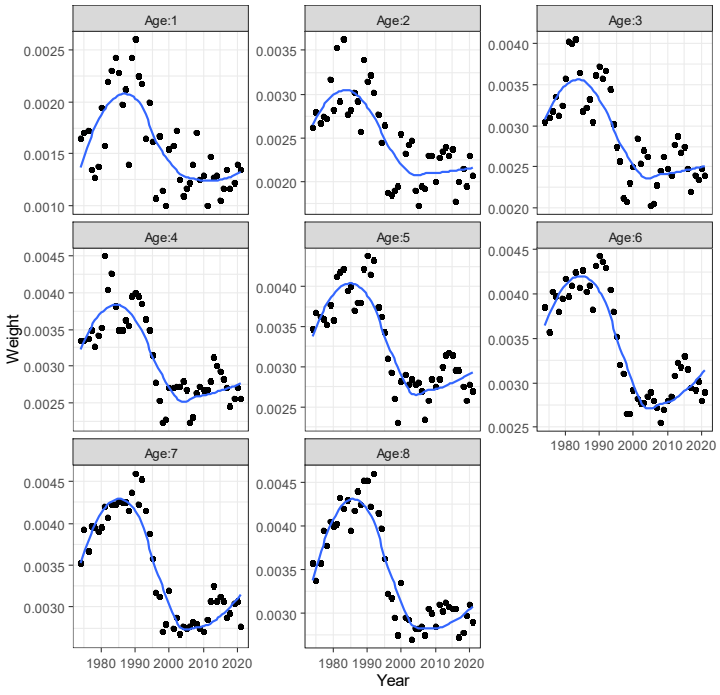


Figure 0-4. Sprat mean weight at age in the catch (and in the sea) as used by WGBFAS. Dots show data points and the blue line is a loess smoother.

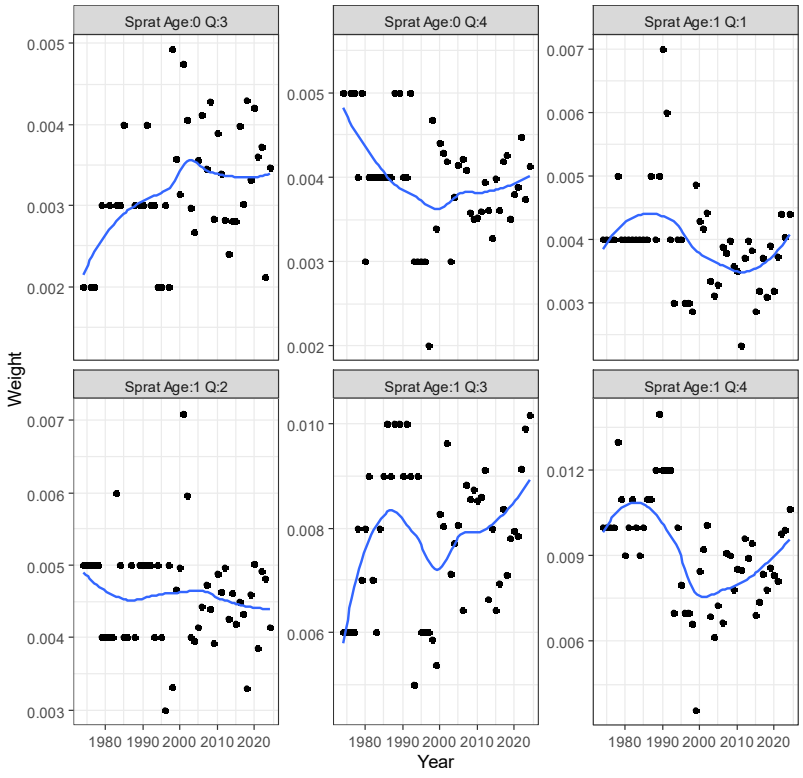


Figure 0-5. Quarterly sprat mean weight at ages 0 and 1 in the catch as provided by WGBFAS. Dots show data points and the blue line is a loess smoother.

Survey data

Survey data are copied from the single-species assessment

	NAME	YEARS	AGES	ALFA AND BETA	SOURCE
1	Acoustic in Oct.	2000-2024	1–7	0.0-0.33 (Q4)	WGBFAS 2025
2	Acoustic_in_May	2001–2018	1–7	0.25–0.50 (Q2)	WGBFAS 2025
3	Acoustic in Oct	1991-2008	1-7	0.0-0.33 (Q4)	WGBFAS 2025
4	LAT_RUS_acoustic shifted	2010-2025	1–1	0.0-0.0 (Q1)	WGBFAS 2025

Biological data

Proportion mature and M at age data are copied from single-species data. M1 is assumed to be 0.05 per quarter for all ages.

External predators

In the 2019 key run, cod was for the first time in the Baltic SMS treated as an "external predator". This means that the stock numbers are given by input, extracted from the ICES WGBFAS, Stock-Synthesis 3 (SS3) assessment for the stock. The SS3 assessment provides cod stock numbers and mean weight by 2-cm length classes for the main length classes. These data were aggregated into length classes used by SMS stomach contents data.

The SS3 results from the 2019, 2022 and 2023 assessments are almost identical (Figure

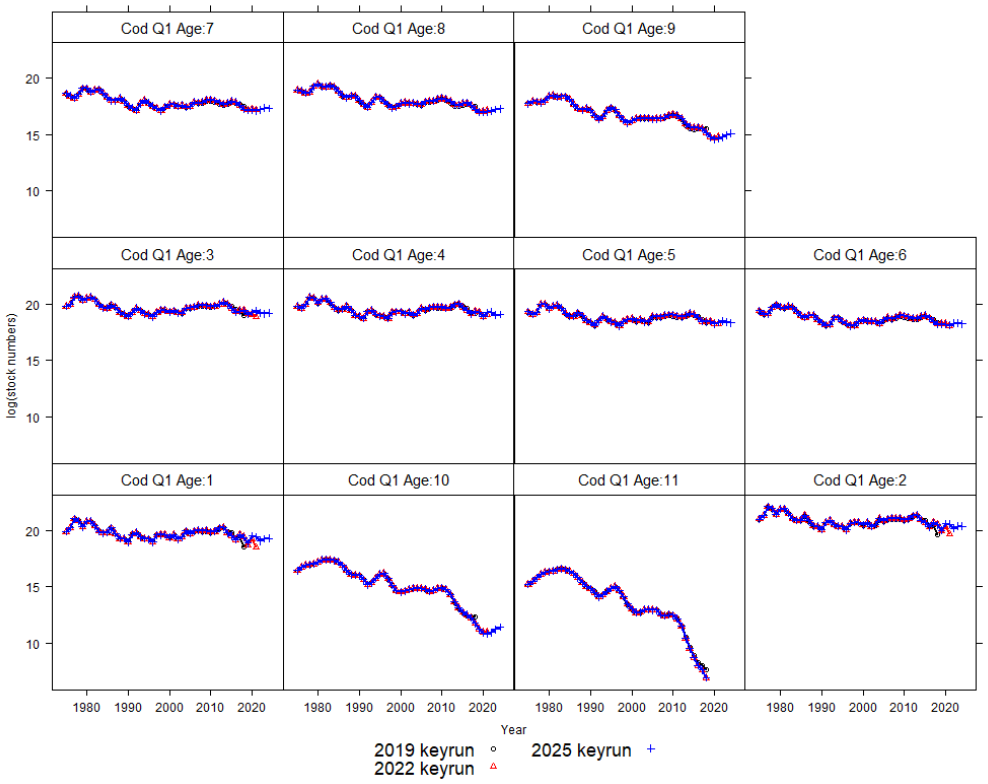


Figure 0-1. Stock numbers at age of cod estimated by the ICES SS3 assessments in 2019, 2022 and 2023.

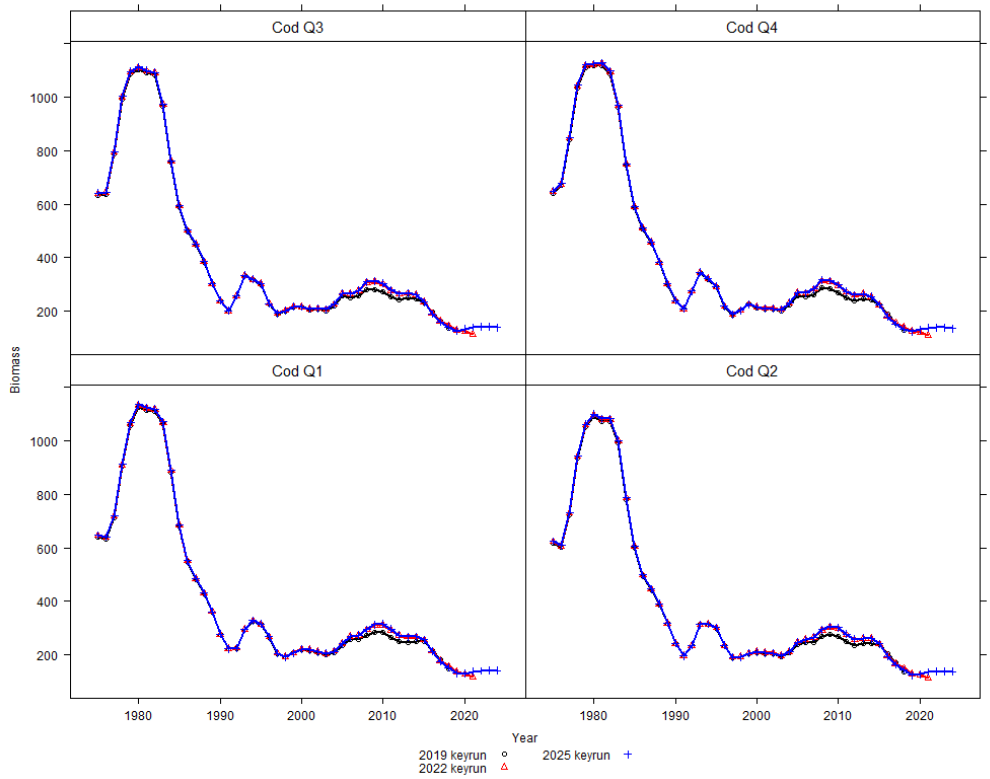


Figure 0-2. Biomass at age of cod estimated by the ICES SS3 assessments in 2019, 2022 and 2023.

Diet and ration data

The 2025 key run uses the same diet as the 2022 data (described below)

Fish stomach data

Two major datasets of cod stomach contents are available:

- "Old": International sampled stomach content data, 1977-1992, Individual stomachs were pooled by cod size class before analysis. The recorded sizes of both predator and are given by wide size classes, e.g. sprat by the size classes 5-10-15 cm, for the oldest data in the time series.
- "New": Individually compiled stomach sampled by mainly Latvia in the period (1963) 1974-2014. Predator and prey sizes are by cm or mm.

"old" pooled stomach data

An international database of Baltic cod stomach contents contains data from 62 427 cod collected during 1977–1994. The collation of national stomach content data sets into one set for multispecies assessment has mainly been done by DIFRES (now DTU Aqua) and the results were published in ICES papers (e.g. ICES 1991/J:30; ICES 1989/J:2; ICES 1990/Assess:25 and ICES 1993/J:11). The stomach contents data are available at "exchange format" from ICES (www.ices.dk).

The "old" data stomach contents data are recorded by year, quarter, predator, predator size class, prey and prey size class. Most stomachs were pooled within a haul and predator size class before analysis, such that diet data from individual fish are scarce.

For part of the time series, data were only provided (pooled) by country and ICES sub-division (SD), such that the variation between hauls could not be analysed.

"new" individually sampled stomach data

More than one hundred thousand stomachs of cod from the Eastern Baltic Sea have been sampled by trawling between 1963 and 2014, by mainly the Latvian institute. Sampling (Figure 0-1) covered the distributional area of the Eastern Baltic cod population (Bagge, 1994) except in the period 1995 to 2004, where sampling was limited to the north-eastern part. Stomach contents are provided by individual fish. Prey items in the stomachs were recorded at the highest possible taxonomic resolution with total mass, and, where identifiable, number of individuals and lengths per prey taxon. Prey sizes are given by mm or cm. Predator length was also recorded and in later years also predator weight (Huwer et al., 2014;). The stomach data are available at ICES (www.ices.dk), however the ICES download facility has not been working for the last 2 years!.

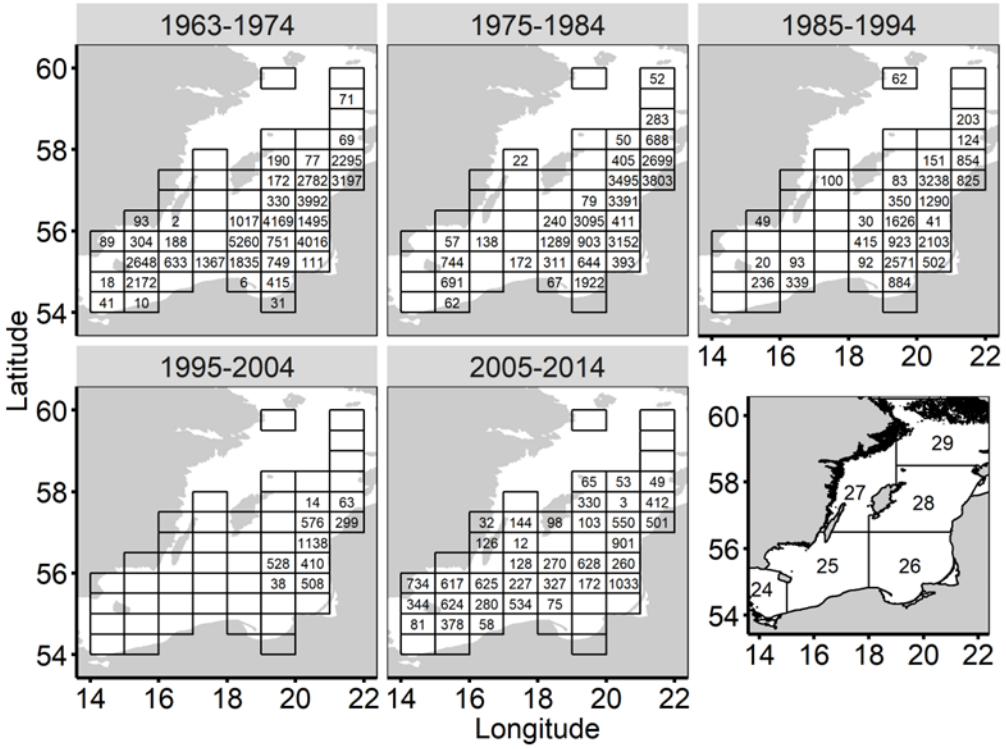


Figure 0-1. ICES sub-divisions (bottom right panel) and stomach sampling coverage: number of *Gadus morhua* stomachs by ICES statistical rectangle for each period specified on top of each panel. Source Neuenfeldt et al., 2020.

Pre-processing of data

In the Latvian data before 2000, prey weight is not given per prey item, but per prey species and digestion stage. To assign weight to each prey item, a length weight relation was first made for each prey species and digestion stage, based on stomach data from predators with only one prey item of a given species. Secondly, these length weight relations were used to assign a weight to each prey with size information. The sum of these weights cannot exceed the total recorded prey weight for the individual stomach. If the sum exceeded the total prey weight (of both sizes and un-sized preys), the mean weight of the sized preys were downscaled and prey items with no size

information was removed. If the sum was smaller than the recorded total weight, and the stomach included preys without size information, the difference in weight was assigned to the prey with no size information.

Compilation of stomach contents data into diet

The compilation of the individual stomach samples from trawl hauls into average diet of the Eastern Baltic Sea basically follows the technique given by ICES (1993). The compilation of stomachs contents for the 2022 key run was done using the FishStomach R-package (available from https://github.com/MortenVinther/FishStomachs).

The compilation of stomach contents data into diet data includes the following steps:

1. Read and check data from agreed exchange format

This step reads data from the ICES exchange format for stomach contents. The "new" data were initially transformed into a format similar to the ICES format as part of the paper Neuenfeldt et al. 2020. The function "read_exchange_data" from the FishStomach package was applied for reading the data in ICES exchange format; more information about the function can be found in the man-pages and vignettes of FishStomachs)

2. Pre-processing of "new" data

Weight by prey items were allocated where total prey weight was the only weight available – see previous section. FishStomach function prey_w_from_pooled_weight was applied.

3. Non assessment species into "other" prey species

Prey species other than herring, sprat and the unidentified mix of those ("Clupeidae" and "Clupeoides") were allocated to "other" species. As cod is an "other predator" cod preys are also included in "other". Function group_prey_species was used for this step.

3. Assign size classes for predators and preys

The length of predators of preys were from the "new" data were recorded in mm or cm the sizes were assigned to sizes classes using the length classes provided in Table 0-2. A wider size range, based on the initial applied size classes (e.g. 5-10-15 cm for preys) were applied for the "old" data set, or for combinations of the "old" and "new" data. Functions put_size_class_on_prey and put_size_class_on_predators were used.

4. Bias correct to take into account regurgitated stomachs within a sample unit

The "old" data compiled as a pooled stomachs, e.g. stomach contents from all 40-50 cm cod in a haul. For each stomach pool, data include the information on the number of a) empty stomachs; b) stomach with skeleton remains only; c) stomach with food and d) stomach with food, but regurgitated. In most cases, stomachs within a haul are pooled at the time of sampling for each predator size class. Only stomach contents from the feeding, non-regurgitated stomachs were recorded and later bulked to save time. In the calculation of the average stomach content, it was assumed that the regurgitated stomachs had similar stomach content as the (valid) feeding fish. The "new" data set does not include information on the presence of regurgitated stomachs, and no corrections were made. Function bias_correct_regurgitated was used.

5. Aggregate stomachs contents within sample_id and size classes.

Average stomach contents by prey species and size class are calculated for all stomachs within the same sample_id (by haul) and predator size class. Function aggregate_within_sample was used.

6. Stratification for allocation of missing data, and calculation of population diet

Strata are used to group data for calculation of mean proportions of e.g. the size distribution of a given prey, and for raising estimated stomach contents to population diet. Three strata were defined as: 1. stratum_time: year and quarter; 2. stratum_sub_area; ICES subdivision (i.e, SD25, SD26 and SD 28) and 3. stratum_area: ICES subdivision, same as above. Function add_strata was used.

7. Allocation of partly identified prey species to known prey species distribution within strata.

Non species identified Clupeoides were allocated to the proportion of identified herring and sprat, using a series of allocation keys calculated by groups of defined from combinations of the defined strata. First allocation keys are derived from data derived by the most "local" area and time period (e.g. within the same predator size class, year-quarter combination and SD), but if sufficient data could not be found for construction of allocation keys, the search area for data was expanded e.g. using data from the same year irrespective of quarter.

8. Allocation of identified prey species with missing size information to known prey sizes within strata.

Size information was allocated to prey items without a size measurement using the same approach as in step 7, with use of "local" allocation data as far as possible. Function redist_unidentified_prey_lengths was used.

9. Calculation of population diet.

The population diet should reflect the average diet of the predators within the model area. First, the average stomach content in weight of the individual prey and prey size classes was calculated by ICES SD as a weighted mean of the available samples within the SD, weighted by the number of fish in the sample. The absolute weights were then transformed into relative weights (proportions) of the individual prey and prey size classes. Finally the population diet was calculated as a weighted average of the prey proportion by the three SDs, weighted by the number of stomachs sampled within the SD. Function calc_population_stom was used.

This approach assumes that sampling of the stomachs largely follows the spatial abundance of the predator. This has however, not been the case in all years (see Figure 0-1).

Diet data for the 2019 key run were compiled following the same steps, using an early version of the same R-code as included in FishStomach, however, the diet dataset from the 2019 and the 2022 key run came out different (Figure 0-2). Differences are due to both a change in methodology (and coding) and in the approach taken to allocate partly identified prey items to fully identified preys or to the group of other prey. The raising procedure of stomach contents to population diet data has also changed.

Due to time limitations, the diet of the population was estimated based on the assumption that the observed stomach contents give an unbiased estimate of the diet. This is in contrast to the assumptions used for the estimation of food ration, as outlined in section below, where it is assumed that the stomach evacuation rate (and thereby

the food intake) depends on e.g. the energy contents of the stomach contents (as described by FishStomach function bias_correct_energy_etc).

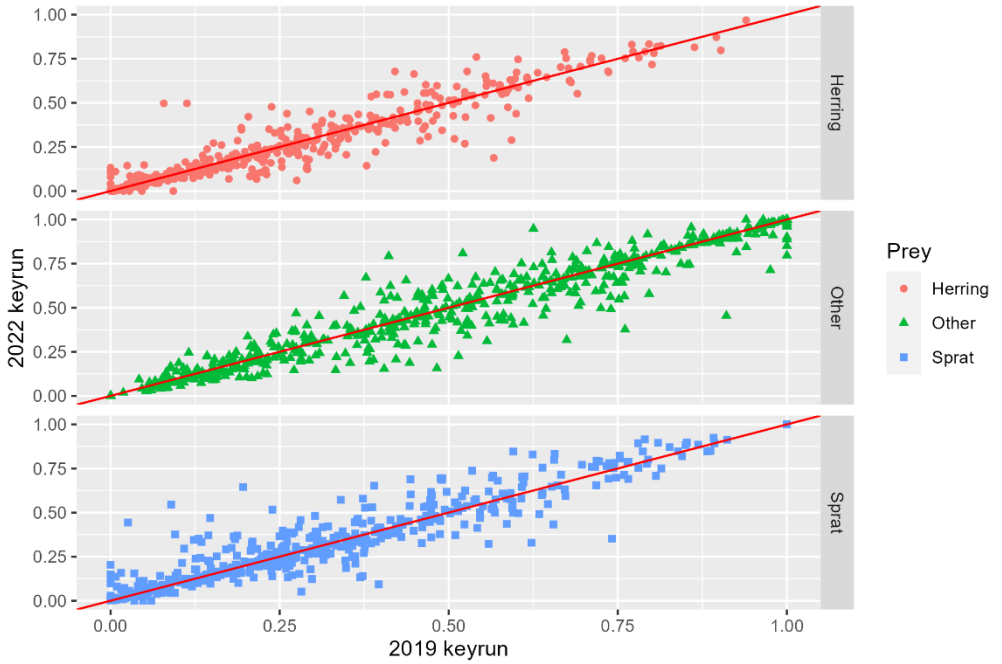


Figure 0-2 Comparison of prey proportions in the diet data used in the 2019 key run (x-axis) and in the 2022 key run (y-axis). Each dot represents the proportion of the prey in a diet observation (year, quarter and cod (predator) size class).

Estimation of food ration from stomach contents data

Average daily energy consumption rates C (kJ d^{-1}) were estimated using the cylinder gastric evacuation rate model (Andersen and Beyer, 2005a, b) by year and 1-cm predator length group for cod between 20 and 80 cm total length, amounting to 109 000 stomachs in this size range from the stomach database. Ambient temperature T was assumed constant at 5°C, corresponding roughly to the average temperature experienced by cod in the Baltic Sea (Righton *et al.*, 2010). Although cod experience varying temperature throughout the year, only significant trends in average temperature regime for the cod in their preferred habitat might potentially bias our analyses. Such trends have not been shown for the Baltic Sea. We assumed constant energy densities E_i for benthic prey (3.5 kJ g^{-1}) and consumed fishes (*Clupea harengus L.* (herring) and sprat 5.5 kJ g^{-1}, cod 4.0 kJ g^{-1} ;Pedersen and Hislop, 2001). E denotes the average energy densities (kJ g^{-1}) of the individually observed total stomach contents S (g). Using the principle that consumption rate C (kJ d^{-1}) on average over population and time equals evacuation rate (Pennington, 1985), and knowing cod total length L (cm) and the basic evacuation rate parameter ρ_0 = 2.43 x 10^{-3}, we used the parametrization of the cylinder model for cod presented in Andersen (2012):

$$C = 24\,\rho_0 L^{1.30} e^{0.083T} E^{0.15} \sqrt{S} \qquad (1)$$

In order to consider recent changes in cod consumption rate, the relationship between average quarterly consumption rate and total length (a priori parametrized as $C=aL^b$

with C the average quarterly consumption rate and L total length) was estimated separately for three different periods.

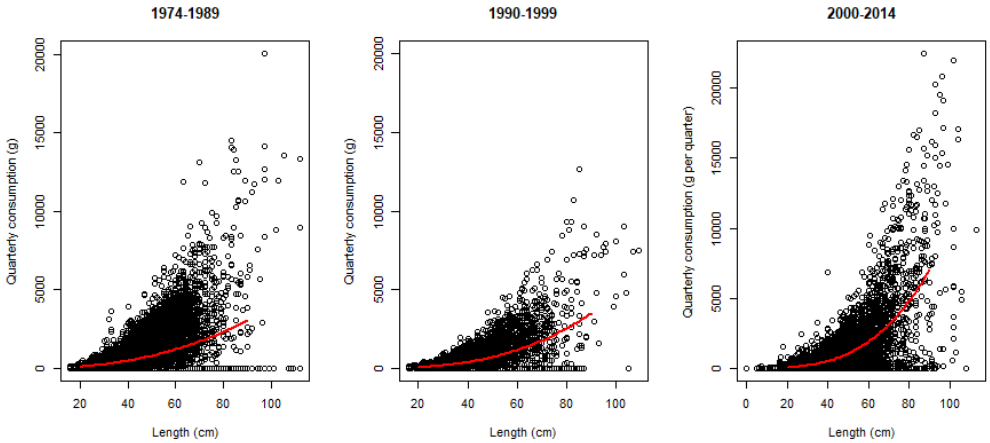


Figure 0-3. Scatterplots of cod total length and estimated quarterly consumption rate. The consumption rate has been estimated separately for 1974-1989, 1990-1999 and 2000-2014 in order to account for recent changes in cod consumption (Neuenfeldt et al., 2020).

Table 0-1. Parameter estimates for the consumption rate model, $C = aL^b$.

PERIOD	PARAMETER	ESTIMATE	STD. ERROR
1974-1989	*a*	0.10367	0.01184
	b	2.28617	0.02834
1990-1999	*a*	0.017408	0.003971
	b	2.713702	0.054565
2000-2014	*a*	0.003230	0.000354
	b	3.243353	0.025560

The stomach data do not include 2015-2021. For this reason, the 2000-2014 estimates were applied for 2015-2024, too.

Subsequently, average quarterly consumption was multiplied by 4 to give average ye arly consumption and then distributed over quarters according to the distribution ke ys given in Table 0-2. Even though specified in the stock annex for the 2019 key run, t he quarterly consumptions were (by an error) not redistributed over quarters, such th at the same consumption was applied for all quarters.

Table 0-2. Proportion of annual consumption by quarter of the year for different periods and size groups. The key was generated using all years (to account for only few data in the 3rd quarter). The length of cod (l.start and l.stop) reflect spawners and non-spawners.

year.start	year.stop	l.start	l.stop	q1_prop	q2_prop	q3_prop	q4_prop
1974	1989	15	30	0.27	0.23	0.25	0.25
1974	1989	31	120	0.22	0.16	0.30	0.32
1990	1999	15	30	0.24	0.22	0.27	0.27
1990	1999	31	120	0.21	0.19	0.31	0.29
2000	2019	15	30	0.30	0.16	0.16	0.38
2000	2019	31	120	0.38	0.19	0.11	0.32

Age length keys

Age length keys (ALK) are used by SMS to transform stock number at age into stock numbers at length for the calculation of predation mortality. Length at age is derived from weight at age in the sea using a length-weight relation. The length distribution for each age is derived from the coefficient of variation (CV) of the mean length at age as estimated from age and length observation from the BITS survey, quarter 1 and 4, 2000-2018. A year and quarter independent CV of mean length at age was derived from the estimated values by quarter (Table 0-1). These CV's (row "Used" in Table 0-1) are afterwards used to produce a length distribution around the mean length for a given age in a given year and quarter, assuming a normal distributed length distribution for each age.

Table 0-1. Coefficient of variation of mean length at age derived from survey data

Species	Quarter	Age								
		0	1	2	3	4	5	6	7	8
Clupea harengus	1	NA	0.14	0.09	0.11	0.13	0.13	0.13	0.13	0.12
	4	0.12	0.10	0.14	0.16	0.16	0.15	0.13	0.13	0.11
	Used	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13
Gadus morhua	1	NA	0.32	0.24	0.21	0.18	0.17	0.17	0.16	0.19
	4	0.34	0.25	0.22	0.18	0.18	0.17	0.18	0.18	0.18
	Used	0.34	0.25	0.23	0.20	0.18	0.18	0.18	0.18	0.18
Sprattus sprattus	1	NA	0.12	0.08	0.09	0.08	0.08	0.07	0.08	0.08
	4	0.10	0.08	0.08	0.08	0.07	0.07	0.06	0.07	0.07
	Used	0.10	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08

The total number of fish by length classes (Table 0-2) are finally calculated as the sum of contributions from each ages. The chosen length classes depend on the length classes used in the stomach data. The "new", individual sampled stomach data (see section 0) have used length classes by cm and mm, however boarder length classes were used due to the low number of stomachs sampled in the individual year and quarter combinations.

The "old" pooled stomach data (see section 0) used larger size classes, e.g. 5-10-15 cm for sprat, in the first years of sampling. This means that the applied length classes used in the SMS configuration depends on the actual used stomach data sets used. As an example, the length classes get wider than outlined in Table 0-2, when both the "old" and "new" stomach data are used. When both the "old" and "new" stomach data are

used, length classes are defined for each individual year, reflecting the widest length class in the particular year.

Table 0-2. Default length classes used for stomach data and ALK.

Species	Lower length (mm)	Species	Lower length (mm)	Species	Lower length (mm)
Gadus morhua	50	Clupea harengus	50	Sprattus sprattus	50
	100		70		60
	150		85		70
	200		100		80
	250		120		90
	300		140		100
	350		160		110
	400		180		120
	500		200		130
	600		220		140
	700		240		
			260		

Predator–prey overlap

The stock area for predator cod (SD 24-32 + part of SD 23) does not completely overlap with the stock areas for herring (SD 25–29 and 32, excluding the Gulf of Riga) and sprat (SD 22-32). This will not matter, given the normally applied assumption in fisheries assessment models that individuals of the population redistributes instantaneously, such that local fishing or predation do not affect the stock distribution.

Predator–prey species overlap is a quarter dependent parameter used in the calculation of food suitability (see equation 8 in Appendix 1). By default, the spatial overlap is set to one, but it is also estimated within SMS for a few combinations, where the "quarter effect" was estimated significantly different from 1.0.

Length–weight relations

Conversions from lengths into weights are used for some SMS configurations. The used parameters values are shown below.

Table 0-1. Length (mm) weight (kg) relation for herring and sprat (W=a*l^b)

SPECIES	A	B
Herring	2.997653e-09	3.136964
Sprat	3.670895e-09	3.107974

The l-w relations were estimated from BITS Q1 & Q4 data, 2000-2018 (minus 2004 data with errors). There is a statistical significant quarter effect in condition (parameter a), however this is ignored for use in SMS, until data for Quarter 2 and 3 data become available.

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Model configuration

The configuration of the SMS model aims firstly to mimic the results from ICES single-species assessment models when SMS is run in single-species mode (no estimation of predation mortality) using the same annual M values as the single-species assessment, and secondly to configure options for estimation of predation mortality.

Appendix 2 presents the SMS configuration (option files) used for the 2025 key run. This configuration is the same as used for the 2022 key, except the data update 2022-2024 and changes in survey data as applied at the ICES benchmark in 2023.

Fishing mortality

SMS uses a separable F-model (see equation 3 of the model description, in Appendix 1) while the ICES single-species assessments use SS3 (herring) and SAM (sprat)s supplementary models) for herring and sprat. Further differences; SMS is using quarterly time steps while the ICES assessment are using annual time steps.

A comparison of output from the two assessments shows quite similar results for herring (Figure 0-1). The "SMS-single sp." run uses the model configuration from the 2024 key run, run without estimation of predation moratlity.. The comparison for sprat (Figure 0-2) show that F and SSB have the same trend, but the levels are different between the ICES version and the SMS. SSB is estimated the 1st January in SMS but at spawning time in the ICES assessment (the proportion of M and F before spawning is set to 40%) which may explain the two levels of SSB estimated. SMS estimates consistently a lower F for sprat. This is because approximately 50% of the annual yield of sprat is taken in the first quarter (Figure 3.1 3). Within SMS, with quarterly time steps, such catch distribution means that fishing is removing most of the individuals, before the natural mortality decreased the stock considerably. This will result in a lower F compared to the ICES-single run, with annual time steps, where F and M is assumed constant over the full year.

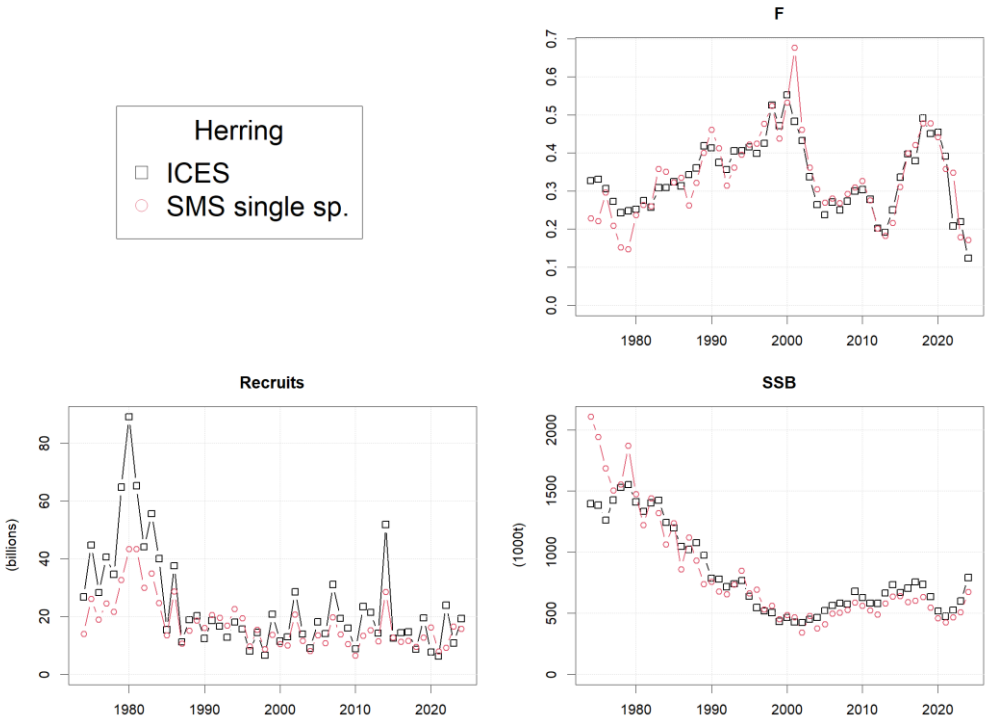


Figure 0-1. Comparison of the herring assessment results from SMS assessment using fixed M (from ICES assessment) and the ICES single species SS3 assessment. Recruitment is at age 0 for the SMS assessment and age 1 for the ICES assessment.

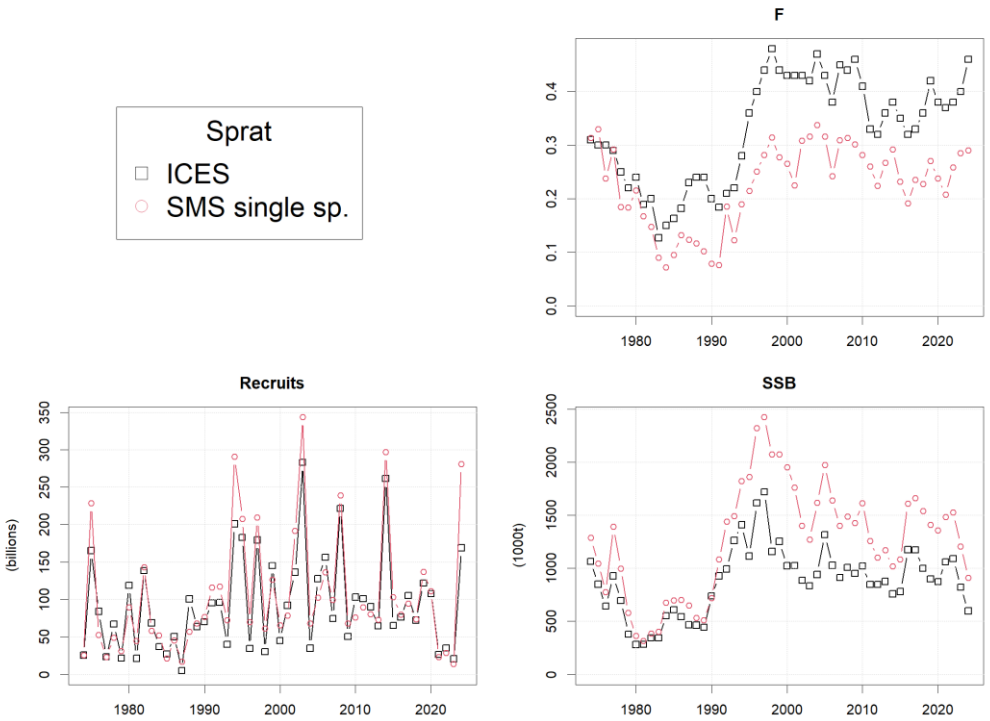


Figure 0-2. Comparison of the sprat assessment results from SMS assessment using fixed M (from ICES assessment) and the ICES single species SAM assessment.

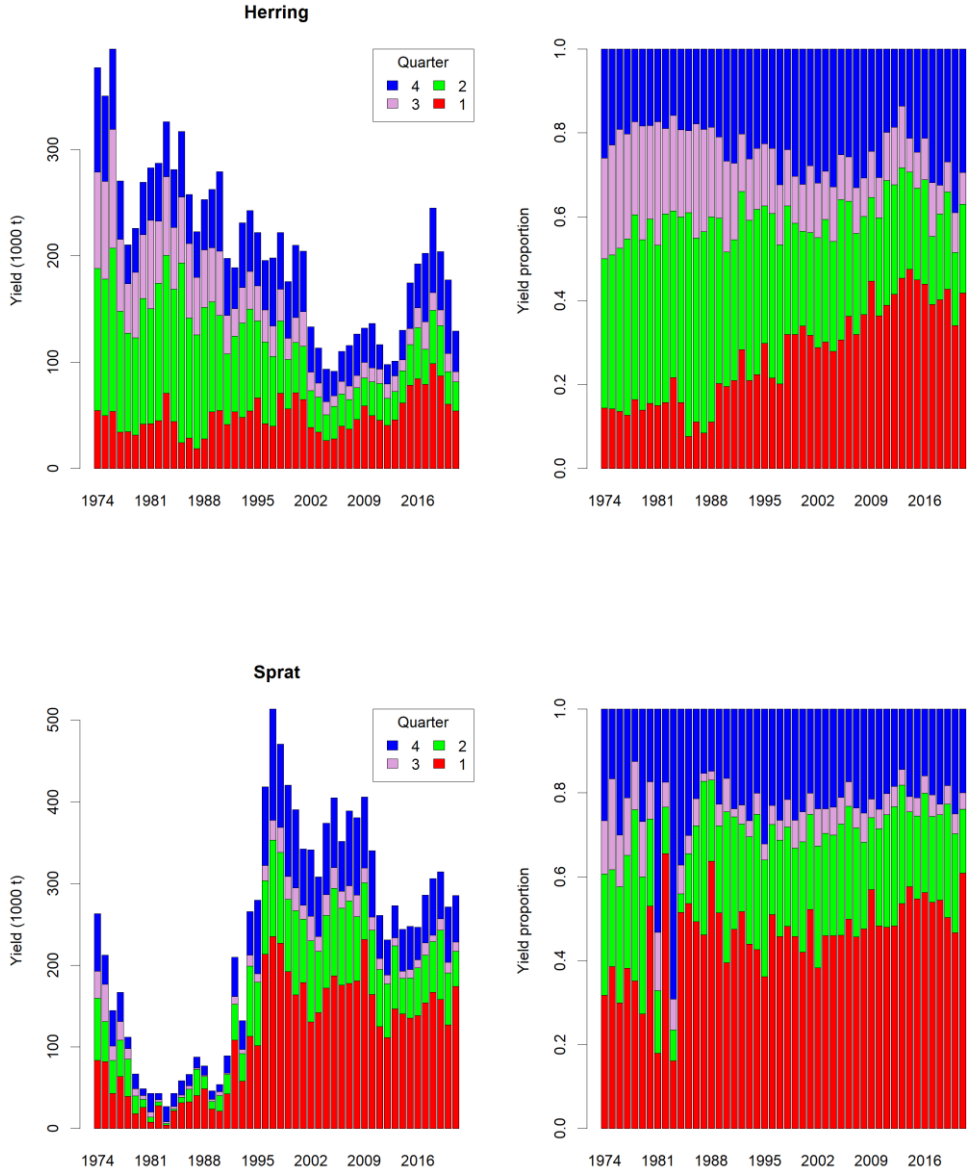


Figure 0-3 Yield by year and quarter in absolute terms and as proportions for herring and sprat.

SMS assumes that season and age effects in the model for F are constant within a specified range of years. For herring the 2021 key run used: 1974-1988, 1989-2004 and 2005-2021. The last period was extended, now (2005-2024). The sam extension of the last year range was also done for sprat.

Configuring predation mortality options

The 2025 key run applied the same configuration of the predation model as applied by the 2022 key run.

Background

The SMS model has three main options for size preferences of predators (see equations 11, 12 and 13 in the description of SMS model, Appendix 1):

1. Log normal size selection: a predator has a preferred prey size ratio and a prey twice as big as the preferred size is as attractive as another half the prey size. The preferred size ration and its variance are estimated by SMS.
2. Uniform size selection: a size preference at 1 within the range of the observed size ratio and 0 outside that ratio.
3. Constraint uniform size selection: as Uniform size selection, but the size preference ratio is constrained to exclude "outliers" in the observed size ratio.

The "Constraint uniform size selection" option was chosen for the 2012 key run. The new stomach data available for the 2019 key run include more detailed data (prey length by cm group, while the old data set has prey length by 5 cm for most years) and a SMS run using the "log normal size selection" gave actually a better model fit than both the "Uniform size selection" and "Constraint uniform size selection". Therefore, the "log normal size selection" and was chosen option for the since the 2019 key run.

Other issues

The SMS model, input and output for several key runs for both the Baltic and the North Sea can be found at Github https://github.com/ices-eg/wg_WGSAM. This Github also store the SMS source code and R scripts for preparing, running and presenting results from a SMS run.

Results of the Eastern Baltic Sea SMS key run

The few changes for the 2025 key run in model configuration and changes in input data for the benchmarked species herring and sprat input data, resulted in minor changes of the output (recruitment, mean F and SSB) compared to 2022. The estimated predation mortalities (M2) from the 2025 key run are also similar and consistent with the M2 values from the previous key run.

Key run summary sheet

Area	Eastern Baltic Sea
Model name	SMS
Type of model	Age–length structured statistical estimation model
Run year	2025
Predatory species	Cod
Prey species	Herrnig, Sprat
Time range	1974–2024.
Time step	Quarterly
Area structure	Eastern Baltic Sea, ICES sub-divisions 25-29 excl Gulf of Riga
Stomach data	Cod: 1974-2014, ~60,000 stomachs
Purpose of key run	Making historic data on natural mortality available and multispecies dynamics
Model changes since last key run in 2022	All time-series updated (2022-2024) and updated in accordance to data used by the ICES Benchmark 2023 for herring and sprat. Diet data were copied from the the previous key run. The configuration was unchanged compared to the 2022 key run.
Output available at	WGSAM Sharepoint/data/EBaltic_SMS_key_run and https://github.com/ices-eg/wg_WGSAM

Further details in	Report of the Working Group on Multispecies Assessment Methods 2025 (WGSAM, 2025)

Results of the 2025 key run

Model diagnostics

The population dynamics of all species except 'external predators' were estimated within the model. The key-run converged and the uncertainties of parameters and key output variables were obtained from the inverse Hessian matrix. Key diagnostics (Table 0-1) show a reasonable fit for catch ("sqrt(catch variance) ~ CV:") and survey indices ("sqrt(Survey variance) ~ CV:") data. Catch and survey data fit better for herring than for sprat. The same can be seen from the catch at age residual plots (Figure 0-2). Herring has, in general, smaller residuals than sprat, but herring residuals show a more clustered distribution with periods of either positive or negative residuals. The survey residuals show in some cases a "year effect" with either all positive or all negative residuals within a year. This is often seen when the survey indices are based on an acoustic measurements.

The residual plot of stomach contents, Figure 0-4, shows a quite randomly distributed residuals for sprat. Model estimates of the stomach contents of herring seems generally higher than observed values in the period after 1990, while the opposite pattern is seen for "other food". The same picture is seen in the boxplots of residuals (Figure 0-5), where the upper two rows of the plot show generally positive residuals for herring and generally negative residuals for "other food" from 1990 onwards. The bias in residuals by quarter seems limited (third row of Figure 0-5). The residual pattern is not independent of predator size (fourth row of Figure 0-5). The model overestimates the stomach contents of herring for the medium sized cod, and underestimate the stomach contents of sprat for the largest cod. This might be a result of a size dependent spatial distribution of cod.

The one step ahead residuals (Trijoulet, et al., 2023) are shown in Figure 0-6. They differ slightly from the standardised residual (Figure 0-4), but the overall pattern is the same for the two kinds of residuals.

Table 0-1. SMS key run model diagnostics.

```
September 11, 2025 12:51:12   run time:35 seconds

objective function (negative log likelihood):  1623.08
Number of parameters: 336
Number of observations used in likelihood: 15864
Maximum gradient: 7.00064e-07
Akaike information criterion (AIC):   3918.16

Number of observations used in the likelihood:
                          Catch    CPUE     S/R Stomach     Sum
Species: 1, Cod               0       0       0    1599    3198
Species: 2, Herring        1632     133      51       0    3632
Species: 3, Sprat          1428     394      51       0    3746
Sum                        6120    1054     204    3198   15864


objective function weight:
                         Catch  CPUE   S/R     Stom.    Stom N.
Species: 1, Cod          0.00  0.00  0.00       1.00       0.00
Species: 2, Herring      1.00  1.00  0.05       0.00       0.00
Species: 3, Sprat        1.00  1.00  0.05       0.00       0.00
```

unweighted objective function contributions:

	Catch	CPUE	SSB/R	stomach	Sum
Cod	0.0	0.0	0.0	-331.1	-331
Herring	596.9	62.8	40.1	0.0	700
Sprat	1092.3	198.4	36.1	0.0	1327
Sum	1689.1	261.2	76.2	-331.1	1696

contribution by fleet:

Species:2, Herring

Herring Int Acoustic Oct.	total:	62.830	mean:	0.472

Species:3, Sprat

Sprat acoustic Oct-2000-24	total:	55.889	mean:	0.420
Sprat acoustic May-2001-24	total:	90.871	mean:	0.564
Sprat acoustic Oct-1991-08	total:	40.865	mean:	0.486
Sprat acoustic Oct Shifted	total:	10.725	mean:	0.670

F, Year effect:

	sp. 2	sp. 3
1974:	1.000	1.000
1975:	0.972	1.000
1976:	1.308	0.694
1977:	0.924	0.798
1978:	0.673	0.484
1979:	0.655	0.461
1980:	1.068	0.519
1981:	1.184	0.373
1982:	1.147	0.316
1983:	1.551	0.187
1984:	1.503	0.152
1985:	1.380	0.201
1986:	1.457	0.284
1987:	1.127	0.270
1988:	1.394	0.266
1989:	1.000	0.249
1990:	1.141	0.212
1991:	1.029	0.219
1992:	0.786	0.578
1993:	0.901	0.395
1994:	0.988	0.648
1995:	1.058	0.750
1996:	1.058	0.894
1997:	1.186	1.031
1998:	1.300	1.164
1999:	1.077	1.035
2000:	1.320	1.000
2001:	1.685	0.848
2002:	1.143	1.160
2003:	0.886	1.211
2004:	0.739	1.307
2005:	1.000	1.221
2006:	1.037	0.945
2007:	0.995	1.195
2008:	1.081	1.212
2009:	1.139	1.169
2010:	1.196	1.088
2011:	1.013	1.021
2012:	0.741	0.891
2013:	0.674	1.049
2014:	0.813	1.156
2015:	1.180	0.930
2016:	1.537	0.764
2017:	1.625	0.933
2018:	1.838	0.899
2019:	1.835	1.063
2020:	1.678	0.922
2021:	1.330	0.803
2022:	1.276	0.998
2023:	0.650	1.119
2024:	0.620	1.190

F, season effect:

```
Herring
age: 1
    1974-1988:   0.033 0.071 0.115 0.250
    1989-2004:   0.070 0.055 0.078 0.250
    2005-2024:   0.188 0.079 0.047 0.250
age: 2
    1974-1988:   0.112 0.406 0.176 0.250
    1989-2004:   0.176 0.210 0.095 0.250
    2005-2024:   0.282 0.159 0.045 0.250
age: 3 - 8
    1974-1988:   0.139 0.621 0.307 0.250
    1989-2004:   0.222 0.366 0.137 0.250
    2005-2024:   0.365 0.249 0.087 0.250

Sprat
age: 1
    1974-1999:   0.073 0.044 0.035 0.250
    2000-2024:   0.322 0.129 0.051 0.250
age: 2 - 7
    1974-1999:   0.426 0.279 0.068 0.250
    2000-2024:   0.644 0.329 0.058 0.250

F, age effect:
--------------
                0      1      2      3      4      5      6      7      8
Herring
1974-1988:  0.000  0.092  0.145  0.151  0.166  0.195  0.195  0.195  0.195
1989-2004:  0.000  0.188  0.243  0.294  0.377  0.491  0.491  0.491  0.491
2005-2024:  0.000  0.097  0.151  0.170  0.237  0.342  0.342  0.342  0.342
Sprat
1974-1999:  0.000  0.093  0.162  0.240  0.247  0.247  0.247  0.247
2000-2024:  0.000  0.113  0.144  0.186  0.201  0.201  0.201  0.201


Exploitation pattern (scaled to mean F=1)
-----------------------------------------
                       0      1      2      3      4      5      6      7      8
Herring
1974-1988 season 1:      0  0.013  0.070  0.090  0.099  0.116  0.116  0.116  0.116
          season 2:      0  0.028  0.253  0.402  0.443  0.521  0.521  0.521  0.521
          season 3:  0.000  0.045  0.110  0.199  0.219  0.257  0.257  0.257  0.257
          season 4:  0.000  0.099  0.156  0.162  0.178  0.210  0.210  0.210  0.210

1989-2004 season 1:      0  0.033  0.106  0.162  0.207  0.271  0.271  0.271  0.271
          season 2:      0  0.026  0.127  0.267  0.342  0.446  0.446  0.446  0.446
          season 3:  0.000  0.036  0.057  0.100  0.128  0.167  0.167  0.167  0.167
          season 4:  0.000  0.116  0.151  0.183  0.234  0.305  0.305  0.305  0.305

2005-2024 season 1:      0  0.071  0.164  0.239  0.334  0.481  0.481  0.481  0.481
          season 2:      0  0.030  0.092  0.163  0.228  0.328  0.328  0.328  0.328
          season 3:  0.000  0.018  0.026  0.057  0.080  0.115  0.115  0.115  0.115
          season 4:  0.000  0.094  0.145  0.164  0.229  0.329  0.329  0.329  0.329

Sprat
1974-1999 season 1:      0  0.027  0.276  0.409  0.420  0.420  0.420  0.420
          season 2:      0  0.016  0.181  0.268  0.275  0.275  0.275  0.275
          season 3:  0.000  0.013  0.044  0.065  0.067  0.067  0.067  0.067
          season 4:  0.000  0.093  0.162  0.240  0.247  0.247  0.247  0.247

2000-2024 season 1:      0  0.145  0.370  0.477  0.516  0.516  0.516  0.516
          season 2:      0  0.058  0.189  0.243  0.263  0.263  0.263  0.263
          season 3:  0.000  0.023  0.033  0.043  0.047  0.047  0.047  0.047
          season 4:  0.000  0.112  0.144  0.185  0.200  0.200  0.200  0.200


sqrt(catch variance) ~ CV:
--------------------------

Herring
 1       0.555
 2       0.326
 3       0.326
 4       0.326
 5       0.326
 6       0.326
 7       0.326
 8       0.326

Sprat
```

	season			
age	1	2	3	4
1	0.632	0.753	0.928	0.605
2	0.567	0.576	0.458	0.375
3	0.420	0.407	0.589	0.330
4	0.420	0.407	0.589	0.330
5	0.420	0.407	0.589	0.330
6	0.501	0.487	0.868	0.672
7	0.501	0.487	0.868	0.672

Survey catchability:

Herring	age 0	age 1	age 2	age 3	age 4	age 5	age 6	age 7
Herring Int Acoustic Oct.		1.125	1.753	2.719	4.170	4.170	4.170	4.170
Sprat								
Sprat acoustic Oct-2000-24		0.849	1.042	1.350	1.386	1.207	1.207	1.207
Sprat acoustic May-2001-24		0.281	0.665	0.963	1.192	0.820	0.820	0.820
Sprat acoustic Oct-1991-08		0.425	0.875	1.115	0.994	0.979	0.979	0.979
Sprat acoustic Oct Shifted		0.585						

sqrt(Survey variance) ~ CV:

Herring	age 0	age 1	age 2	age 3	age 4	age 5	age 6	age 7
Herring Int Acoustic Oct.		0.63	0.38	0.38	0.38	0.38	0.32	0.32
Sprat								
Sprat acoustic Oct-2000-24		0.37	0.37	0.37	0.37	0.37	0.37	0.37
Sprat acoustic May-2001-24		0.43	0.43	0.43	0.43	0.43	0.43	0.43
Sprat acoustic Oct-1991-08		0.39	0.39	0.39	0.39	0.39	0.39	0.39
Sprat acoustic Oct Shifted		0.47						

Average F:

	sp. 2	sp. 3
1974:	0.232	0.253
1975:	0.223	0.250
1976:	0.301	0.175
1977:	0.213	0.203
1978:	0.155	0.122
1979:	0.149	0.114
1980:	0.238	0.127
1981:	0.267	0.093
1982:	0.261	0.078
1983:	0.353	0.047
1984:	0.341	0.038
1985:	0.317	0.051
1986:	0.331	0.071
1987:	0.261	0.068
1988:	0.318	0.067
1989:	0.395	0.063
1990:	0.457	0.054
1991:	0.412	0.056
1992:	0.316	0.147
1993:	0.362	0.101
1994:	0.396	0.164
1995:	0.423	0.191
1996:	0.424	0.227
1997:	0.476	0.262
1998:	0.523	0.297
1999:	0.433	0.263
2000:	0.523	0.256
2001:	0.669	0.217
2002:	0.450	0.295
2003:	0.352	0.308
2004:	0.294	0.335
2005:	0.257	0.312
2006:	0.266	0.241
2007:	0.255	0.304
2008:	0.277	0.308
2009:	0.292	0.298
2010:	0.305	0.276
2011:	0.257	0.258
2012:	0.189	0.226
2013:	0.174	0.268
2014:	0.209	0.293
2015:	0.304	0.238
2016:	0.395	0.195

```
2017:  0.418  0.238
2018:  0.476  0.230
2019:  0.474  0.272
2020:  0.435  0.236
2021:  0.345  0.206
2022:  0.330  0.256
2023:  0.167  0.286
2024:  0.160  0.303

Average M2      age 0  age 1  age 2  age 3  age 4  age 5  age 6  age 7  age 8
Herring         0.251  0.302  0.183  0.142  0.119  0.104  0.093  0.083  0.068
Sprat           0.150  0.327  0.218  0.192  0.180  0.174  0.170  0.170

Recruit-SSB                                 alfa      beta        var      sd
Herring      Geometric mean:              16.807               0.282    0.531
Sprat        Geometric mean:              18.475               0.462    0.679


Multispecies parameters
=======================

stomach content variance model: Dirichlet distribution

Vulnerability pred - prey
-------------------------
           Other-food     Herring    Sprat
Cod             1.000       8.111      3.307

Size selection parameters:
--------------------------           Cod
Size selection model:              log-norm.
Sum prey sizes in likelihood:           yes
Preferred size ratio:                 5.563
Variance of size ratio:               2.589

Other food Suitability slope:
Cod                 0.2993


Stomach variance:    value    internal     max alfa0
Cod                  0.424    0.424        41.406


Predator prey season overlap
----------------------------
Predator:Cod         Other-food  Herring     Sprat
  q:1                 1           1           1
  q:2                 0.435       0.265       1
  q:3                 0.435       0.265       0.314
  q:4                 1.811       1           0.895
```

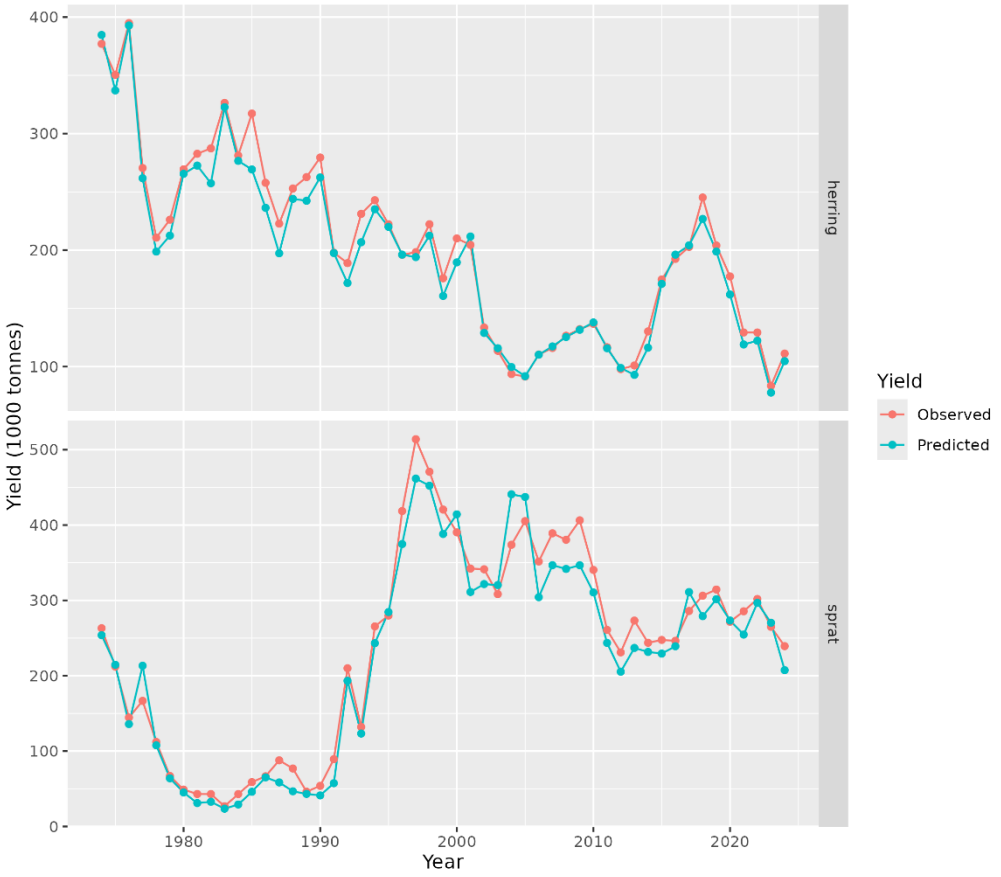


Figure 0-1 Observed and model predicted catch.

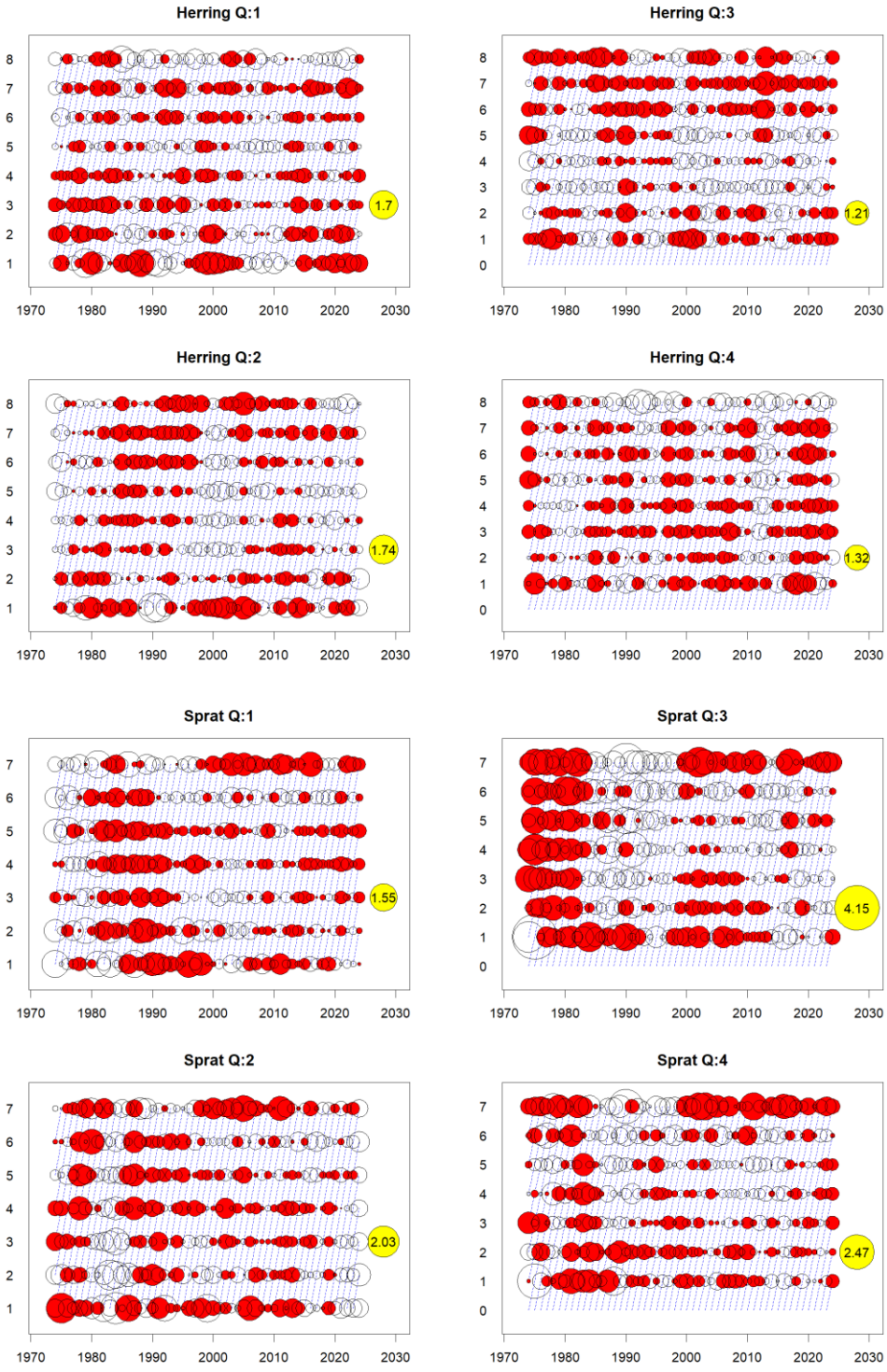


Figure 0-2. Residual plots for catch-at-age observations by species and quarter. Residuals are not standardised. The red dots show that the observed catches are larger than the model estimates. The yellow dots show the largest residual value as a reference for the dot sizes.

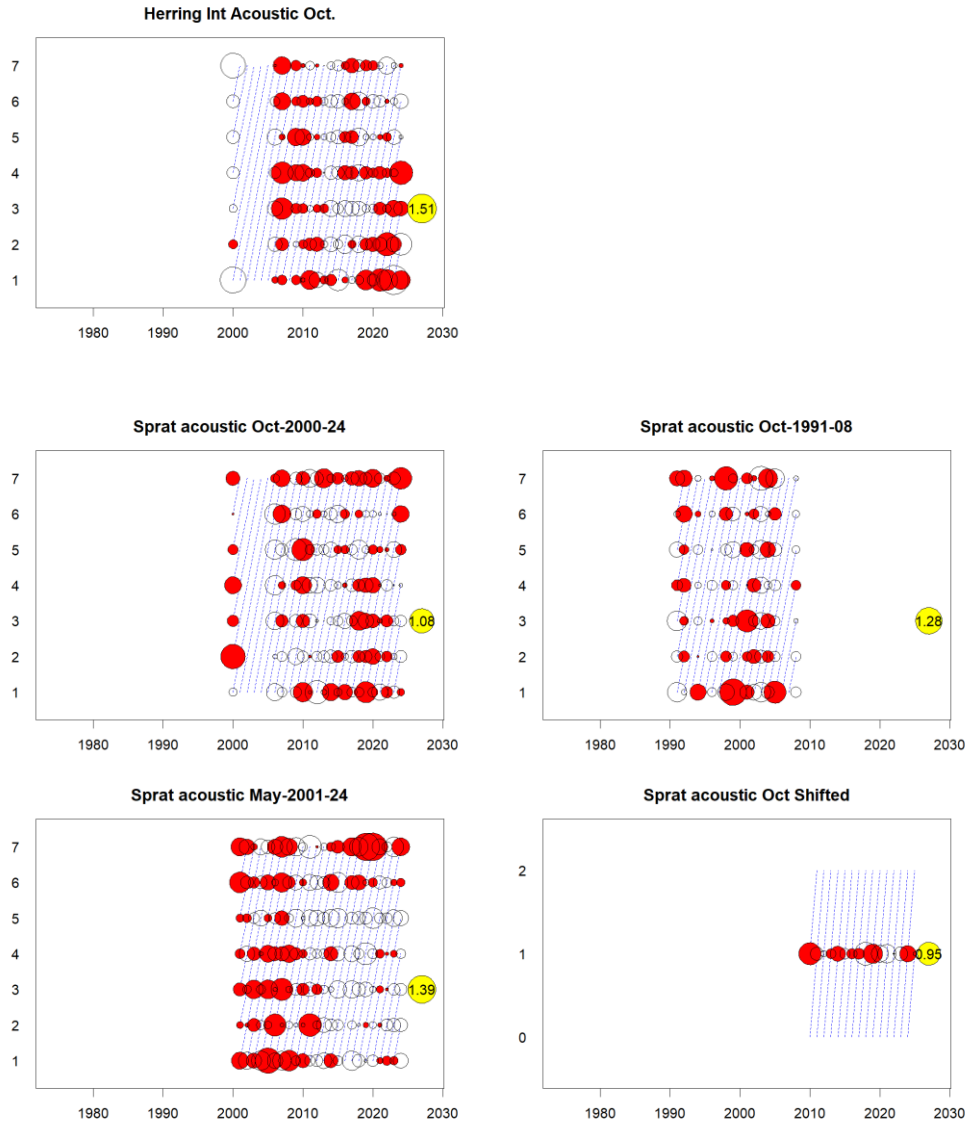


Figure 0-3. Residual plots for survey, Catch per unit effort, at age observations by species and survey. Residuals are not standardised. The red dots show that the observed catches are larger than the model estimates. The yellow dots show the largest residual values per plot.



Figure 0-4. Stomach contents residuals (Standardised residuals). The y-axis shows prey group and predator (cod) size class. The x-axis time period, where the upper panel is sorted by year and quarter, and lower panel sorted by quarter and year. Green dots show that the observed stomach contents are lower than the model estimates.

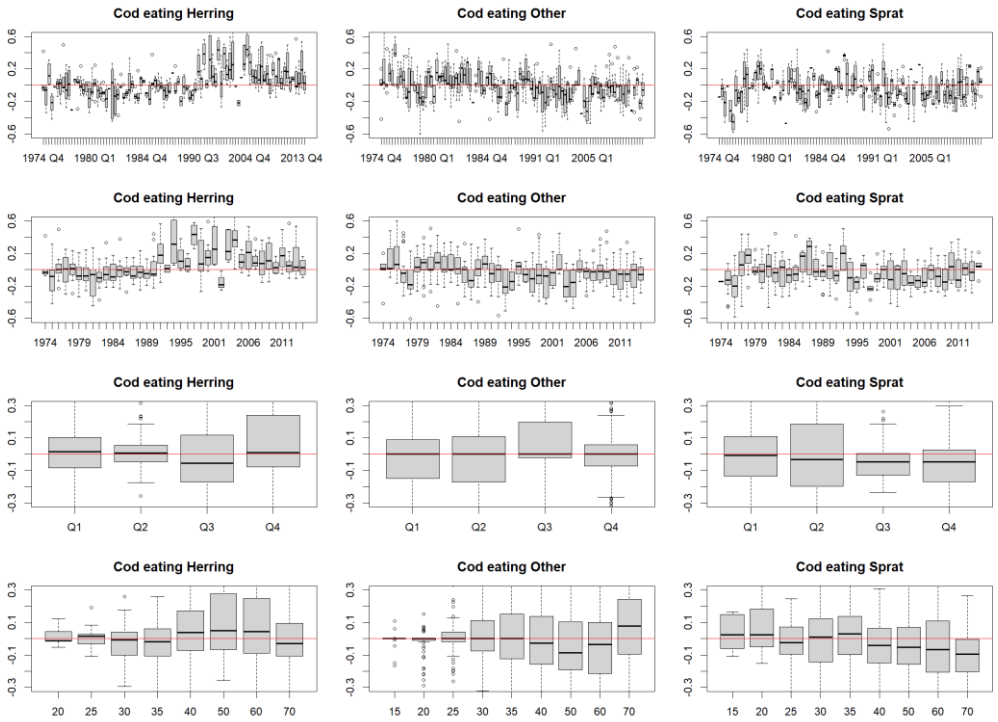


Figure 0-5. Box plot of stomach contents residuals ("Raw"). Upper (first) row shows the boxplots by individual quarter and years, second row by year (quarters combined), third row by quarter and fourth row by cod size class.



Figure 0-6 Stomach contents residuals (One step ahead residuals, see Trijoulet, et al., 2023). The y-axis shows prey group and predator (cod) size class. The x-axis time

period, where the upper panel is sorted by year and quarter. Green dots show that the observed stomach contents are lower than the model estimates.

References

Trijoulet, V., Albertsen, C.M., Kristensen, K., Legault C.M., Miller T.J., Nielsen, A. (2023). Model validation for compositional data in stock assessment models: Calculating residuals with correct properties. Fisheries Research, Volume 257: 106487 DOI: 10.1016/j.fishres.2022.106487

Stock summary and M2 results

The stock summaries are presented in Figure 0-7 (herring) and Figure 0-8 (sprat).

The estimated predation mortalities (M2) are shown in details in Figure 0-9 and Figure 0-10. Total natural mortality M=M1+M2 are tabulated in tables given below. Figure 0-11 shows the same data using the same scale on the y-axis and with an added smoother. The smoothed M values are tabulated in Table 0-2 and 0-3.

A comparison of M2 from this key run with M2 from the 2022 key run show similar values for herring (Figure 0-12) and for sprat (Figure 0-13). There is a slight increase in M2 for both species due to a slight increase in cod biomass (Figure 5.2-2).

Natural mortalities (M=M1+M2) estimated by SMS may be used as input to the ICES stock assessment of herring and sprat. If M values are used, WGSAM does recommend to update the full time-series of M.



Figure 0-7. SMS output for Herring. Catch weight, Recruitment, F, SSB, Biomass removed due to fishery (F), predation by SMS species (M2) and residual natural mortality (M1). The predation mortality (M2) presented by the 0-group (black solid line) is for the second half of the year. The M2 for the rest of the ages are annual values.

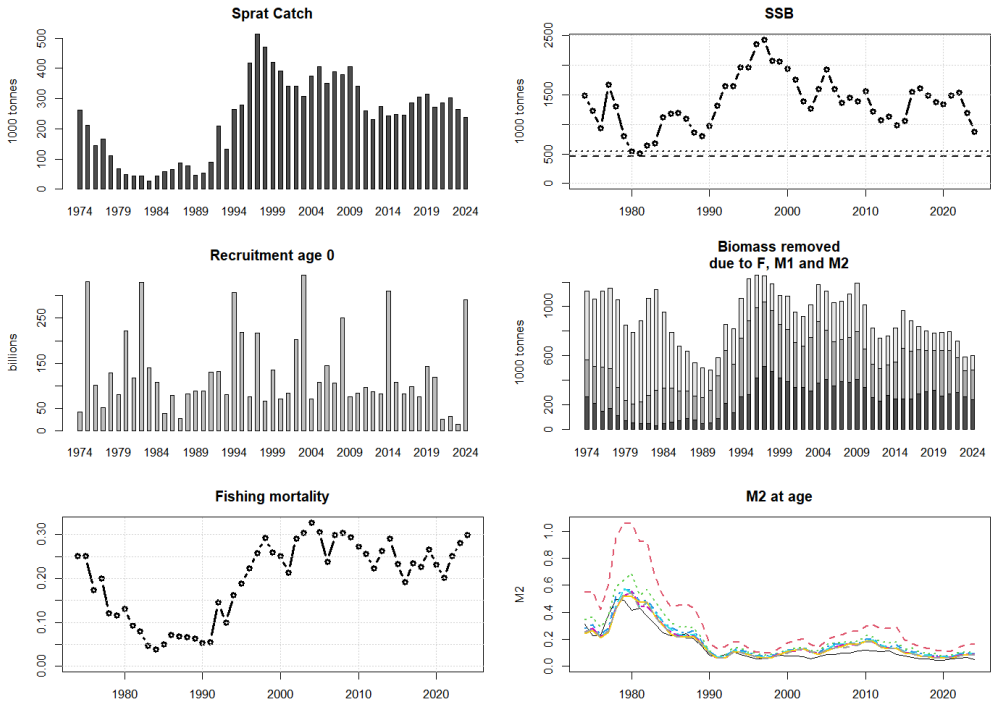


Figure 0-8. SMS output for Sprat. Catch weight, Recruitment, F, SSB, Biomass removed due to fishery (F), predation by SMS species (M2) and residual natural mortality (M1). The predation mortality (M2) presented by the 0-group (black solid line) is for the second half of the year. The M2 for the rest of the ages are annual values.

Table 5.1-2 Herring : Natural mortality (sum of quarterly M1+M2)

Year/Age	0	1	2	3	4	5	6	7	8
1974	0.455	0.437	0.310	0.253	0.233	0.220	0.218	0.203	0.173
1975	**0.389**	**0.482**	**0.340**	**0.276**	**0.253**	**0.239**	**0.238**	**0.223**	**0.189**
1976	**0.380**	**0.416**	**0.306**	**0.258**	**0.239**	**0.227**	**0.226**	**0.213**	**0.182**
1977	**0.517**	**0.465**	**0.327**	**0.273**	**0.253**	**0.239**	**0.237**	**0.222**	**0.190**
1978	**0.716**	**0.688**	**0.387**	**0.343**	**0.323**	**0.301**	**0.275**	**0.260**	**0.229**
1979	**0.797**	**0.868**	**0.415**	**0.353**	**0.340**	**0.326**	**0.318**	**0.283**	**0.234**
1980	**0.691**	**0.877**	**0.520**	**0.417**	**0.370**	**0.375**	**0.326**	**0.302**	**0.267**
1981	**0.702**	**0.794**	**0.500**	**0.393**	**0.340**	**0.309**	**0.309**	**0.276**	**0.236**
1982	0.656	0.812	0.482	0.398	0.336	0.302	0.283	0.280	0.228
1983	0.593	0.729	0.538	0.382	0.357	0.314	0.285	0.267	0.238
1984	0.493	0.627	0.482	0.376	0.303	0.303	0.274	0.249	0.222
1985	0.440	0.538	0.438	0.334	0.282	0.251	0.243	0.229	0.207
1986	0.409	0.492	0.375	0.333	0.263	0.241	0.222	0.209	0.185
1987	0.443	0.499	0.313	0.272	0.255	0.220	0.204	0.193	0.173
1988	0.392	0.507	0.369	0.268	0.255	0.237	0.213	0.195	0.174
1989	0.302	0.424	0.288	0.290	0.242	0.215	0.204	0.185	0.166
1990	0.195	0.287	0.208	0.187	0.195	0.169	0.162	0.156	0.148

1991	0.158	0.228	0.190	0.165	0.149	0.159	0.141	0.145	0.135
1992	0.189	0.237	0.193	0.170	0.146	0.138	0.147	0.135	0.131
1993	0.237	0.296	0.246	0.208	0.192	0.174	0.164	0.171	0.150
1994	0.208	0.303	0.251	0.222	0.194	0.185	0.173	0.158	0.157
1995	0.181	0.267	0.229	0.212	0.195	0.184	0.179	0.167	0.164
1996	0.157	0.233	0.212	0.192	0.183	0.175	0.167	0.162	0.150
1997	0.143	0.210	0.195	0.178	0.169	0.161	0.156	0.152	0.147
1998	0.167	0.216	0.188	0.175	0.161	0.154	0.148	0.146	0.135
1999	0.191	0.244	0.204	0.184	0.174	0.162	0.152	0.150	0.139
2000	0.199	0.324	0.232	0.218	0.207	0.198	0.183	0.174	0.172
2001	0.208	0.342	0.245	0.213	0.208	0.194	0.186	0.183	0.180
2002	0.195	0.356	0.262	0.226	0.203	0.195	0.185	0.178	0.182
2003	0.162	0.310	0.208	0.189	0.178	0.172	0.164	0.156	0.153
2004	0.187	0.280	0.255	0.196	0.182	0.164	0.159	0.152	0.146
2005	0.229	0.336	0.286	0.256	0.209	0.186	0.170	0.162	0.153
2006	0.232	0.368	0.249	0.246	0.231	0.207	0.179	0.169	0.160
2007	0.246	0.379	0.267	0.241	0.220	0.209	0.180	0.168	0.153
2008	0.252	0.404	0.269	0.246	0.225	0.199	0.208	0.182	0.170
2009	0.282	0.420	0.296	0.247	0.237	0.210	0.193	0.208	0.185
2010	0.301	0.466	0.337	0.275	0.239	0.233	0.214	0.199	0.197
2011	0.280	0.474	0.307	0.281	0.241	0.218	0.212	0.194	0.195
2012	0.261	0.422	0.234	0.241	0.209	0.184	0.174	0.163	0.153
2013	0.277	0.401	0.254	0.195	0.200	0.176	0.162	0.156	0.149
2014	0.224	0.404	0.260	0.208	0.175	0.181	0.161	0.155	0.149
2015	0.205	0.330	0.219	0.193	0.174	0.159	0.158	0.149	0.145
2016	0.181	0.304	0.271	0.189	0.177	0.166	0.156	0.152	0.145
2017	0.162	0.268	0.208	0.193	0.159	0.153	0.144	0.136	0.132
2018	0.155	0.246	0.188	0.158	0.156	0.137	0.135	0.133	0.127
2019	0.139	0.237	0.175	0.162	0.146	0.146	0.133	0.130	0.127
2020	0.138	0.223	0.172	0.153	0.146	0.137	0.138	0.127	0.126
2021	0.155	0.227	0.177	0.164	0.151	0.147	0.139	0.137	0.132
2022	0.169	0.257	0.196	0.181	0.164	0.159	0.149	0.147	0.141
2023	0.174	0.279	0.206	0.177	0.166	0.155	0.155	0.148	0.138
2024	0.139	0.274	0.173	0.166	0.154	0.141	0.138	0.136	0.129

Table 5.1-3 Sprat : Natural mortality (sum of quarterly M1+M2)

Year/Age	0	1	2	3	4	5	6	7
1974	0.412	0.750	0.545	0.480	0.454	0.454	0.441	0.451
1975	0.328	0.757	0.567	0.505	0.479	0.479	0.464	0.474
1976	0.334	0.620	0.486	0.438	0.420	0.420	0.411	0.418
1977	0.480	0.811	0.572	0.486	0.465	0.465	0.453	0.462
1978	0.592	1.154	0.787	0.722	0.643	0.633	0.623	0.623
1979	0.587	1.266	0.840	0.772	0.776	0.719	0.723	0.738
1980	0.512	1.257	0.888	0.762	0.745	0.756	0.718	0.736
1981	0.533	1.124	0.720	0.679	0.641	0.644	0.671	0.623

1982	0.472	1.121	0.771	0.688	0.669	0.642	0.670	0.678
1983	0.412	0.868	0.682	0.615	0.596	0.582	0.570	0.568
1984	0.349	0.725	0.600	0.529	0.523	0.507	0.501	0.500
1985	0.327	0.638	0.522	0.488	0.474	0.455	0.440	0.448
1986	0.327	0.654	0.491	0.465	0.439	0.424	0.418	0.414
1987	0.344	0.660	0.489	0.443	0.425	0.421	0.420	0.409
1988	0.298	0.630	0.481	0.466	0.435	0.419	0.416	0.405
1989	0.244	0.520	0.409	0.380	0.375	0.366	0.363	0.359
1990	0.180	0.374	0.311	0.307	0.300	0.294	0.296	0.290
1991	0.167	0.331	0.272	0.269	0.264	0.262	0.261	0.261
1992	0.187	0.347	0.282	0.269	0.266	0.262	0.261	0.259
1993	0.206	0.382	0.340	0.324	0.314	0.310	0.307	0.301
1994	0.184	0.378	0.336	0.319	0.307	0.305	0.301	0.301
1995	0.168	0.333	0.303	0.301	0.293	0.289	0.287	0.286
1996	0.156	0.304	0.293	0.279	0.277	0.272	0.271	0.271
1997	0.152	0.297	0.280	0.275	0.266	0.259	0.259	0.257
1998	0.167	0.307	0.285	0.280	0.277	0.269	0.267	0.268
1999	0.179	0.336	0.304	0.293	0.293	0.292	0.285	0.281
2000	0.174	0.375	0.317	0.318	0.313	0.309	0.307	0.299
2001	0.179	0.390	0.333	0.320	0.321	0.315	0.317	0.317
2002	0.170	0.403	0.342	0.337	0.331	0.330	0.330	0.330
2003	0.153	0.363	0.315	0.308	0.308	0.303	0.307	0.308
2004	0.168	0.340	0.314	0.295	0.289	0.291	0.289	0.290
2005	0.187	0.392	0.360	0.347	0.324	0.319	0.314	0.320
2006	0.189	0.421	0.372	0.366	0.357	0.339	0.333	0.334
2007	0.196	0.433	0.377	0.360	0.359	0.357	0.344	0.333
2008	0.198	0.459	0.379	0.370	0.360	0.365	0.366	0.348
2009	0.212	0.460	0.382	0.367	0.360	0.356	0.362	0.355
2010	0.221	0.498	0.428	0.399	0.388	0.385	0.382	0.384
2011	0.212	0.508	0.415	0.407	0.392	0.380	0.381	0.376
2012	0.206	0.480	0.378	0.356	0.354	0.345	0.342	0.343
2013	0.216	0.480	0.370	0.341	0.334	0.332	0.331	0.332
2014	0.187	0.484	0.377	0.355	0.337	0.331	0.332	0.337
2015	0.177	0.395	0.326	0.314	0.306	0.300	0.296	0.302
2016	0.164	0.369	0.333	0.307	0.296	0.293	0.290	0.291
2017	0.157	0.344	0.305	0.298	0.284	0.278	0.278	0.277
2018	0.154	0.330	0.291	0.284	0.281	0.273	0.268	0.270
2019	0.145	0.321	0.290	0.278	0.273	0.273	0.262	0.262
2020	0.146	0.309	0.272	0.268	0.263	0.260	0.261	0.256
2021	0.155	0.315	0.282	0.273	0.270	0.267	0.261	0.266
2022	0.162	0.341	0.301	0.291	0.284	0.280	0.278	0.278
2023	0.165	0.360	0.304	0.300	0.295	0.287	0.286	0.286
2024	0.147	0.362	0.302	0.295	0.291	0.285	0.282	0.282

Table 0-2. Herring GAM-Smoothed Natural mortality (sum of quarterly M1 (=0.1)+M2)

Year/Age	0	1	2	3	4	5	6	7	8
1974	0.358	0.351	0.275	0.231	0.217	0.205	0.206	0.194	0.167
1975	0.441	0.451	0.314	0.262	0.244	0.232	0.228	0.214	0.183
1976	0.519	0.547	0.350	0.291	0.270	0.257	0.250	0.233	0.199
1977	0.587	0.632	0.385	0.318	0.294	0.279	0.268	0.250	0.213
1978	0.641	0.702	0.416	0.342	0.313	0.298	0.284	0.263	0.224
1979	0.676	0.752	0.442	0.361	0.328	0.312	0.294	0.273	0.233
1980	0.688	0.777	0.462	0.375	0.337	0.318	0.299	0.277	0.237
1981	0.676	0.776	0.475	0.382	0.340	0.319	0.298	0.276	0.237
1982	0.645	0.754	0.480	0.383	0.337	0.313	0.292	0.271	0.233
1983	0.600	0.715	0.477	0.378	0.328	0.303	0.281	0.261	0.227
1984	0.547	0.665	0.464	0.367	0.315	0.288	0.267	0.249	0.218
1985	0.489	0.608	0.441	0.349	0.299	0.271	0.251	0.235	0.207
1986	0.433	0.548	0.408	0.326	0.279	0.253	0.234	0.219	0.195
1987	0.380	0.488	0.369	0.298	0.257	0.233	0.216	0.203	0.183
1988	0.331	0.431	0.326	0.269	0.236	0.214	0.199	0.188	0.171
1989	0.287	0.379	0.284	0.242	0.215	0.197	0.184	0.174	0.160
1990	0.251	0.334	0.249	0.218	0.198	0.183	0.172	0.163	0.152
1991	0.222	0.298	0.223	0.201	0.186	0.173	0.164	0.156	0.146
1992	0.202	0.272	0.209	0.192	0.179	0.168	0.159	0.153	0.144
1993	0.189	0.255	0.203	0.188	0.176	0.166	0.158	0.152	0.144
1994	0.181	0.245	0.204	0.189	0.176	0.166	0.159	0.154	0.145
1995	0.177	0.242	0.209	0.191	0.177	0.168	0.162	0.156	0.148
1996	0.175	0.244	0.215	0.194	0.179	0.170	0.164	0.159	0.150
1997	0.174	0.249	0.218	0.196	0.181	0.172	0.165	0.160	0.153
1998	0.173	0.258	0.221	0.196	0.182	0.173	0.165	0.161	0.154
1999	0.174	0.268	0.222	0.197	0.183	0.174	0.166	0.161	0.156
2000	0.176	0.281	0.224	0.198	0.185	0.175	0.166	0.161	0.157
2001	0.181	0.295	0.227	0.200	0.188	0.177	0.167	0.161	0.158
2002	0.189	0.310	0.233	0.206	0.192	0.180	0.169	0.163	0.160
2003	0.199	0.325	0.241	0.214	0.199	0.185	0.173	0.166	0.161
2004	0.211	0.340	0.251	0.224	0.206	0.191	0.177	0.169	0.163
2005	0.225	0.356	0.262	0.234	0.214	0.196	0.182	0.173	0.165
2006	0.238	0.372	0.272	0.243	0.221	0.201	0.187	0.177	0.167
2007	0.250	0.388	0.280	0.250	0.226	0.205	0.190	0.180	0.169
2008	0.260	0.404	0.286	0.254	0.228	0.207	0.192	0.182	0.170
2009	0.268	0.418	0.288	0.254	0.228	0.207	0.193	0.182	0.171
2010	0.271	0.428	0.288	0.251	0.225	0.205	0.191	0.181	0.170

2011	0.271	0.432	0.284	0.245	0.219	0.201	0.188	0.178	0.169
2012	0.265	0.428	0.277	0.237	0.212	0.195	0.183	0.174	0.166
2013	0.254	0.413	0.267	0.226	0.203	0.188	0.177	0.168	0.162
2014	0.237	0.386	0.254	0.214	0.193	0.179	0.170	0.161	0.156
2015	0.216	0.352	0.239	0.201	0.182	0.170	0.161	0.154	0.149
2016	0.194	0.315	0.224	0.189	0.171	0.162	0.153	0.146	0.142
2017	0.174	0.280	0.210	0.178	0.162	0.154	0.146	0.140	0.136
2018	0.157	0.252	0.198	0.169	0.155	0.148	0.140	0.135	0.131
2019	0.146	0.234	0.190	0.164	0.151	0.144	0.137	0.132	0.128
2020	0.142	0.229	0.185	0.162	0.150	0.143	0.137	0.132	0.127
2021	0.143	0.234	0.183	0.163	0.151	0.143	0.138	0.134	0.128
2022	0.148	0.246	0.183	0.166	0.154	0.145	0.141	0.137	0.130
2023	0.156	0.263	0.184	0.170	0.158	0.148	0.145	0.141	0.134
2024	0.165	0.282	0.187	0.175	0.163	0.152	0.149	0.146	0.137

Table 0-3. Sprat GAM-Smoothed Natural mortality (sum of quarterly M1 (=0.1)+M2).

Year/Age	0	1	2	3	4	5	6	7
1974	0.342	0.604	0.475	0.420	0.393	0.394	0.376	0.388
1975	0.393	0.744	0.551	0.490	0.462	0.460	0.446	0.456
1976	0.439	0.875	0.623	0.555	0.528	0.523	0.512	0.520
1977	0.479	0.988	0.686	0.612	0.585	0.578	0.570	0.576
1978	0.509	1.075	0.736	0.658	0.631	0.621	0.616	0.620
1979	0.525	1.126	0.767	0.687	0.661	0.648	0.646	0.648
1980	0.525	1.134	0.775	0.696	0.671	0.657	0.656	0.656
1981	0.507	1.098	0.761	0.685	0.661	0.646	0.645	0.644
1982	0.478	1.029	0.729	0.658	0.636	0.620	0.619	0.616
1983	0.440	0.939	0.684	0.620	0.599	0.583	0.581	0.578
1984	0.398	0.840	0.630	0.575	0.556	0.540	0.537	0.533
1985	0.357	0.743	0.574	0.527	0.509	0.495	0.491	0.487
1986	0.320	0.659	0.520	0.481	0.464	0.451	0.447	0.443
1987	0.288	0.587	0.469	0.437	0.423	0.411	0.407	0.403
1988	0.260	0.528	0.423	0.398	0.385	0.375	0.372	0.367
1989	0.237	0.478	0.384	0.364	0.352	0.344	0.341	0.337
1990	0.218	0.437	0.351	0.336	0.325	0.319	0.316	0.312
1991	0.203	0.403	0.326	0.315	0.305	0.300	0.298	0.294
1992	0.191	0.375	0.310	0.301	0.292	0.288	0.285	0.282
1993	0.182	0.353	0.301	0.293	0.285	0.281	0.279	0.275
1994	0.175	0.336	0.297	0.289	0.282	0.278	0.276	0.273
1995	0.170	0.325	0.296	0.288	0.282	0.278	0.276	0.273

1996	0.167	0.320	0.298	0.289	0.284	0.280	0.277	0.275
1997	0.165	0.320	0.300	0.291	0.286	0.282	0.280	0.279
1998	0.164	0.326	0.302	0.293	0.289	0.285	0.283	0.282
1999	0.163	0.335	0.304	0.296	0.292	0.288	0.287	0.287
2000	0.164	0.346	0.308	0.300	0.297	0.292	0.291	0.292
2001	0.167	0.358	0.314	0.306	0.302	0.298	0.297	0.298
2002	0.170	0.371	0.322	0.314	0.309	0.305	0.304	0.304
2003	0.174	0.383	0.332	0.324	0.318	0.314	0.313	0.312
2004	0.179	0.394	0.343	0.334	0.328	0.324	0.322	0.320
2005	0.185	0.406	0.355	0.345	0.338	0.334	0.332	0.328
2006	0.191	0.419	0.367	0.356	0.348	0.344	0.341	0.336
2007	0.196	0.433	0.378	0.365	0.356	0.352	0.348	0.343
2008	0.201	0.449	0.386	0.371	0.362	0.357	0.354	0.349
2009	0.206	0.465	0.392	0.375	0.365	0.360	0.358	0.353
2010	0.208	0.479	0.395	0.376	0.366	0.361	0.359	0.356
2011	0.209	0.487	0.394	0.373	0.363	0.358	0.357	0.355
2012	0.207	0.487	0.389	0.367	0.357	0.352	0.351	0.351
2013	0.202	0.477	0.379	0.357	0.347	0.342	0.341	0.342
2014	0.194	0.454	0.364	0.343	0.334	0.329	0.328	0.329
2015	0.184	0.423	0.346	0.328	0.318	0.313	0.312	0.314
2016	0.173	0.388	0.327	0.311	0.303	0.298	0.295	0.297
2017	0.163	0.356	0.310	0.296	0.288	0.284	0.280	0.282
2018	0.154	0.329	0.295	0.284	0.277	0.273	0.268	0.270
2019	0.149	0.314	0.287	0.277	0.271	0.266	0.262	0.263
2020	0.147	0.311	0.284	0.276	0.269	0.265	0.261	0.261
2021	0.148	0.318	0.285	0.278	0.272	0.268	0.264	0.264
2022	0.151	0.331	0.290	0.283	0.278	0.274	0.270	0.271
2023	0.156	0.350	0.297	0.291	0.286	0.281	0.279	0.279
2024	0.161	0.371	0.306	0.299	0.296	0.290	0.288	0.289

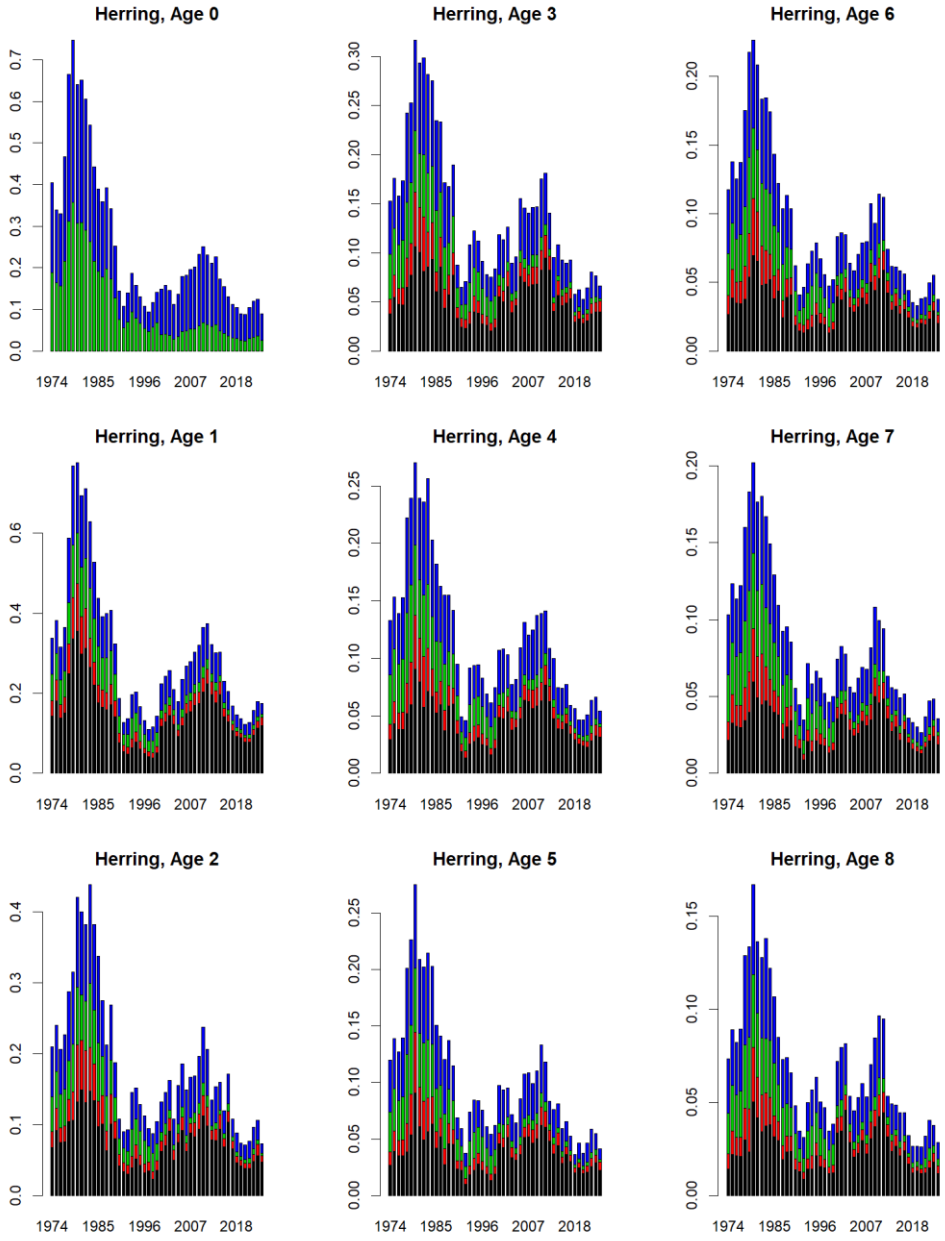


Figure 0-9. Annual predation mortality (M2) of herring the colours show M2 by quarter (green Q3, blue Q4, black Q1 and red Q2).

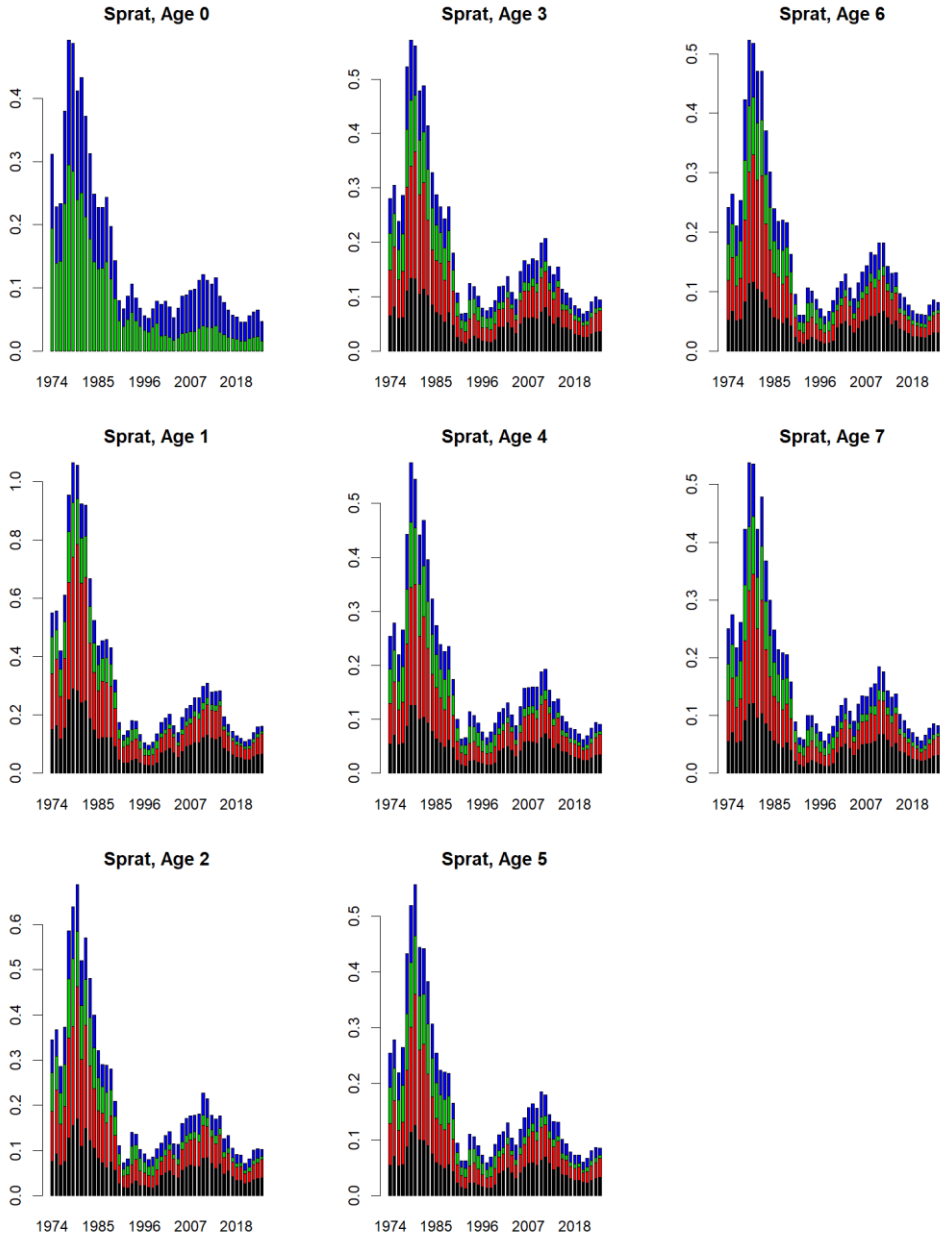
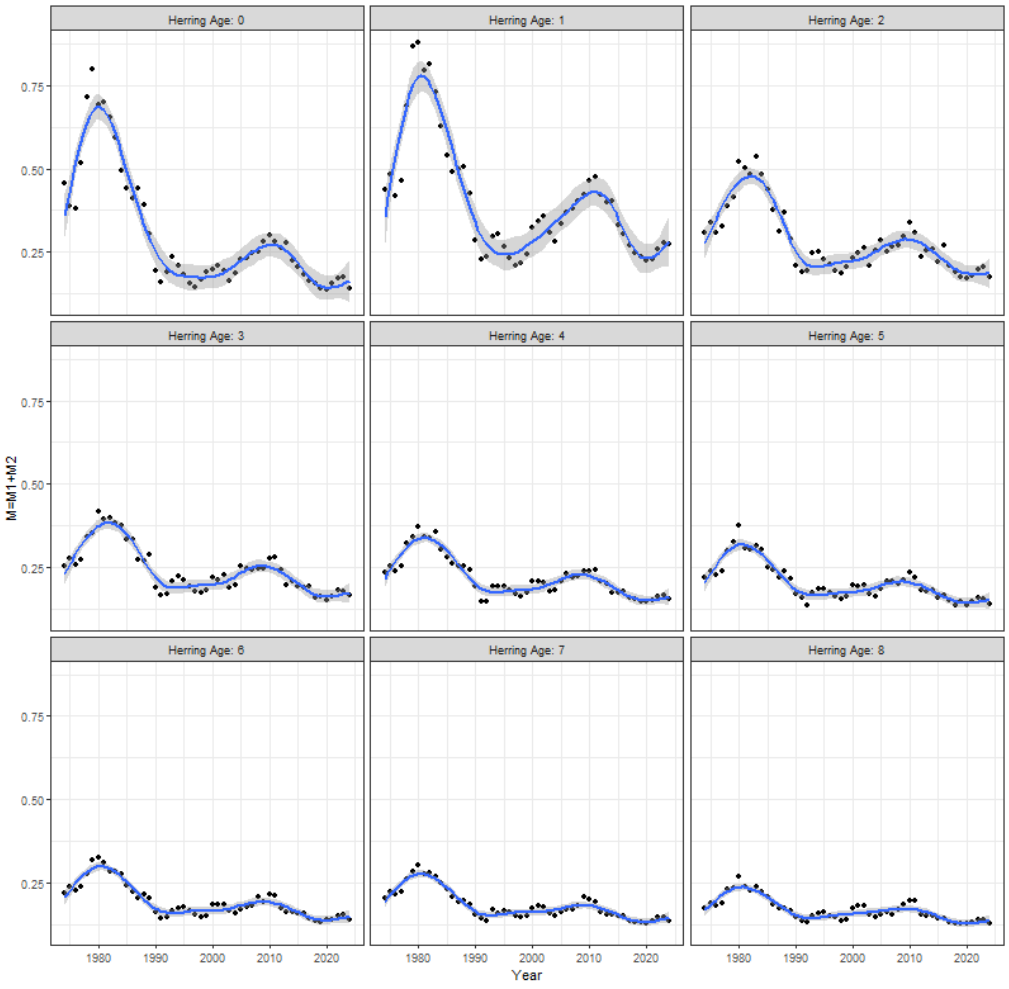


Figure 0-10. Annual predation mortality (M2) of herring the colours show M2 by quarter (green Q3, blue Q4, black Q1 and red Q2).


Herring Age: 0
Herring Age: 1
Herring Age: 2
Herring Age: 3
Herring Age: 4
Herring Age: 5
Herring Age: 6
Herring Age: 7
Herring Age: 8
M=M1+M2
0.75
0.50
0.25
1980
1990
2000
2010
2020
Year

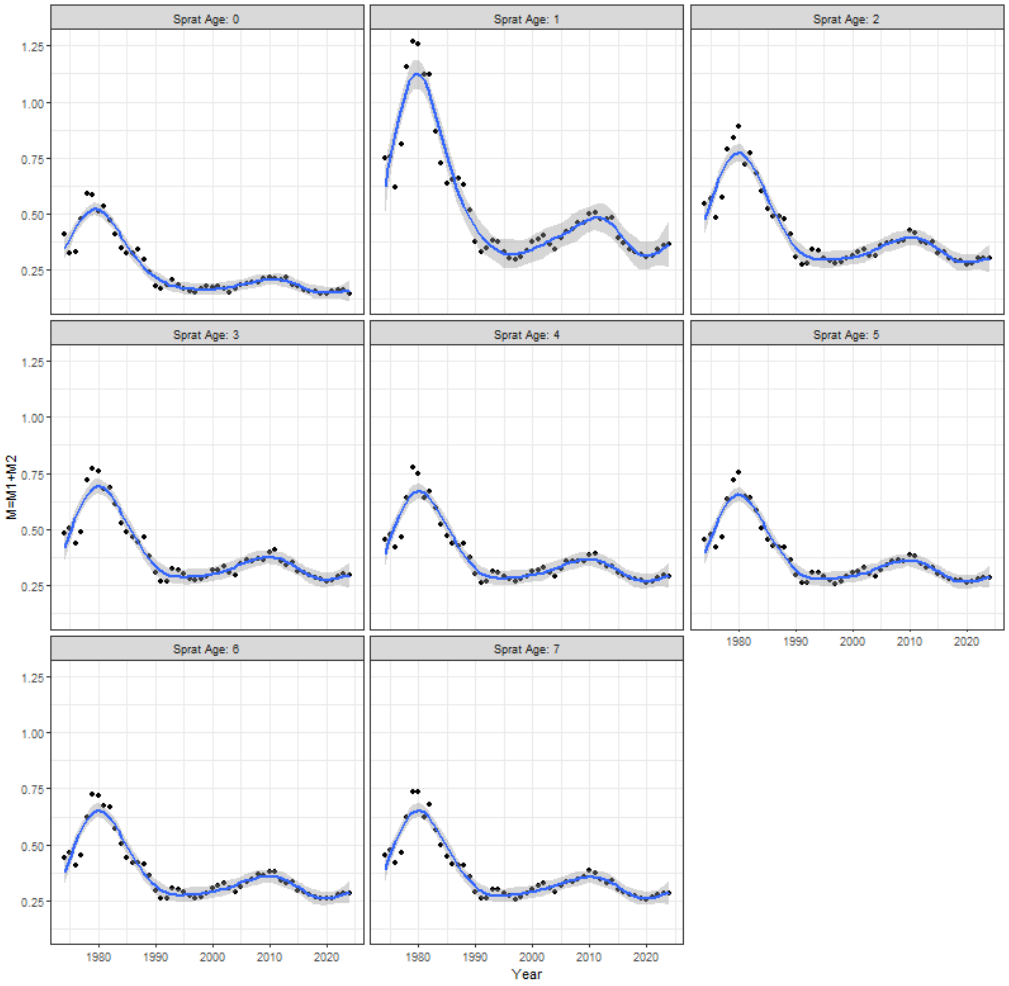


Figure 0-11. Annual natural mortalities (M=M1+M2) by species and age. Black dots are the sum of quarterly M1 and M2; the blue line is a gam spline estimate.



Figure 0-12. Herring. Comparison of predation mortality (M2) estimated by the most recent key runs.

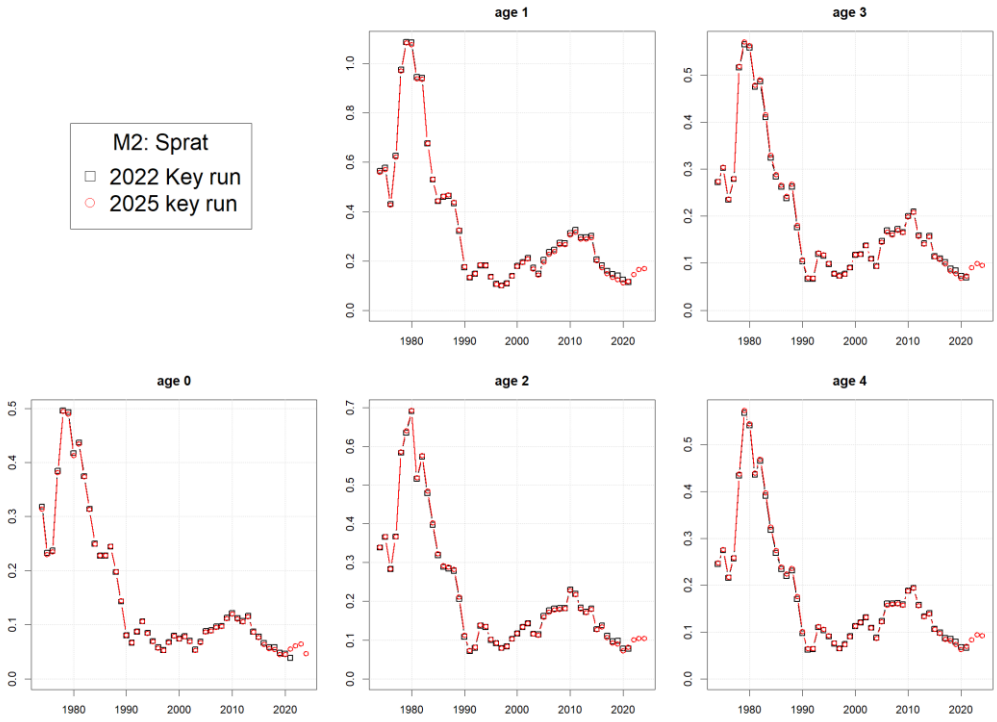


Figure 0-13. Sprat. Comparison of predation mortality (M2) estimated by the most recent key runs.

M2 for herring ensemble approach

The ICES single stock assessment of Central Baltic herring uses an ensemble approach with three levels of natural mortalities (M) based on the 2022 key run:

1. M1 for herring equal 0.1 (the key run configuration)
2. M1 for herring equal 0.2
3. M derived from a SMS run with a lower cod consumption rate, calculated from the 10% percentile of the parameters "a" and "b" used to calculate cod consumption (consumption= a *W^b), see section XX. M1 equal to 0.1.

In this section, the three estimates of M are recalculated using the 2025 key run as basis. Figure 0-14 shows a comparison of the estimated M2 values from the three configurations. The key run (M1=0.1) estimates the highest M2, followed by M2 estimated with the assumption of M=0.1. A lower cod consumption in combination with M1=0.1 estimates the lowest M2 values.

The three levels of M is estimated consistently between the most recent two key runs (Figure 0-15).

Calculated values of M (the sum of M1 and M2) are presented in Table 0-4 and Table 0-6. The smoothed M values are shown in Table 0-5 and Table 0-7

Table 0-4 Herring ensemble approach, M1 equal 0.2 : Natural mortality (sum of quarterly M1+M2)

Year/Age	0	1	2	3	4	5	6	7	8
1974	0.439	0.466	0.361	0.316	0.300	0.289	0.288	0.277	0.254
1975	0.382	0.504	0.385	0.334	0.316	0.304	0.303	0.292	0.265
1976	0.378	0.451	0.360	0.321	0.306	0.296	0.295	0.285	0.261
1977	0.496	0.492	0.378	0.334	0.317	0.306	0.305	0.293	0.267
1978	0.671	0.675	0.424	0.388	0.372	0.355	0.334	0.322	0.297
1979	0.752	0.829	0.447	0.396	0.386	0.375	0.368	0.340	0.301
1980	0.653	0.842	0.536	0.449	0.411	0.414	0.375	0.355	0.327
1981	0.665	0.770	0.519	0.431	0.387	0.362	0.361	0.335	0.303
1982	0.623	0.788	0.504	0.435	0.384	0.357	0.341	0.338	0.297
1983	0.570	0.717	0.552	0.421	0.400	0.366	0.342	0.327	0.304
1984	0.481	0.634	0.508	0.417	0.357	0.357	0.334	0.314	0.292
1985	0.436	0.561	0.474	0.386	0.342	0.316	0.310	0.299	0.281
1986	0.406	0.520	0.420	0.385	0.327	0.309	0.294	0.283	0.264
1987	0.438	0.526	0.368	0.335	0.321	0.292	0.279	0.270	0.255
1988	0.392	0.534	0.415	0.330	0.320	0.305	0.286	0.272	0.255
1989	0.314	0.463	0.347	0.349	0.309	0.288	0.278	0.264	0.249
1990	0.222	0.351	0.284	0.267	0.273	0.252	0.247	0.242	0.236
1991	0.191	0.303	0.271	0.250	0.237	0.246	0.231	0.234	0.226
1992	0.218	0.310	0.273	0.255	0.235	0.228	0.236	0.226	0.223
1993	0.260	0.360	0.316	0.285	0.271	0.257	0.248	0.254	0.237
1994	0.235	0.366	0.321	0.296	0.273	0.265	0.256	0.244	0.243
1995	0.212	0.336	0.303	0.289	0.274	0.265	0.261	0.251	0.249
1996	0.192	0.309	0.291	0.273	0.265	0.259	0.253	0.248	0.239

1997	**0.180**	**0.290**	**0.276**	**0.262**	**0.255**	**0.248**	**0.243**	**0.241**	**0.237**
1998	**0.201**	**0.294**	**0.271**	**0.260**	**0.248**	**0.242**	**0.237**	**0.236**	**0.227**
1999	**0.221**	**0.317**	**0.283**	**0.266**	**0.258**	**0.248**	**0.240**	**0.238**	**0.229**
2000	**0.227**	**0.383**	**0.304**	**0.292**	**0.283**	**0.275**	**0.264**	**0.257**	**0.255**
2001	**0.234**	**0.397**	**0.314**	**0.288**	**0.283**	**0.272**	**0.266**	**0.263**	**0.260**
2002	**0.222**	**0.407**	**0.326**	**0.297**	**0.278**	**0.272**	**0.264**	**0.258**	**0.261**
2003	**0.194**	**0.368**	**0.282**	**0.267**	**0.258**	**0.253**	**0.247**	**0.241**	**0.239**
2004	**0.216**	**0.343**	**0.323**	**0.273**	**0.262**	**0.248**	**0.243**	**0.239**	**0.233**
2005	**0.252**	**0.389**	**0.346**	**0.321**	**0.283**	**0.264**	**0.252**	**0.246**	**0.238**
2006	**0.254**	**0.415**	**0.316**	**0.312**	**0.300**	**0.281**	**0.258**	**0.251**	**0.244**
2007	**0.265**	**0.423**	**0.329**	**0.308**	**0.291**	**0.282**	**0.259**	**0.250**	**0.239**
2008	**0.270**	**0.443**	**0.330**	**0.311**	**0.294**	**0.274**	**0.281**	**0.260**	**0.251**
2009	**0.297**	**0.457**	**0.352**	**0.312**	**0.304**	**0.283**	**0.269**	**0.281**	**0.262**
2010	**0.313**	**0.494**	**0.385**	**0.334**	**0.305**	**0.300**	**0.285**	**0.274**	**0.271**
2011	**0.295**	**0.501**	**0.360**	**0.338**	**0.306**	**0.288**	**0.283**	**0.269**	**0.270**
2012	**0.278**	**0.458**	**0.302**	**0.307**	**0.281**	**0.262**	**0.254**	**0.246**	**0.238**
2013	**0.292**	**0.442**	**0.319**	**0.272**	**0.275**	**0.257**	**0.245**	**0.241**	**0.236**
2014	**0.246**	**0.445**	**0.324**	**0.281**	**0.256**	**0.261**	**0.245**	**0.240**	**0.235**
2015	**0.233**	**0.385**	**0.292**	**0.270**	**0.255**	**0.244**	**0.243**	**0.236**	**0.232**
2016	**0.212**	**0.365**	**0.337**	**0.268**	**0.258**	**0.249**	**0.242**	**0.238**	**0.233**
2017	**0.195**	**0.336**	**0.285**	**0.272**	**0.245**	**0.240**	**0.233**	**0.226**	**0.224**
2018	**0.190**	**0.318**	**0.269**	**0.244**	**0.243**	**0.227**	**0.226**	**0.224**	**0.219**
2019	**0.176**	**0.310**	**0.258**	**0.247**	**0.235**	**0.235**	**0.224**	**0.222**	**0.219**
2020	**0.174**	**0.298**	**0.255**	**0.240**	**0.235**	**0.228**	**0.228**	**0.219**	**0.219**
2021	**0.189**	**0.301**	**0.259**	**0.249**	**0.238**	**0.235**	**0.229**	**0.227**	**0.224**
2022	**0.201**	**0.326**	**0.274**	**0.262**	**0.248**	**0.244**	**0.237**	**0.235**	**0.230**
2023	**0.205**	**0.344**	**0.282**	**0.258**	**0.250**	**0.241**	**0.241**	**0.236**	**0.228**
2024	**0.175**	**0.341**	**0.256**	**0.250**	**0.241**	**0.231**	**0.228**	**0.226**	**0.221**

Table 0-5 Herring ensemble, Smoothed M, M1 equal 0.2

Year/Age	0	1	2	3	4	5	6	7	8
1974	**0.354**	**0.393**	**0.333**	**0.298**	**0.287**	**0.277**	**0.278**	**0.269**	**0.248**
1975	**0.428**	**0.478**	**0.364**	**0.323**	**0.309**	**0.299**	**0.296**	**0.285**	**0.261**
1976	**0.498**	**0.559**	**0.395**	**0.347**	**0.330**	**0.319**	**0.314**	**0.300**	**0.274**
1977	**0.560**	**0.632**	**0.423**	**0.369**	**0.349**	**0.338**	**0.329**	**0.314**	**0.285**
1978	**0.609**	**0.691**	**0.449**	**0.388**	**0.365**	**0.353**	**0.341**	**0.325**	**0.294**
1979	**0.641**	**0.734**	**0.470**	**0.404**	**0.377**	**0.364**	**0.350**	**0.333**	**0.300**
1980	**0.652**	**0.756**	**0.487**	**0.415**	**0.385**	**0.370**	**0.354**	**0.336**	**0.304**
1981	**0.642**	**0.756**	**0.499**	**0.422**	**0.387**	**0.370**	**0.353**	**0.335**	**0.304**
1982	**0.615**	**0.738**	**0.504**	**0.423**	**0.384**	**0.365**	**0.348**	**0.331**	**0.301**
1983	**0.576**	**0.706**	**0.502**	**0.420**	**0.378**	**0.357**	**0.340**	**0.324**	**0.296**
1984	**0.528**	**0.665**	**0.492**	**0.411**	**0.368**	**0.346**	**0.329**	**0.314**	**0.289**
1985	**0.478**	**0.618**	**0.474**	**0.397**	**0.355**	**0.332**	**0.316**	**0.303**	**0.280**
1986	**0.429**	**0.568**	**0.448**	**0.378**	**0.339**	**0.318**	**0.302**	**0.290**	**0.271**

1987	**0.382**	**0.518**	**0.415**	**0.356**	**0.322**	**0.302**	**0.289**	**0.278**	**0.262**
1988	**0.339**	**0.471**	**0.380**	**0.333**	**0.305**	**0.288**	**0.275**	**0.266**	**0.253**
1989	**0.302**	**0.427**	**0.346**	**0.310**	**0.289**	**0.274**	**0.264**	**0.256**	**0.245**
1990	**0.270**	**0.390**	**0.317**	**0.292**	**0.276**	**0.263**	**0.254**	**0.247**	**0.239**
1991	**0.246**	**0.360**	**0.296**	**0.278**	**0.266**	**0.256**	**0.248**	**0.242**	**0.234**
1992	**0.229**	**0.339**	**0.285**	**0.271**	**0.261**	**0.252**	**0.245**	**0.240**	**0.233**
1993	**0.218**	**0.325**	**0.282**	**0.269**	**0.259**	**0.251**	**0.245**	**0.240**	**0.233**
1994	**0.212**	**0.318**	**0.283**	**0.270**	**0.259**	**0.251**	**0.246**	**0.241**	**0.234**
1995	**0.208**	**0.316**	**0.288**	**0.272**	**0.261**	**0.253**	**0.248**	**0.244**	**0.236**
1996	**0.207**	**0.318**	**0.293**	**0.275**	**0.262**	**0.255**	**0.250**	**0.246**	**0.239**
1997	**0.206**	**0.322**	**0.296**	**0.276**	**0.264**	**0.256**	**0.251**	**0.247**	**0.240**
1998	**0.206**	**0.329**	**0.297**	**0.276**	**0.264**	**0.257**	**0.251**	**0.247**	**0.241**
1999	**0.206**	**0.337**	**0.297**	**0.276**	**0.265**	**0.257**	**0.250**	**0.246**	**0.242**
2000	**0.208**	**0.347**	**0.298**	**0.276**	**0.265**	**0.258**	**0.250**	**0.246**	**0.243**
2001	**0.211**	**0.358**	**0.300**	**0.278**	**0.267**	**0.259**	**0.251**	**0.246**	**0.244**
2002	**0.217**	**0.369**	**0.303**	**0.281**	**0.270**	**0.261**	**0.252**	**0.247**	**0.244**
2003	**0.226**	**0.381**	**0.309**	**0.287**	**0.275**	**0.264**	**0.254**	**0.249**	**0.245**
2004	**0.236**	**0.393**	**0.317**	**0.295**	**0.281**	**0.268**	**0.258**	**0.251**	**0.247**
2005	**0.248**	**0.405**	**0.325**	**0.302**	**0.286**	**0.272**	**0.261**	**0.254**	**0.248**
2006	**0.259**	**0.418**	**0.333**	**0.310**	**0.292**	**0.276**	**0.264**	**0.257**	**0.249**
2007	**0.269**	**0.431**	**0.340**	**0.315**	**0.295**	**0.279**	**0.267**	**0.259**	**0.250**
2008	**0.278**	**0.443**	**0.344**	**0.318**	**0.297**	**0.280**	**0.269**	**0.260**	**0.251**
2009	**0.284**	**0.455**	**0.346**	**0.318**	**0.296**	**0.280**	**0.269**	**0.260**	**0.252**
2010	**0.288**	**0.464**	**0.345**	**0.315**	**0.294**	**0.278**	**0.268**	**0.259**	**0.251**
2011	**0.287**	**0.467**	**0.342**	**0.311**	**0.290**	**0.275**	**0.265**	**0.257**	**0.250**
2012	**0.283**	**0.464**	**0.337**	**0.304**	**0.284**	**0.271**	**0.262**	**0.254**	**0.248**
2013	**0.273**	**0.452**	**0.329**	**0.296**	**0.277**	**0.265**	**0.257**	**0.250**	**0.245**
2014	**0.259**	**0.431**	**0.319**	**0.286**	**0.270**	**0.259**	**0.251**	**0.245**	**0.241**
2015	**0.241**	**0.403**	**0.308**	**0.277**	**0.261**	**0.252**	**0.245**	**0.239**	**0.236**
2016	**0.223**	**0.373**	**0.297**	**0.267**	**0.254**	**0.246**	**0.239**	**0.234**	**0.231**
2017	**0.205**	**0.345**	**0.286**	**0.259**	**0.247**	**0.240**	**0.234**	**0.229**	**0.226**
2018	**0.191**	**0.322**	**0.277**	**0.253**	**0.241**	**0.236**	**0.230**	**0.225**	**0.223**
2019	**0.182**	**0.308**	**0.270**	**0.249**	**0.238**	**0.233**	**0.228**	**0.224**	**0.221**
2020	**0.178**	**0.304**	**0.266**	**0.247**	**0.237**	**0.232**	**0.227**	**0.223**	**0.220**
2021	**0.179**	**0.308**	**0.264**	**0.248**	**0.238**	**0.232**	**0.228**	**0.225**	**0.221**
2022	**0.183**	**0.317**	**0.264**	**0.250**	**0.241**	**0.234**	**0.230**	**0.227**	**0.222**
2023	**0.190**	**0.331**	**0.265**	**0.253**	**0.244**	**0.236**	**0.233**	**0.230**	**0.225**
2024	**0.197**	**0.346**	**0.266**	**0.257**	**0.247**	**0.239**	**0.237**	**0.234**	**0.227**

Table 0-6 Herring ensemble, 10% percentile of the parameter a and b for food consumptions of cod (lim_10): Natural mortality (sum of quarterly M1+M2).

Year/Age	0	1	2	3	4	5	6	7	8
1974	0.420	0.390	0.274	0.224	0.207	0.196	0.194	0.182	0.157
1975	0.361	0.432	0.301	0.244	0.224	0.212	0.211	0.199	0.170
1976	0.357	0.377	0.274	0.231	0.215	0.204	0.202	0.192	0.165
1977	0.489	0.423	0.294	0.245	0.227	0.215	0.214	0.201	0.172
1978	0.680	0.625	0.344	0.304	0.286	0.267	0.244	0.231	0.204
1979	0.758	0.788	0.365	0.309	0.298	0.286	0.278	0.248	0.207
1980	0.653	0.793	0.457	0.363	0.322	0.326	0.284	0.263	0.233
1981	0.663	0.717	0.441	0.344	0.296	0.270	0.270	0.242	0.208
1982	0.616	0.733	0.422	0.347	0.293	0.263	0.247	0.244	0.201
1983	0.555	0.654	0.473	0.332	0.310	0.273	0.247	0.232	0.208
1984	0.457	0.561	0.424	0.327	0.263	0.263	0.239	0.217	0.194
1985	0.407	0.480	0.386	0.292	0.246	0.219	0.213	0.201	0.182
1986	0.378	0.438	0.330	0.292	0.230	0.212	0.196	0.185	0.165
1987	0.410	0.446	0.275	0.239	0.225	0.195	0.181	0.172	0.156
1988	0.361	0.452	0.324	0.235	0.224	0.208	0.188	0.174	0.156
1989	0.280	0.379	0.254	0.256	0.214	0.191	0.181	0.166	0.150
1990	0.146	0.218	0.164	0.151	0.156	0.140	0.136	0.131	0.127
1991	0.122	0.181	0.155	0.139	0.128	0.135	0.123	0.126	0.120
1992	0.144	0.187	0.157	0.142	0.127	0.122	0.127	0.120	0.118
1993	0.175	0.225	0.190	0.165	0.154	0.143	0.137	0.141	0.128
1994	0.153	0.227	0.192	0.173	0.154	0.149	0.141	0.133	0.132
1995	0.135	0.203	0.177	0.166	0.155	0.148	0.145	0.137	0.136
1996	0.119	0.181	0.167	0.153	0.148	0.143	0.138	0.135	0.128
1997	0.110	0.167	0.157	0.146	0.140	0.135	0.132	0.129	0.127
1998	0.127	0.171	0.152	0.144	0.136	0.131	0.127	0.126	0.119
1999	0.142	0.188	0.162	0.149	0.143	0.136	0.129	0.128	0.121
2000	0.184	0.291	0.207	0.195	0.186	0.177	0.165	0.158	0.156
2001	0.193	0.307	0.219	0.191	0.186	0.174	0.168	0.165	0.162
2002	0.180	0.319	0.233	0.201	0.181	0.174	0.166	0.160	0.163
2003	0.150	0.278	0.186	0.170	0.160	0.156	0.149	0.143	0.140
2004	0.174	0.252	0.230	0.177	0.165	0.150	0.145	0.140	0.134
2005	0.213	0.301	0.255	0.228	0.187	0.167	0.154	0.148	0.140
2006	0.216	0.330	0.222	0.219	0.206	0.185	0.161	0.153	0.146
2007	0.230	0.340	0.238	0.214	0.196	0.186	0.162	0.152	0.140
2008	0.235	0.363	0.238	0.218	0.200	0.179	0.185	0.164	0.154
2009	0.263	0.377	0.262	0.219	0.210	0.187	0.173	0.185	0.166
2010	0.281	0.417	0.298	0.243	0.211	0.206	0.190	0.178	0.175
2011	0.261	0.425	0.270	0.247	0.213	0.193	0.188	0.173	0.174
2012	0.244	0.379	0.208	0.214	0.186	0.165	0.157	0.148	0.140
2013	0.258	0.361	0.226	0.176	0.179	0.160	0.148	0.143	0.137
2014	0.206	0.362	0.231	0.186	0.158	0.164	0.147	0.142	0.137
2015	0.190	0.295	0.196	0.173	0.157	0.145	0.145	0.137	0.133
2016	0.168	0.273	0.242	0.170	0.160	0.150	0.143	0.139	0.133
2017	0.151	0.242	0.188	0.174	0.146	0.141	0.134	0.127	0.124
2018	0.145	0.224	0.172	0.146	0.144	0.128	0.127	0.125	0.120
2019	0.131	0.217	0.160	0.149	0.136	0.136	0.125	0.123	0.120

2020	**0.130**	**0.204**	**0.158**	**0.142**	**0.137**	**0.129**	**0.130**	**0.120**	**0.120**
2021	**0.147**	**0.209**	**0.163**	**0.152**	**0.140**	**0.137**	**0.130**	**0.129**	**0.125**
2022	**0.159**	**0.235**	**0.179**	**0.166**	**0.151**	**0.147**	**0.139**	**0.137**	**0.132**
2023	**0.164**	**0.256**	**0.188**	**0.162**	**0.153**	**0.143**	**0.144**	**0.138**	**0.129**
2024	**0.131**	**0.252**	**0.159**	**0.153**	**0.143**	**0.132**	**0.129**	**0.127**	**0.122**

Table 0-7. Herring ensemble, 10% percentile of the parameter a and b for food consumptions of cod (lim_10,) :Smoothed M.

Year/Age	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
1974	**0.333**	**0.314**	**0.247**	**0.208**	**0.196**	**0.186**	**0.187**	**0.176**	**0.153**
1975	**0.412**	**0.406**	**0.279**	**0.234**	**0.219**	**0.207**	**0.205**	**0.192**	**0.166**
1976	**0.487**	**0.493**	**0.311**	**0.258**	**0.240**	**0.228**	**0.222**	**0.208**	**0.179**
1977	**0.553**	**0.571**	**0.340**	**0.281**	**0.259**	**0.247**	**0.237**	**0.221**	**0.190**
1978	**0.605**	**0.635**	**0.367**	**0.301**	**0.275**	**0.262**	**0.250**	**0.232**	**0.199**
1979	**0.639**	**0.681**	**0.389**	**0.317**	**0.287**	**0.273**	**0.258**	**0.240**	**0.205**
1980	**0.650**	**0.704**	**0.407**	**0.328**	**0.295**	**0.278**	**0.262**	**0.243**	**0.209**
1981	**0.639**	**0.703**	**0.418**	**0.334**	**0.297**	**0.278**	**0.260**	**0.242**	**0.208**
1982	**0.609**	**0.682**	**0.423**	**0.335**	**0.294**	**0.273**	**0.255**	**0.237**	**0.205**
1983	**0.565**	**0.647**	**0.420**	**0.330**	**0.286**	**0.264**	**0.246**	**0.229**	**0.200**
1984	**0.513**	**0.600**	**0.409**	**0.320**	**0.275**	**0.252**	**0.234**	**0.218**	**0.192**
1985	**0.456**	**0.546**	**0.388**	**0.305**	**0.260**	**0.237**	**0.220**	**0.206**	**0.183**
1986	**0.400**	**0.488**	**0.358**	**0.284**	**0.243**	**0.221**	**0.205**	**0.193**	**0.173**
1987	**0.346**	**0.431**	**0.321**	**0.259**	**0.224**	**0.204**	**0.190**	**0.179**	**0.162**
1988	**0.295**	**0.375**	**0.282**	**0.233**	**0.205**	**0.187**	**0.175**	**0.165**	**0.152**
1989	**0.250**	**0.323**	**0.243**	**0.208**	**0.186**	**0.171**	**0.161**	**0.153**	**0.142**
1990	**0.211**	**0.278**	**0.209**	**0.186**	**0.170**	**0.158**	**0.150**	**0.143**	**0.134**
1991	**0.180**	**0.241**	**0.183**	**0.169**	**0.157**	**0.148**	**0.141**	**0.135**	**0.129**
1992	**0.158**	**0.214**	**0.167**	**0.158**	**0.149**	**0.141**	**0.136**	**0.131**	**0.125**
1993	**0.142**	**0.196**	**0.160**	**0.152**	**0.144**	**0.138**	**0.133**	**0.129**	**0.124**
1994	**0.133**	**0.186**	**0.159**	**0.150**	**0.143**	**0.137**	**0.133**	**0.129**	**0.124**
1995	**0.129**	**0.183**	**0.162**	**0.151**	**0.143**	**0.138**	**0.134**	**0.131**	**0.126**
1996	**0.128**	**0.186**	**0.167**	**0.154**	**0.145**	**0.140**	**0.136**	**0.133**	**0.128**
1997	**0.131**	**0.195**	**0.173**	**0.157**	**0.148**	**0.142**	**0.138**	**0.135**	**0.131**
1998	**0.136**	**0.208**	**0.179**	**0.161**	**0.151**	**0.146**	**0.141**	**0.137**	**0.134**
1999	**0.143**	**0.224**	**0.185**	**0.165**	**0.156**	**0.149**	**0.143**	**0.140**	**0.137**
2000	**0.151**	**0.241**	**0.192**	**0.171**	**0.161**	**0.153**	**0.146**	**0.142**	**0.140**
2001	**0.161**	**0.260**	**0.199**	**0.177**	**0.166**	**0.158**	**0.150**	**0.145**	**0.143**
2002	**0.173**	**0.277**	**0.207**	**0.184**	**0.172**	**0.162**	**0.153**	**0.148**	**0.145**
2003	**0.185**	**0.293**	**0.216**	**0.193**	**0.179**	**0.167**	**0.157**	**0.151**	**0.147**
2004	**0.198**	**0.308**	**0.226**	**0.201**	**0.186**	**0.172**	**0.161**	**0.155**	**0.149**
2005	**0.211**	**0.322**	**0.235**	**0.210**	**0.192**	**0.177**	**0.165**	**0.158**	**0.151**
2006	**0.223**	**0.335**	**0.243**	**0.217**	**0.197**	**0.181**	**0.168**	**0.160**	**0.152**
2007	**0.234**	**0.348**	**0.249**	**0.222**	**0.201**	**0.183**	**0.171**	**0.162**	**0.153**
2008	**0.243**	**0.362**	**0.253**	**0.225**	**0.202**	**0.184**	**0.172**	**0.163**	**0.154**

2009	**0.249**	**0.374**	**0.255**	**0.224**	**0.202**	**0.184**	**0.172**	**0.163**	**0.154**
2010	**0.252**	**0.383**	**0.254**	**0.221**	**0.199**	**0.182**	**0.171**	**0.162**	**0.154**
2011	**0.252**	**0.387**	**0.251**	**0.216**	**0.195**	**0.179**	**0.168**	**0.160**	**0.153**
2012	**0.247**	**0.384**	**0.245**	**0.210**	**0.189**	**0.174**	**0.165**	**0.157**	**0.150**
2013	**0.236**	**0.371**	**0.237**	**0.201**	**0.181**	**0.169**	**0.160**	**0.153**	**0.147**
2014	**0.221**	**0.347**	**0.226**	**0.191**	**0.173**	**0.162**	**0.154**	**0.147**	**0.143**
2015	**0.201**	**0.317**	**0.214**	**0.181**	**0.165**	**0.155**	**0.147**	**0.141**	**0.137**
2016	**0.181**	**0.284**	**0.202**	**0.171**	**0.156**	**0.148**	**0.141**	**0.135**	**0.132**
2017	**0.162**	**0.253**	**0.190**	**0.162**	**0.149**	**0.142**	**0.135**	**0.130**	**0.127**
2018	**0.147**	**0.228**	**0.180**	**0.155**	**0.143**	**0.137**	**0.131**	**0.126**	**0.124**
2019	**0.137**	**0.213**	**0.173**	**0.151**	**0.140**	**0.134**	**0.129**	**0.124**	**0.121**
2020	**0.133**	**0.209**	**0.169**	**0.150**	**0.139**	**0.133**	**0.128**	**0.124**	**0.121**
2021	**0.135**	**0.214**	**0.168**	**0.150**	**0.140**	**0.134**	**0.129**	**0.126**	**0.122**
2022	**0.140**	**0.226**	**0.168**	**0.153**	**0.143**	**0.135**	**0.132**	**0.128**	**0.123**
2023	**0.147**	**0.241**	**0.169**	**0.156**	**0.146**	**0.138**	**0.135**	**0.132**	**0.126**
2024	**0.156**	**0.259**	**0.171**	**0.160**	**0.150**	**0.141**	**0.139**	**0.136**	**0.129**

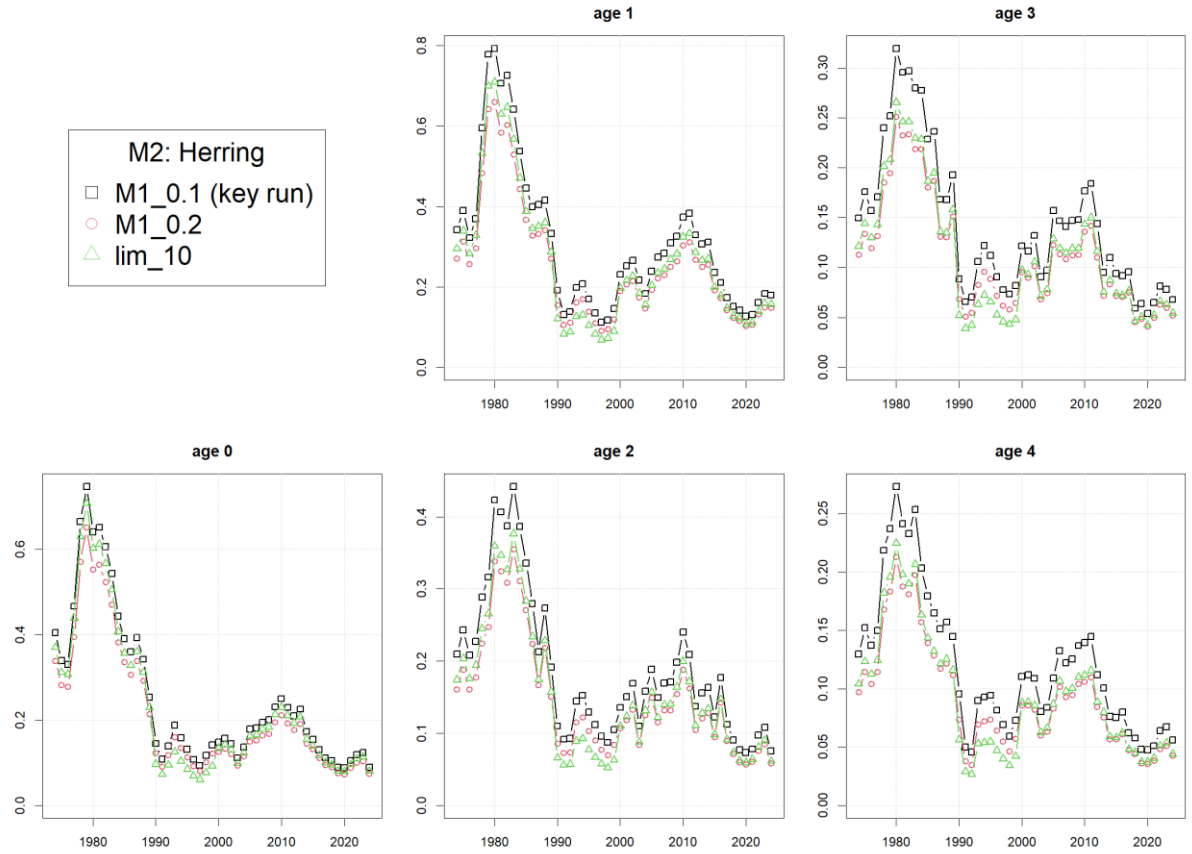


Figure 0-14. Herring ensemble. M2 values for herring using M1=0.1 (same as in the key run), M1=0.2 and with cod consumption calculated from 10% percentile of the parameter a and b for food consumptions of cod (lim_10)

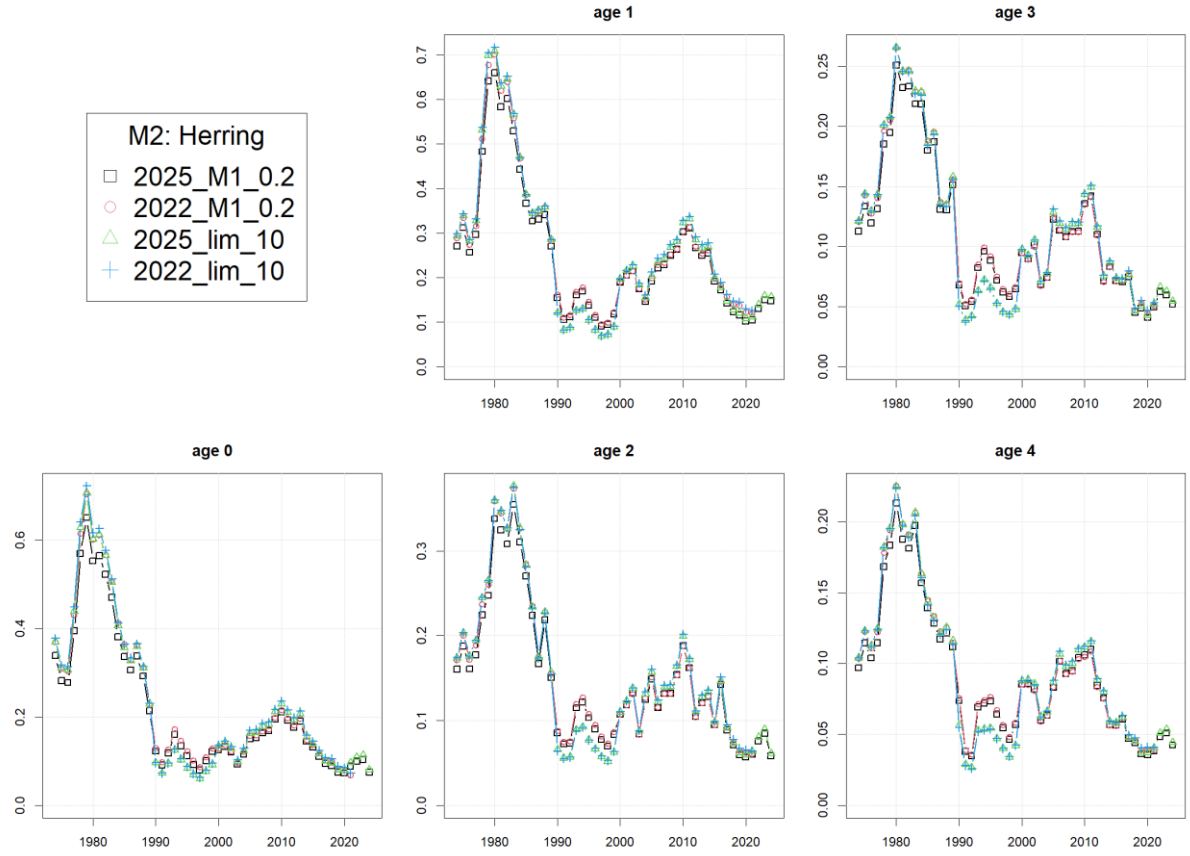


Figure 0-15. Herring ensemble. M2 values for herring based on the key runs in 2022 and 2025, except using M1=0.2 for herring and with cod consumption calculated from 10% percentile of the parameter a and b for food consumptions of cod (lim_10)

Uncertainties of parameters and output

SMS estimate the uncertainties of selected output variables using the Hessian matrix and the delta-method approximation. Most variables like stock number and F for dynamic species are estimated within the model, while other variables like the stock numbers of the "external predators" cod are assumed known without errors. With cod as the only predator, this combination of estimated and assumed "known" variables will certainly lead to an underestimate of the uncertainties of e.g. predation mortality. Therefore, the uncertainties estimated from the Hessian matrix are not presented in details.

An example of estimated uncertainties is presented in the below figures. The confidence interval seems too tight!

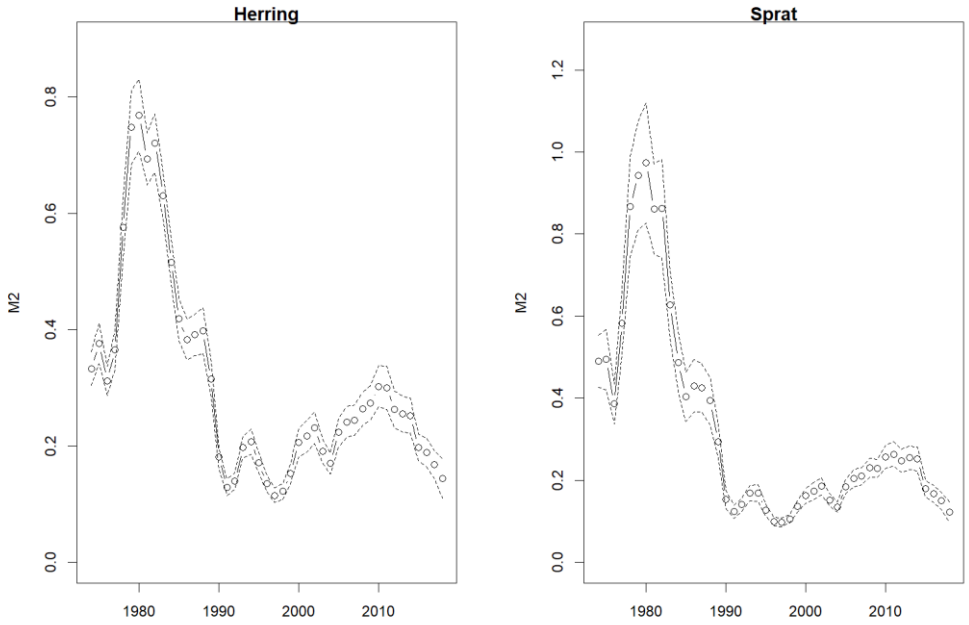


Figure 0-16. Values of M2 and 95% confidence interval (+- 2*standard deviation) for age 1 of herring and sprat. From the 2019 key run.

Sensitivity test

To get a better idea on true uncertainties several sensitivity runs were carried out as mainly as part of the **2019 and 2022 key run**. The list below gives an overview of the sensitivity analysis made. The retrospective analysis was repeated using the most recent key run analysis, and the sensitivity of M1 was made on the 2022 and 2025 key runs.

1. Retrospective analysis (5 year peel of all input data). 2025 key run.
2. Sensitivity to stomach data (old vs. new stomach data set). 2019 key run.
3. Sensitivity to stomach data (aggregation stomach data over a 5 or 10 years period). 2019 key run.
4. Sensitivity towards using different assumptions for size selection. 2019 key run.
5. Sensitivity towards using or not using an overlap index for Other Food. 2019 key run.
6. Sensitivity towards consumption rates. 2019 key run.
7. Sensitivity towards residual mortality (M1). 2015 key run.
8. Comparison of the 2012, 2019 and 2022 key run.
9. Comparison with the Gadget model run. 2019 key run.

Retrospective analysis (5 year peel of all input data)

The retrospective analysis shows variable estimates of recruitment, SSB and F for the terminal years in the time series, (Figure 0-1). Comparison with the same kind of output for the ICES assessment (WGBFAS, 2025) reveals however a similar variability in the ICES single species assessment output.

The retrospective analysis shows a consistent estimate of predation mortalities (Figure 0-2). This consistent estimate is probably also because all runs use the same stomach contents data; the last year with stomach data is 2014. As for all other retrospective assessment analysis, values (M2) in the terminal year of the time-series have larger uncertainties.



Figure 0-1. Retrospective analysis for herring and sprat. Summary output.

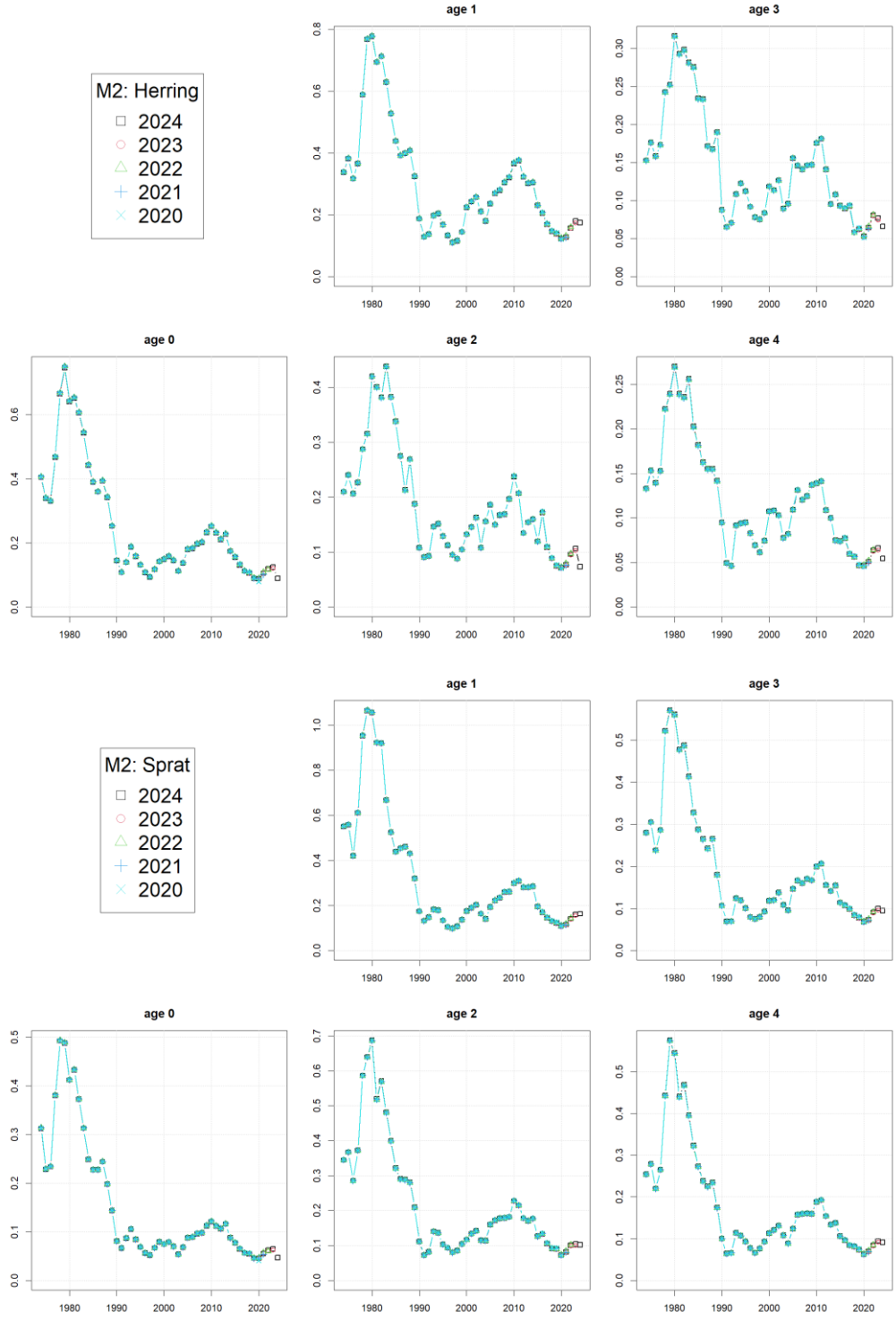


Figure 0-2. Retrospective analysis for herring and sprat, M2 at age.

Sensitivity to stomach data (old vs. new stomach data set)

The choice of stomach contents data, "old", "new" or combined was investigated as part of the 2019 key run. The actual choice has limited effect on the SMS stock summary output (Figure 0-4) or predation mortalities (Figure 0-4)

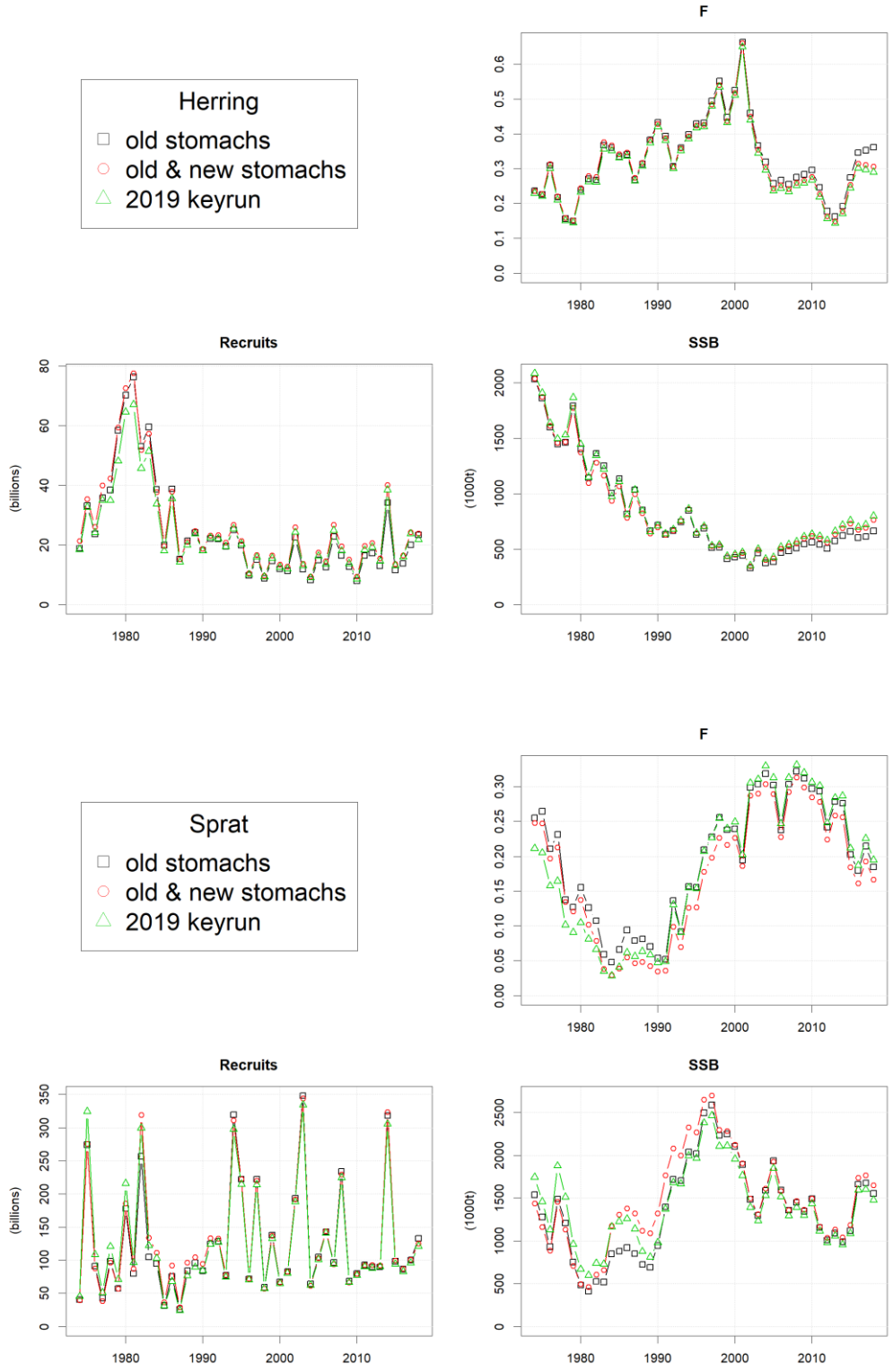


Figure 0-3. Comparison of output from SMS runs with combinations of contents data: "old" pooled stomach data, "new" individually sample stomachs. The key run uses only the "new" data.

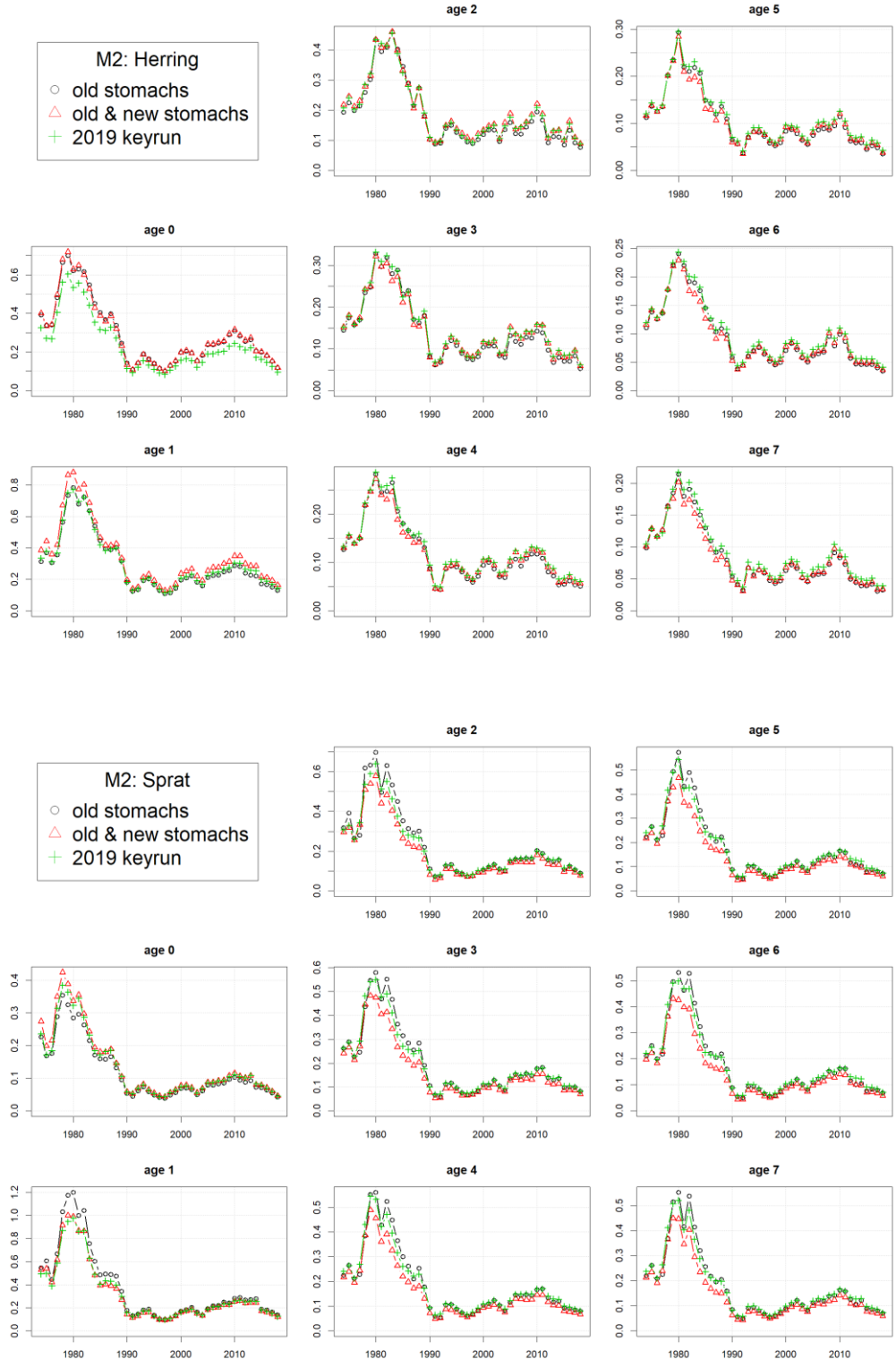


Figure 0-4. Comparison of M2 values from SMS runs with combinations of contents data: "old" pooled stomach data, "new" individually sampled stomachs. The key run uses only the "new" data.

Sensitivity towards using different assumptions for size selection

Three options for predator prey size selection were tried as part of the 2019 key run:

1. Log normal size selection (key run): a predator has a preferred prey size ratio and a prey twice as big as the preferred size is as attractive as another half the prey size. The preferred size ration and its variance are estimated by SMS.
2. Uniform size selection: a size preference at 1 within the range of the observed size ratio and 0 outside that ratio.
3. Constraint uniform size selection: as Uniform size selection, but the size preference ratio is constrained to exclude "outliers" from the observed size ratio, estimated from a quantile regression (Figure 0-5).

.

The main performance statistics of a SMS run for the three size selection models (Table 0-1) show the best model likelihood and AIC for the key run.

Stock summary output (Figure 0-6) and M2 (Figure 0-7) are quite sensitive to the choice of size selection option. It seems as if the "constraint uniform" option excludes interactions from medium sized cod on larger herring (Figure 0-5) such that M2 on herring ages 4-8 becomes very low. The (unconstraint) "uniform" options includes the full observed predator/prey size ratio which results in a higher M2 for the older herring than for the "constraint uniform" option.

The "constraint uniform" option performed well in the 2012 key run, however there is difference in the quality of stomach contents data used in the old and the new 2019 key run. The old key runs made use of stomach contents data with large size classes for predator preys, e.g. sprat 5-10-15 cm, while the new stomach data uses a much smaller size classes, e.g. by cm group for sprat. With wider size classes, the in predator/prey size ratio becomes imprecise, such that the cutting of "outliers" by the "constraint uniform" options had a limited effect. With the new data, the full range of observations should probably be used, if a uniform size selection option is used.

Table 0-1. SMS main performance statistics from a SMS run with the "uniform size selection", "constraint uniform size selection" and the key run.

uniform size selection

```
objective function (negative log likelihood):  -1166.15
Number of parameters: 289
Number of observations used in likelihood: 14892
Akaike information criterion (AIC):   -1754.31

Number of observations used in the likelihood:
unweighted objective function contributions (total):
                Catch    CPUE    S/R   Stom.       Sum
Cod               0.0     0.0     0.0  -173.4      -173
Herring        -667.6  -117.3   -12.5     0.0      -797
Sprat          -100.9  -106.0    -5.5     0.0      -212
Sum            -768.6  -223.3   -18.0  -173.4     -1183
```

constraint uniform size selection

```
objective function (negative log likelihood):  -1110.12
Number of parameters: 289
Number of observations used in likelihood: 14892
Akaike information criterion (AIC):   -1642.24

unweighted objective function contributions (total):
                Catch    CPUE    S/R   Stom.       Sum
Cod               0.0     0.0     0.0  -182.5      -183
Herring        -645.5  -121.1    -8.2     0.0      -775
Sprat           -56.8  -103.5    -5.2     0.0      -166
Sum            -702.4  -224.7   -13.4  -182.5     -1123
```

Log-normal size selection (key run 2019)

```
objective function (negative log likelihood):  -1232.3
Number of parameters: 292
Number of observations used in likelihood: 14892
Akaike information criterion (AIC):   -1880.6

unweighted objective function contributions (total):
                Catch    CPUE    S/R   Stom.       Sum
Cod               0.0     0.0     0.0  -256.2      -256
Herring        -660.4  -118.7    -8.6     0.0      -788
Sprat           -92.3  -104.0    -5.6     0.0      -202
Sum            -752.7  -222.7   -14.2  -256.2     -1246
```

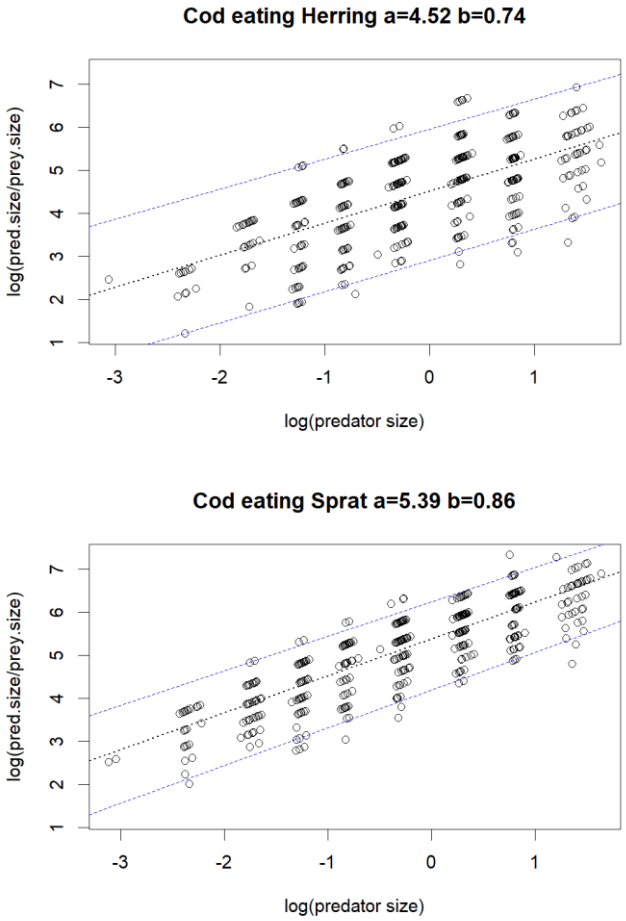


Figure 0-5. Quantile regression with observations of predator and predator/prey sizes. The blue lines show the 2.5% and 97.5 % percentile lines, which defines the "size selection window".

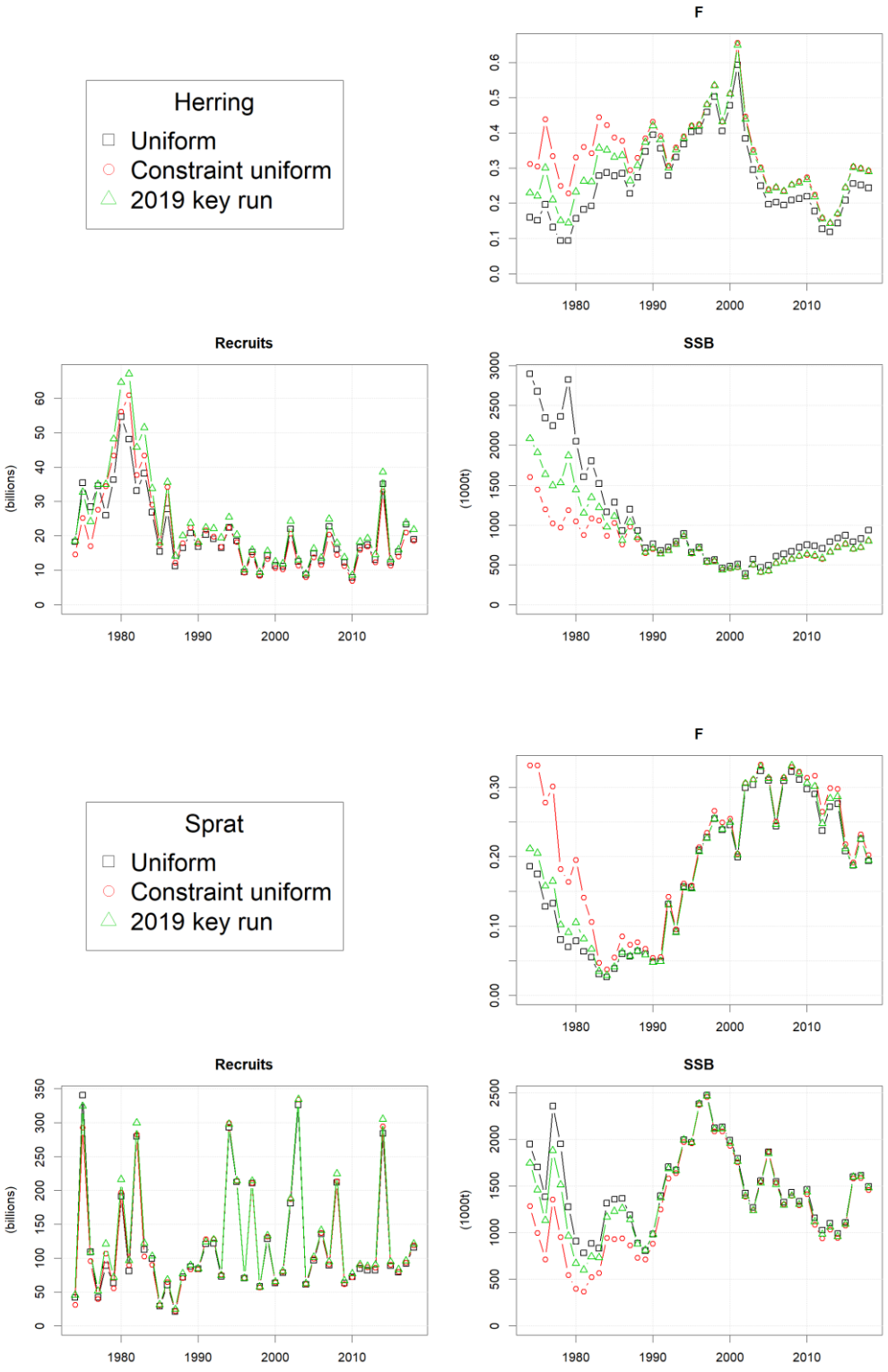


Figure 0-6. Comparison of output from SMS runs with three options for predator prey size selection.

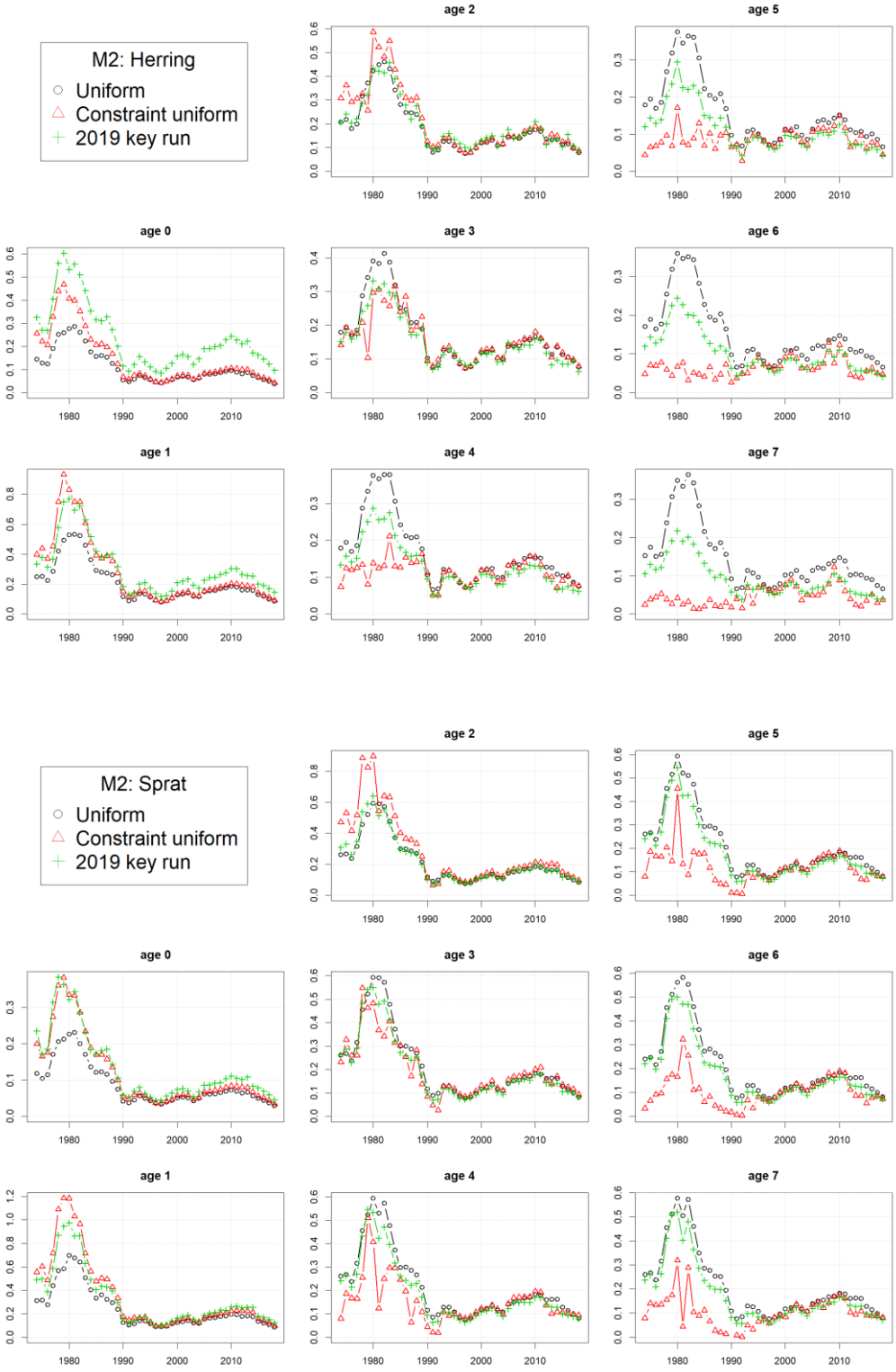


Figure 0-7. Comparison of M2 values from SMS runs with three options for predator prey size selection.

Sensitivity to stomach data (aggregation over a 5 or 10 years period)

Stomachs data are by default aggregated for each combination of year and quarter of the year. For most cases this leads to a rather few stomachs for some of the predator length classes and uncertainties on how partly identified prey items should be assigned. Aggregating stomach data over more years, e.g. 5 or 10 years, provides a larger sample size and a smaller observation uncertainties, but an e.g. ”10 years diet” will not reflect the variability in available food for the individual years. This effect of aggregating stomachs over several years was investigated as part of the 2019 key run.

The average likelihood contribution (Table 0-2) show that likelihood per stomach contents observation becomes better (more negative) when data are aggregated over some years compared to the key run which uses data by year. The best average likelihood for stomach data is obtained using a 5-years aggregation. This may be interpreted that pooling stomach data between year gives a higher precision (more stomachs) of data used by SMS. However, using a very wide year range may negatively affect the fit between “observed” stomach contents and the model estimate of stomach contents calculated for the midpoint of the years used in the data aggregation. Likelihood contributions from For Catch, CPUE and S/R observations are quite the same for the three configurations.

M2 values for the three configurations are differ mainly for age 0 and 1 of herring and sprat (Figure 0-8).

Table 0-2. Objective function contributions (per observation) from SMS models using stomach contents data aggregated over 5, 10 years and from the key run.

5 years aggregation:

	Catch	CPUE	S/R	Stomachs
Cod	0.00	0.00	0.00	-0.28
Herring	-0.46	-0.46	-0.18	0.00
Sprat	-0.08	-0.33	-0.12	0.00

10 years aggregation:

	Catch	CPUE	S/R	Stomachs
Cod	0.00	0.00	0.00	-0.26
Herring	-0.47	-0.45	-0.18	0.00
Sprat	-0.09	-0.33	-0.11	0.00

Key run:

	Catch	CPUE	S/R	Stomachs
Cod	0.00	0.00	0.00	-0.16
Herring	-0.46	-0.47	-0.19	0.00
Sprat	-0.07	-0.33	-0.12	0.00

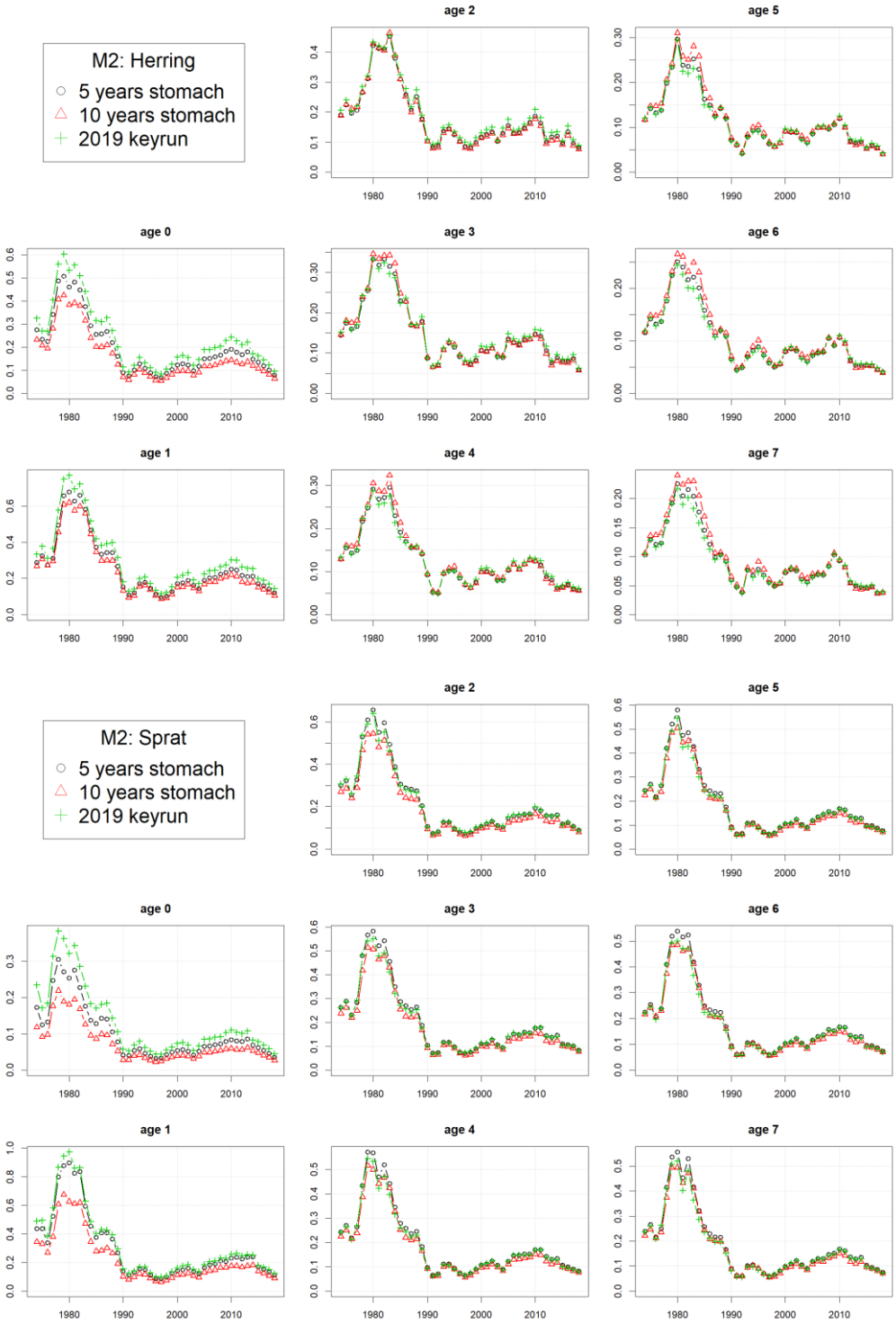


Figure 0-8. M2 estimated from SMS runs using stomach contents data aggregated over 5, 10 years and from the key run.

Sensitivity towards using an overlap index for Other Food

The "other food" prey *Saduria entomon* is an important benthic prey item for cod. The occurrence of this prey depends on the oxygen level at the bottom, which have changed considerably in the model timespan.

A time-series of total area (km^2) of hypoxic bottoms (between 20 and 100 m depth) was used to develop an index for overlap with *Saduria entomon* and other benthic components, assuming ≤1 ml l^{-1} (approx. 1.4 mg l^{-1}) as threshold for oxygen concentration to indicate failure in benthic productivity. With A_h indicating the hypoxic bottom area, the index was defined as $(A_h / maxA_h)^{-1}$, yielding higher values the smaller the hypoxic area was in a given year. We applied a 5-yr running mean. Weighting the areas with the sub-division specific cod distribution did not change the index time series except for the last two years with data, 2013 and 2014.

The overlap index between cod and the prey species herring and sprat was left unchanged (assumed 1 throughout the period)

The performance statistics (Table 0-3) for the runs with input overlap index and the 2019 key run are almost the same, even though the key run has a better total model likelihood. The likelihood contributions from stomach observations are the same for the two models.

Stomach contents residuals (Figure 0-9) are similar to the key run residuals (Figure 0-4) but residuals are actually less clustered in positive and negative residuals when the input overlap index is applied.

Table 0-3. SMS main performance statistics from a SMS run with input overlap index for Other Food and the key run.

Log-normal size selection (2019 key run)

```
objective function (negative log likelihood):  -1232.3
Number of parameters: 292
Number of observations used in likelihood: 14892
Akaike information criterion (AIC):   -1880.6
unweighted objective function contributions (total):
               Catch    CPUE    S/R   Stom.       Sum
Cod              0.0     0.0     0.0  -256.2      -256
Herring       -660.4  -118.7    -8.6     0.0      -788
Sprat          -92.3  -104.0    -5.6     0.0      -202
Sum           -752.7  -222.7   -14.2  -256.2     -1246
```

With input overlap index for Other Food

```
Objective function (negative log likelihood):  -1210.41
Number of parameters: 292
Number of observations used in likelihood: 14892
Akaike information criterion (AIC):   -1836.83

unweighted objective function contributions (total):
               Catch    CPUE    S/R   Stom.       Sum
Cod              0.0     0.0     0.0  -256.3      -256
Herring       -662.2  -116.4   -20.4     0.0      -799
Sprat          -71.9  -102.3    -5.9     0.0      -180
Sum           -734.1  -218.7   -26.3  -256.3     -1235
```

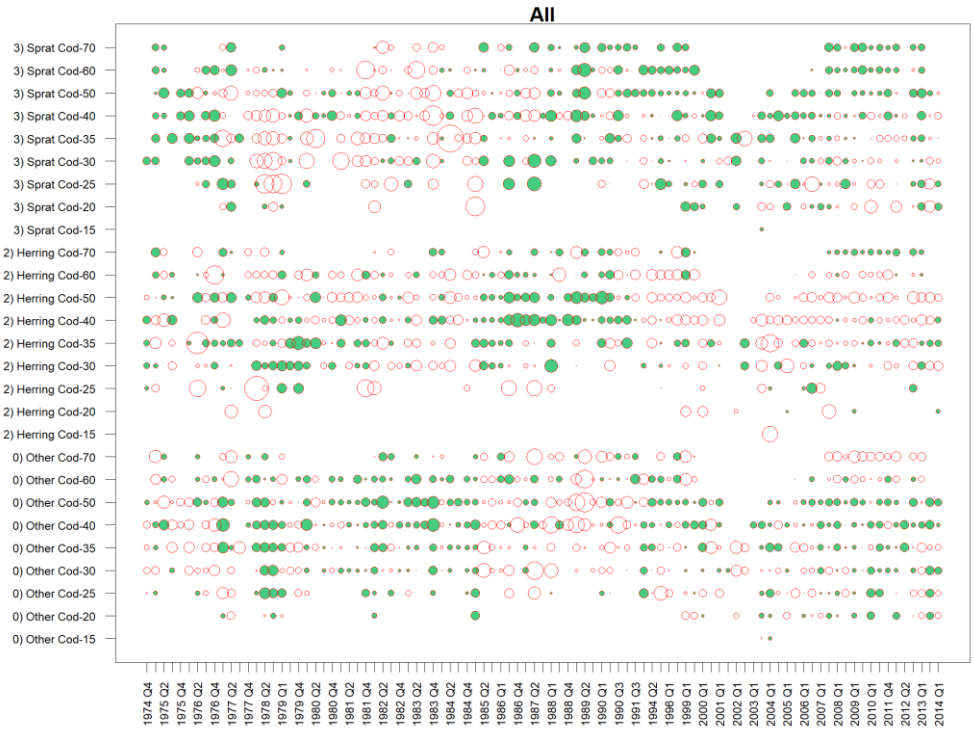


Figure 0-9. Stomach contents residuals ("Dirichlet residuals", Peter Lewy, pers. comm.). The y-axis shows prey group and predator (cod) size class. The x-axis is time period sorted by year and quarter. Green dots show that the observed stomach contents are lower than the model estimate.

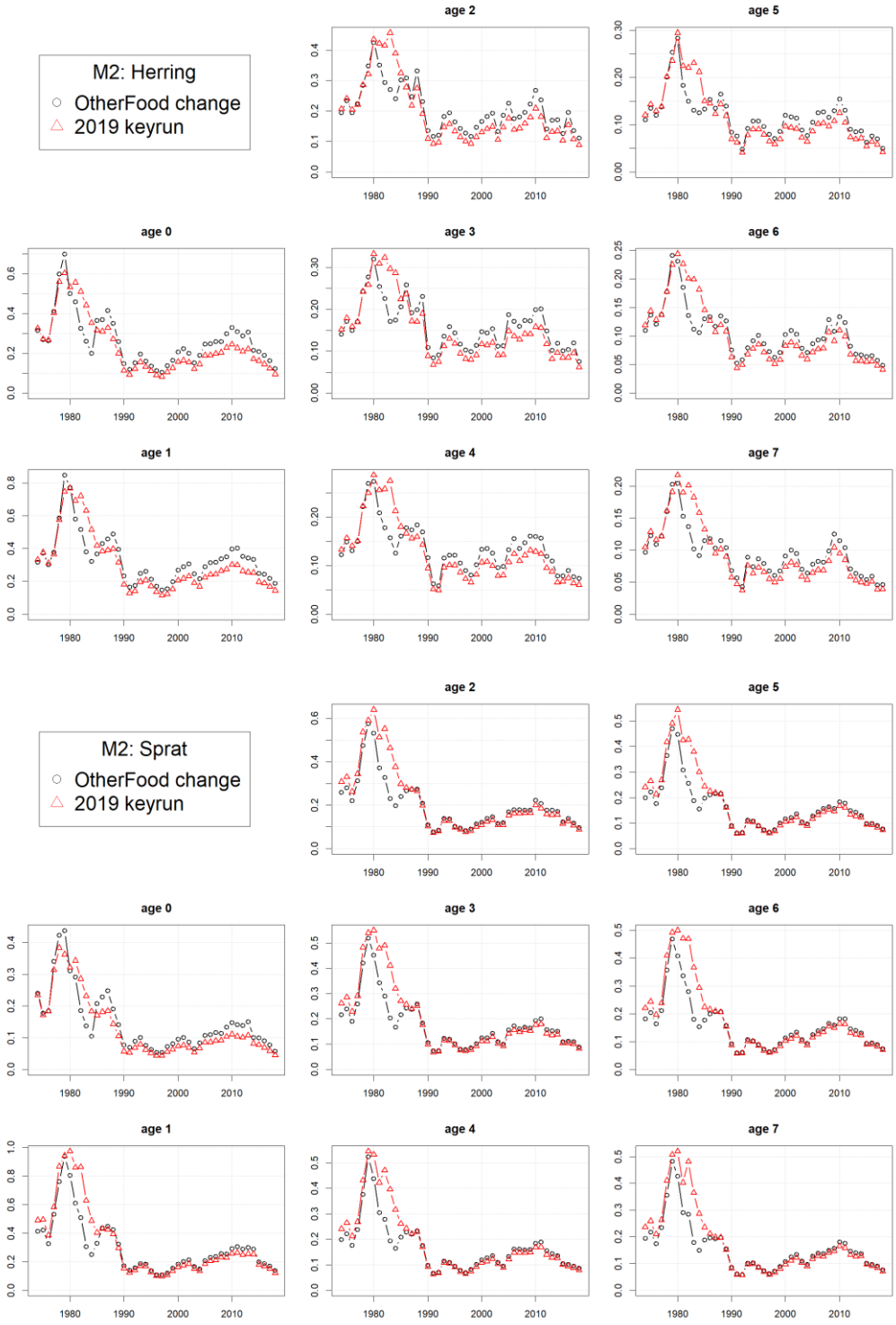


Figure 0-10. M2 estimated from a SMS run with input overlap index for Other Food and from the key run.

Sensitivity towards consumption rates

SMS can estimate a scaling factor for the input consumption rate by species. For cod this was estimated to 0.47, with a standard deviation at 0.07 for the 2019 key run. This run was used to illustrate the sensitivity of M2 to consumption rates.

The performance statistics (Table 0-4 and Table 0-1) show a slightly better fit, when the factor to the input consumption rate is applied. Likelihood contribution from stomach becomes better on the cost of the likelihood for catch at age. A similar exercise using the 2022 key run estimated a ration scaling factor at 0.46 (sd 0.07) and the same shifts in likelihood contributions as shown for the 2019 key run.

M2 values are lower when a considerably lower consumption rate are applied, but the reduction is not linear to the reduction in consumption, as expected (Figure 0-11). The reduction in M2 is larger for herring than for sprat.

Table 0-4 SMS main performance statistics from a SMS run with input overlap index for Other Food and the 2019 key run.

Log-normal size selection (2019 key run)

```
objective function (negative log likelihood):  -1232.3
Number of parameters: 292
Number of observations used in likelihood: 14892
Akaike information criterion (AIC):   -1880.6
unweighted objective function contributions (total):
                Catch    CPUE    S/R   Stom.       Sum
Cod               0.0     0.0     0.0  -256.2      -256
Herring        -660.4  -118.7    -8.6     0.0      -788
Sprat           -92.3  -104.0    -5.6     0.0      -202
Sum            -752.7  -222.7   -14.2  -256.2     -1246
```

With input consumption rates *0.47

```
objective function (negative log likelihood):  -1244.78
Number of parameters: 293
Number of observations used in likelihood: 14892
Akaike information criterion (AIC):   -1903.57

unweighted objective function contributions (total):
                Catch    CPUE    S/R   Stom.       Sum
Cod               0.0     0.0     0.0  -273.5      -273
Herring        -659.6  -119.9   -16.1     0.0      -796
Sprat           -87.2  -103.6    -5.4     0.0      -196
Sum            -746.7  -223.5   -21.5  -273.5     -1265
```

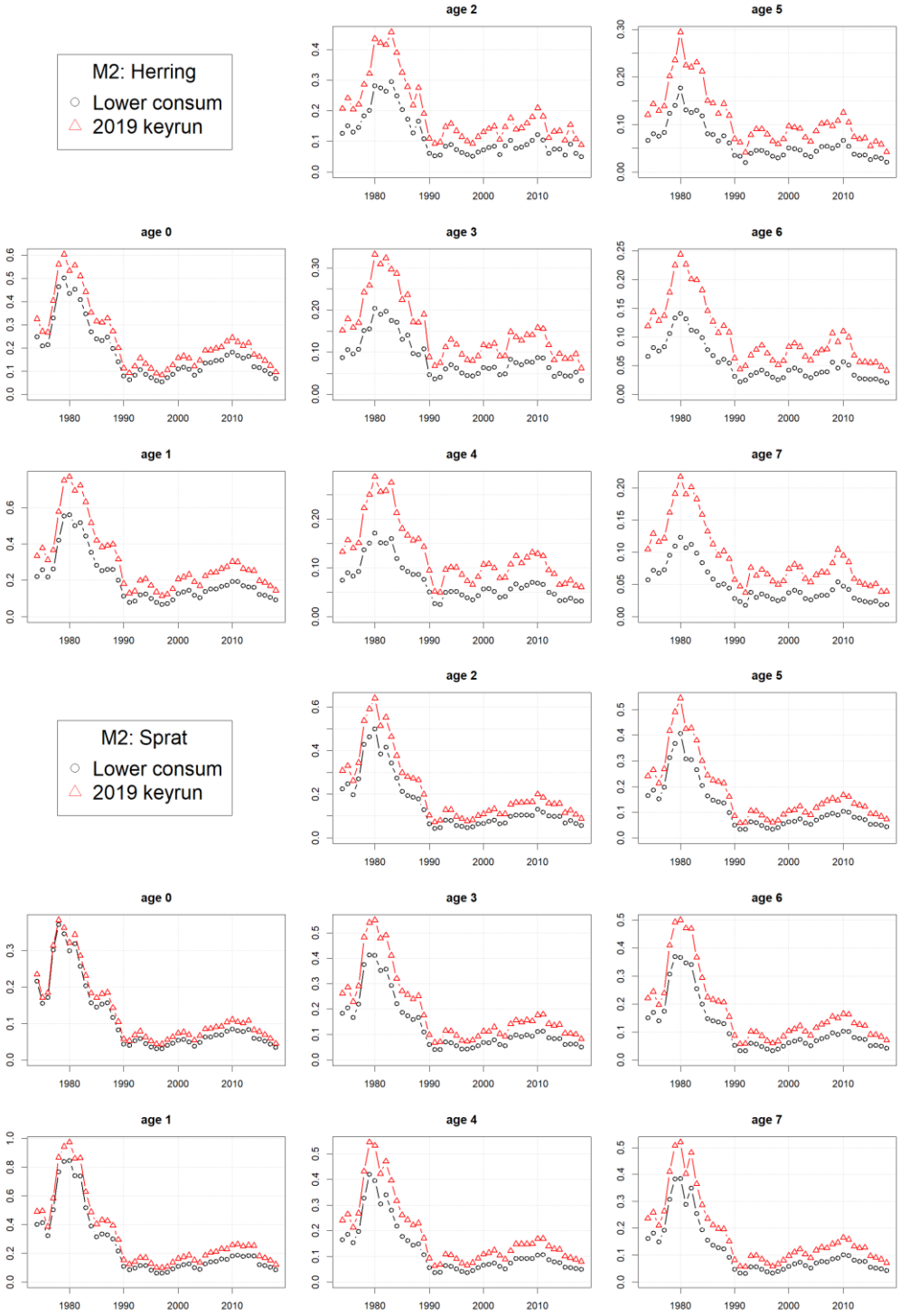


Figure 0-11. M2 estimated from a SMS run with lower (47%) consumption rates the key run.

Sensitivity towards residual natural mortality (M1).

M1 values are assumed to be constant, independent of age and known without errors, however there are no data to actually support the exact values chosen. As part of the 2019 key run it was decided to use 0.1 for herring and 0.2 (annual values) for sprat, with the justification that this are the values used for the same species for the North Sea key run.

To show the sensitivity of the M1 values on the estimated M2 values, an SMS run was made where the M1 values used in the 2022 key run were multiplied 0.5. In the 2025 key run, SMS was run with M1 multiplied by 2. M1 for herring was changed from 0.1 to 0.2 (annual values), and M1 for herring changed from 0.2 to 0.4 (annual values).

Table 0-5. SMS main performance statistics for the baseline key run and a run where M1 values are multiplied by 2.

	CATCH	CPUE	SSB/R	STOMACH	SUM
Baseline key run					
Cod	0	0	0	-331.1	-331
Herring	596.9	62.8	40.1	0	700
Sprat	1092.3	198.4	36.1	0	1327
SUM	1689.1	261.2	76.2	-331.1	1696
Sensitivity run (M1*2)					
Cod	0	0	0	-399.8	-400
Herring	591.8	83.3	11.4	0	687
Sprat	1078.4	190.5	36.5	0	1307
SUM	1670.2	273.8	47.8	-399.8	1592

The model statistics are not affected much by the choice of M1, but the fits for both catch, and stomach data are the best for the sensitivity run (changed M1 values). The model configuration is the same for the two runs, and is chosen on the basis of the key run, which may influence this result.

A higher M1 value results in a higher recruitment, a lower F and a higher SSB for both herring and sprat (Figure 0-12). M2 values (Figure 0-13) are estimated lower with a higher M1 as the stock sizes of the prey stock are higher and cod has the same total consumption irrespective of M1. Cod, is an "other predator" with input stock size is not affected by the change in M1. The sum of M1 and M2 (M), shows overall a decrease at the beginning of the time series and an increase in the second part of the time series for the sensitivity run with higher M1 for both prey species.

To conclude: Estimated values of recruitment, F and SSB depends on the value of M1. The estimated M2 depends on the applied M1 values as well, with the lowest M2 for the highest M1. The sum of M1 and estimated M2 (M) as used in the ICES single species assessment becomes in most cases higher with a higher M1. As the stock size of the predator cod in this run is not affected by the change in M1, the results from the Baltic Sea run may not be valid for e.g. the North Sea, where a change in M1 will also affect the main part of the predators.

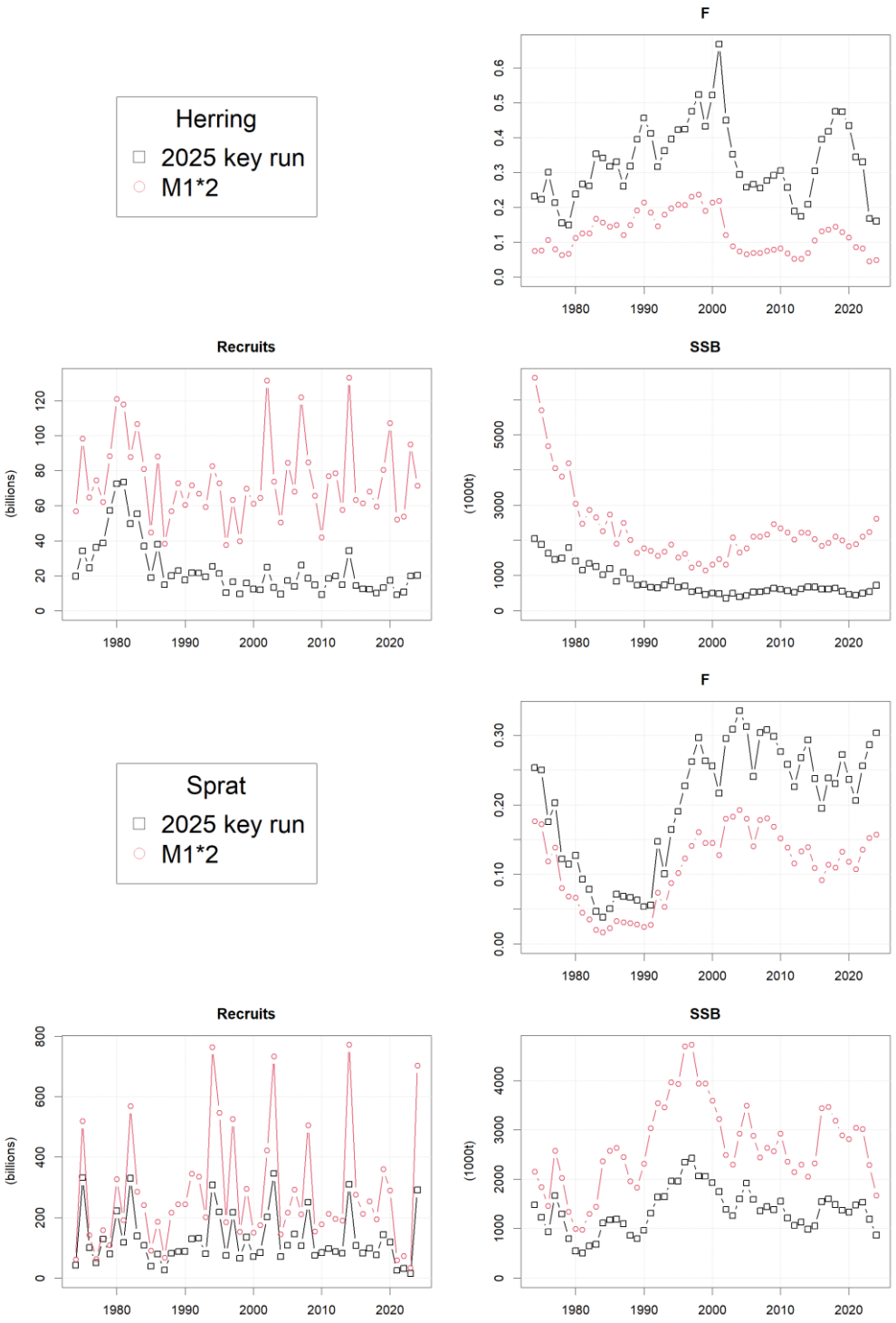


Figure 0-12 Comparison of the 2025 key run and a run with M1 multiplied by 2. Summary output.

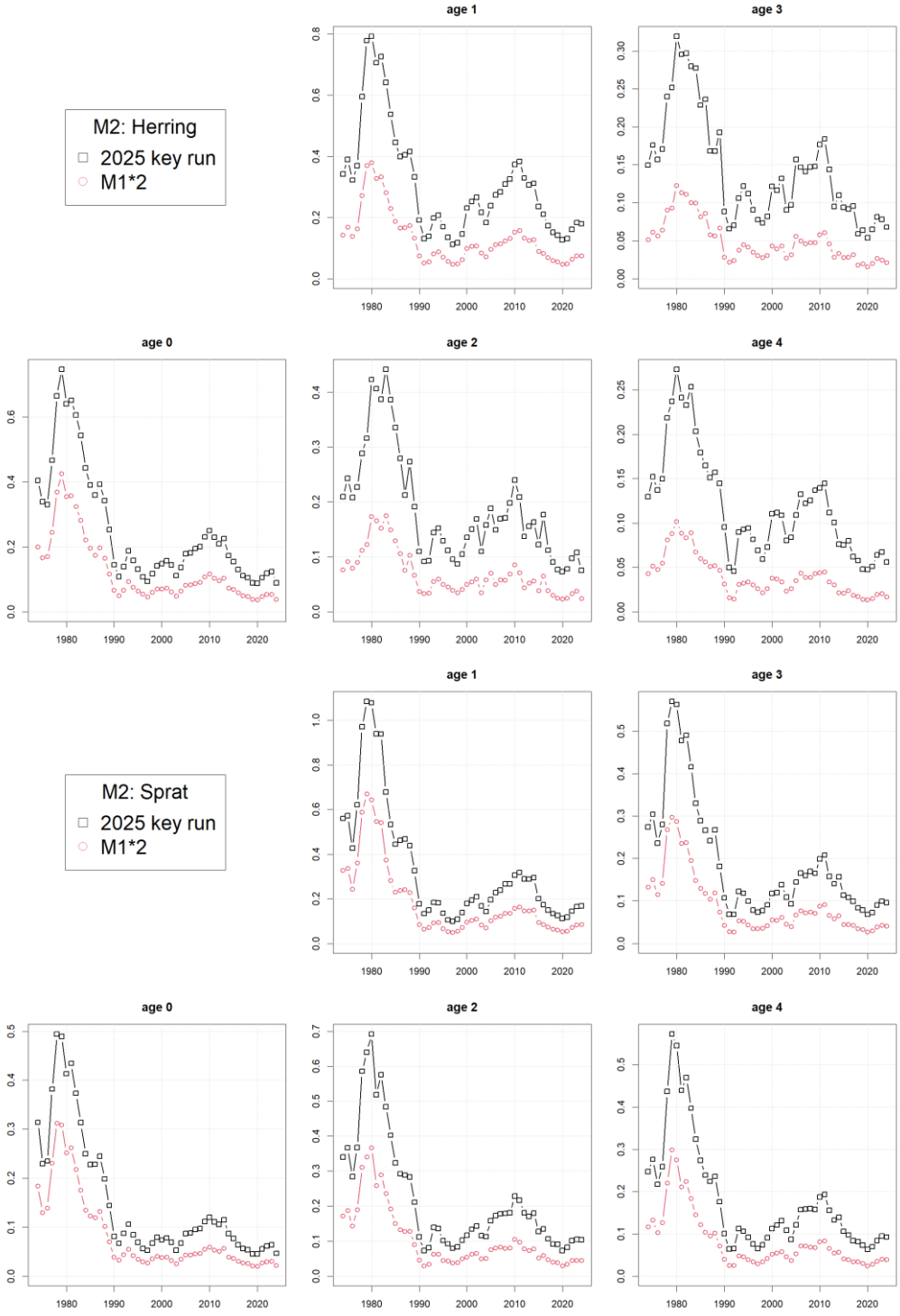


Figure 0-13 Comparison of the 2025 key run and a run with M1 multiplied by 2. Predation mortality (M2).

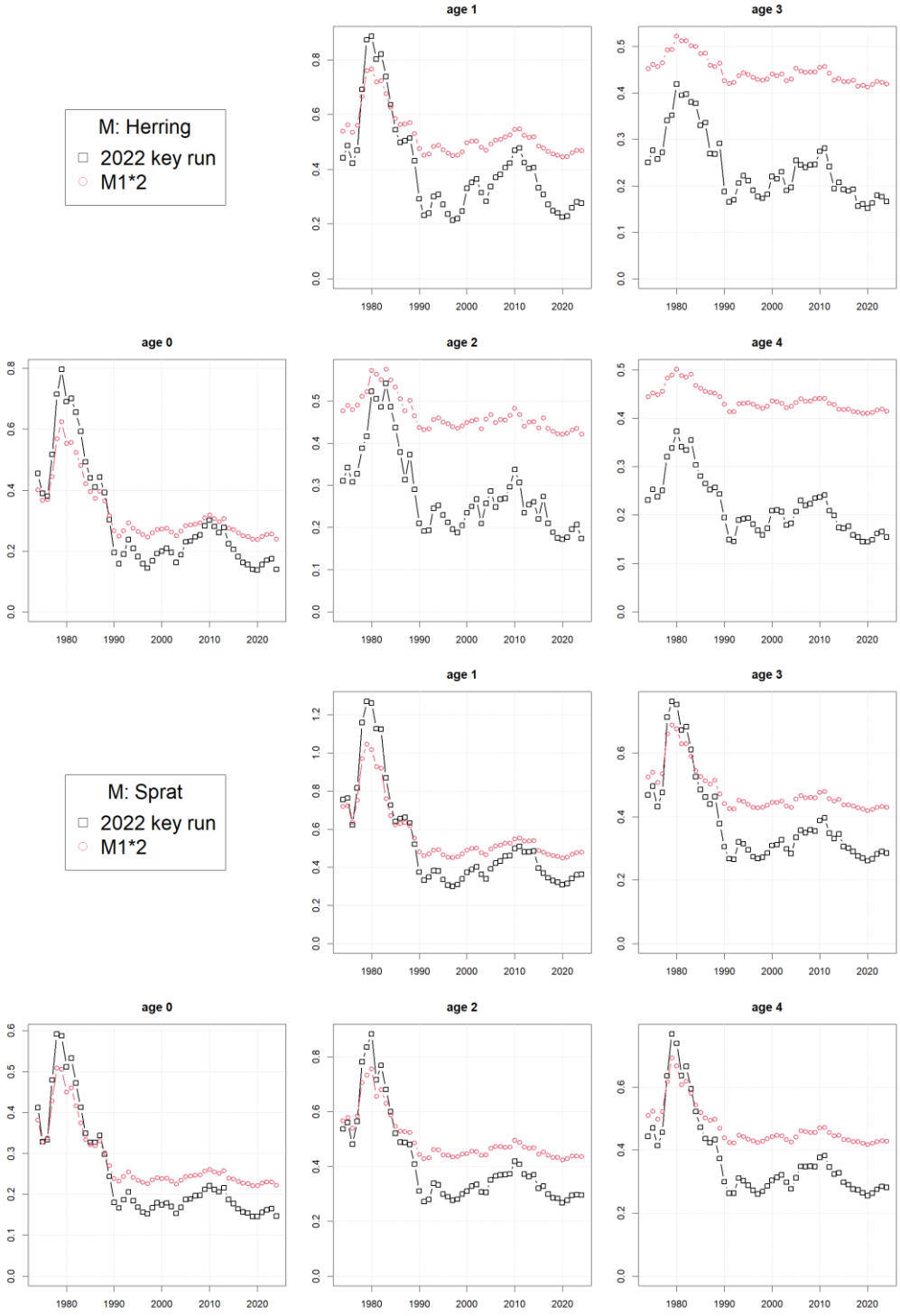


Figure 0-14 Comparison of the 2025 key run and a run with M1 multiplied by 2. Natural Mortality (M = M1 + M2).

Comparison with the old key runs

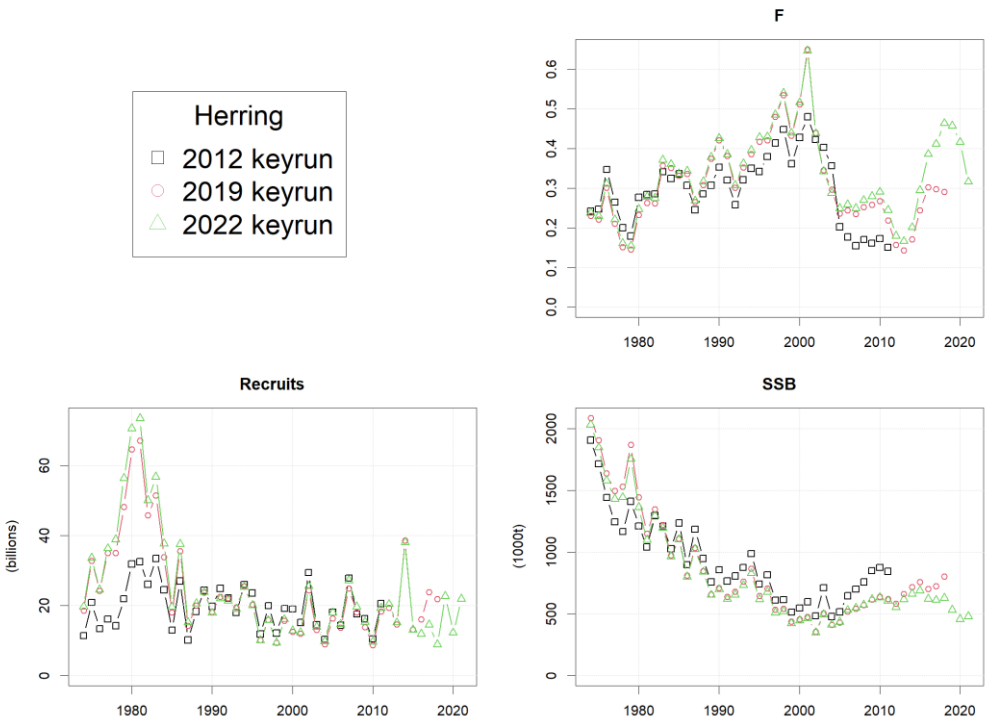
Comparison of the 2012 and 2019 key runs

Even though the 2012 and the 2019 key run are based on different stomach data, different assumption about the only predator species and different M1, the two key runs show quite similar results for the summary output recruitment, SSB and mean F (Figure 0-15). Herring F and SSB are similar, while recruitment is considerably higher in the beginning of the time series in the 2019 key run, probably as an effect of the assumed larger cod stock. For Sprat, the trend in SSB and F is the same in the two runs, but F in the 2019 key run is consistently estimated lower and SSB higher.

The difference in M2 for the two runs is more pronounced, especially for herring (Figure 0-16). Herring M2 is now estimated higher for all ages, and much higher for the first part of the time series. The difference is probably due to the assumption of a larger cod stock (especially of larger cod) in the 2019 key run, and the application of the predator-prey size selection model in the new key run, whereas the old version used a "constraint uniform" size selection. Herring M2 follows better the stock size of cod in the new run which may indicate that the uniform size selection option was not the best choice for the 2012 key run.

Comparison of the 2019 and 2022 key runs.

The results from the 2019 and 2022 are very similar (Figure 0-15 and Figure 0-16), as few changes were made in in model configuration and data other than addition of stock assessment data for the period 2019-2021. The new data have revised F and SSB in the most recent years in the 2019 key run, with the largest changes for herring. This revision is also seen in the ICES single species assessment. Estimated values have changed slightly, with the largest for the M2 of 0-groups of herring and sprat.



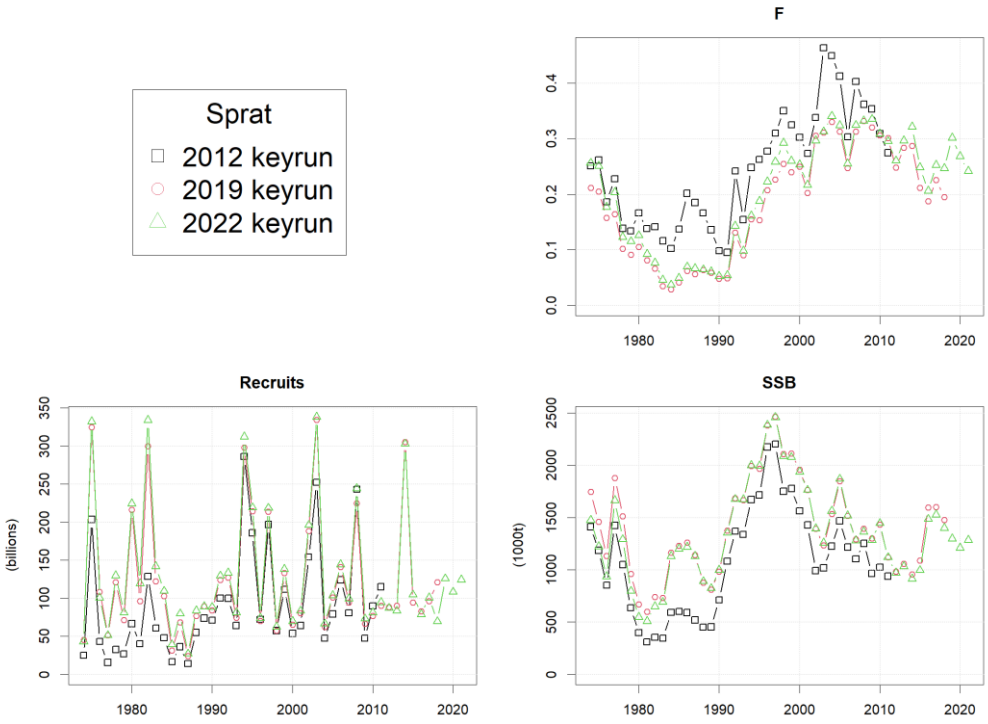


Figure 0-15. Comparison of the 2012, 2019 and 2022 key runs. Summary output.

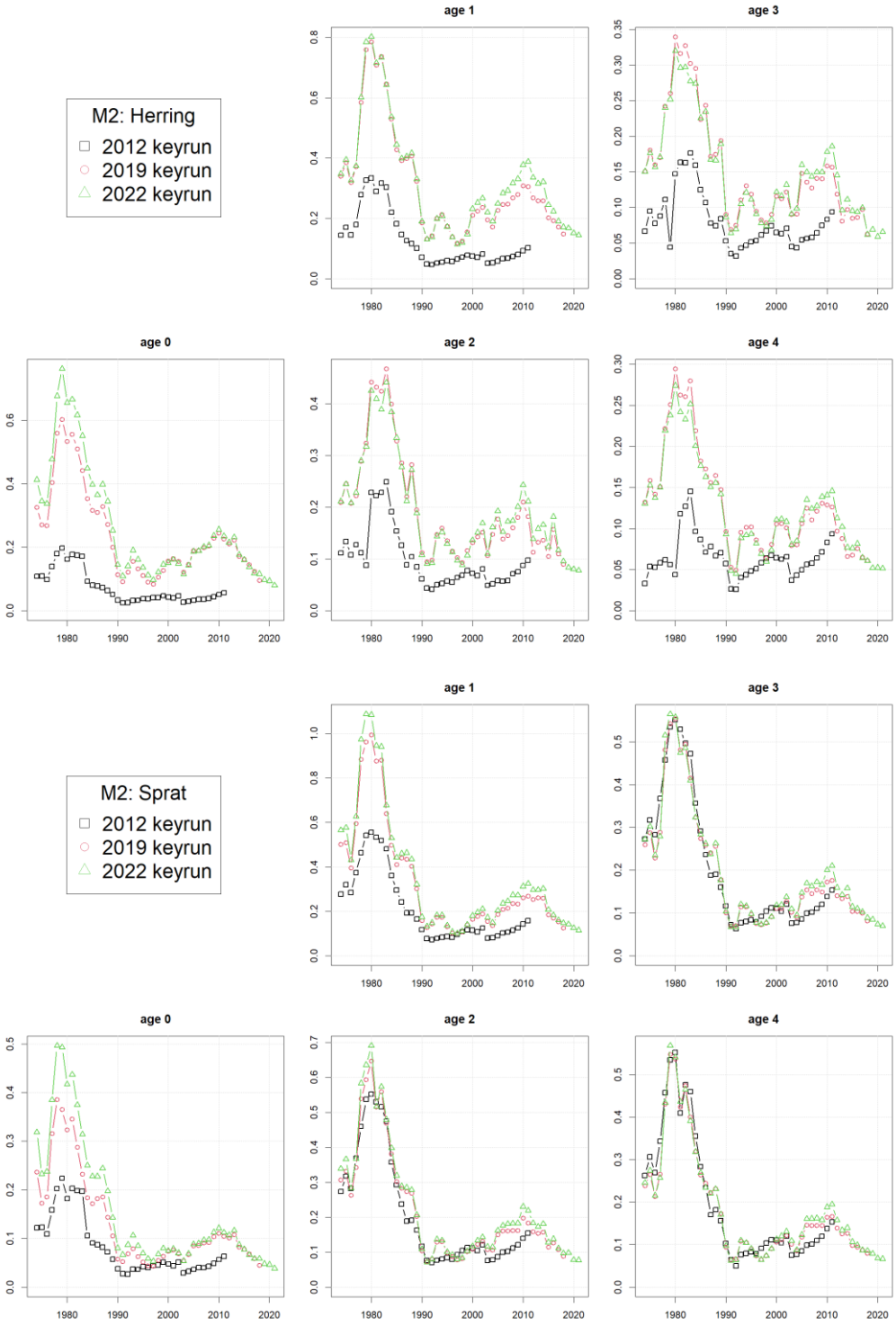


Figure 0-16. Comparison of the 2012, 2019 and 2022 key runs. Predation mortality (M2)

Comparison with the Gadget model run.

A comparison of the SMS key run and the results from the Gadget model evaluated at WGSAM 2019 is presented for herring (Figure 0-17) and for sprat (Figure 0-18). The estimated M values are quite similar.



Figure 0-17. Comparison of M (M=M1+M2) values for herring from the SMS 2019 key run and a Gadget configuration.



Figure 0-18. Comparison of M (M=M1+M2) values for sprat from the 2019 key run and a Gadget configuration.

Conclusion, key runs

2019 key run

WGSAM 2019 discussed and reviewed the changes in input data and the results in detail and concluded that:

The (2019) key run as currently best possible run with SMS to provide natural mortality estimates. WGSAM recommends to use these values as input to single species stock assessments. The full time series should be used and not only an update for the years after the last key run in 2012.

However, there are also clear limitations with the approach and results have been shown to be sensitive to e.g., consumption rates, assumptions regarding M1 and treatment of "Other Food" as well as the size selectivity of cod. In addition, the results depend to a large extent on the outcome of the ICES Eastern Baltic cod assessment. Any bias in this assessment directly influences the predation mortality estimates. Assumptions around other food and constant vulnerabilities may also bias the natural mortality estimates to some extent. Contrarily, the very similar results from the Gadget model run are encouraging and increase the credibility of the provided M time series.

WGSAM (2019) does not recommend using the uncertainty estimates around M as these are underestimated due to the assumption that the cod population is known without error.

2022 key run

WGSAM 2022 reviewed the updated key run for the central Baltic Sea and concluded that it provides the best possible estimates of natural mortality for the Baltic sprat and central Baltic herring stocks. WGSAM recommends using these values as input to single species stock assessments. The full time series should be used and not only an update for the years after the last key run in 2019.

Main aspects of uncertainty and limitations highlighted for the 2019 model remain also for this updated key run.

2024 key run

WGSAM 2024 reviewed the updated key run for the central Baltic Sea and concluded that it provides the best possible estimates of natural mortality for the Baltic sprat and central Baltic herring stocks. WGSAM recommends using these values as input to single species stock assessments. The full time series should be used and not only an update for the years after the last key run in 2022.

Main aspects of uncertainty and limitations highlighted for the 2019 and 2022 model remain also for this updated key run

Identified areas of priority research

WGSAM 2019 recommendation

WGSAM 2019 recommended:

1. More analyses on stomach data to get a better process understanding what is driving the systematic changes in relative stomach contents.

2. A split of Other Food in parts where the time dynamic can be taken into account (e.g., flatfish and *Saduria entomon*) and a part that still needs to be assumed constant in time may be beneficial.

3. The inclusion of spatial dynamics (either directly or via overlap coefficients) may improve the fit to data sources.

4. A run with age 1 as recruits could be tried because input for the 0 group is highly uncertain.

5. Account for the uncertainty in cod numbers in the model.

Some work has been done since 2019 to address these recommendation. The development of the R-package FishStomachs allows a consistent approach to analyse stomach data (**Recommendation 1**). Several papers (e.g. Neuenfeldt et al., 2022) discuss the poor state of the cod in the Eastern Baltic Sea in relation to e.g. food availability, parasite load and effects of environmental changes. A model like SMS is however not adequate to handle such issues.

With respect to **recommendation 5**) the SMS model was extended such that uncertainties of the stock size of "other predators" could be taken into account in e.g. the estimation of the uncertainties on M2 values. The approach uses "observations" of stock size (e.g. 10 samples of stock size drawn from the "real" (WGBFAS estimate) cod stock size with a specified CV. These "observations" are used in the SMS likelihood where the "observed" and "real" data are used to estimate a model stock size. This

resulting uncertainty of the model stock size are afterwards used in the calculation of e.g. M2 values, in a similar way as for all other SMS output. The likelihood contributions from comparing the model stock size with the "real" stock size are used together with the likelihood contributions from the catch, cpue and stomach observations. Preliminary runs with the new version of SMS show that the best likelihood is obtained with a relative poor fit between the model cod stock size and the "real" stock size. The new SMS version has tendency to estimate a considerably lower stock size of cod than the "real" stock size. Said in another way, the likelihood for all (other) observations in SMS become better if the model stock size is decreased, on the cost of a poor fit between model stock size and "real" stock size. More work is needed, before the new version of SMS can be applied.

WGSAM 2022 recommendation

For further work on the Baltic Sea SMS key run, WGSAM 2022 recommends the following:

1. Additional analyses of the cod diet to get a better process understanding of what is driving the systematic changes in relative stomach contents.

2. A split of Other Food in parts where the time dynamic can be considered (e.g., flounder and *Saduria entomon*) directly or via a proxy and a part that still needs to be assumed constant in time may be beneficial. The availability of the different food groups for cod in the Baltic Sea have changed strongly over time (e.g. Haase et al. 2020).

In 2019, the WGSAM reviewers made these two recommendations, on which no work has been done since then. The overestimations of the model in the amount of herring eaten by cod and the underestimations of other food and sprat (all in recent years, Figure 8), indicate that it is still a good idea to consider these recommendations. The recent changes in the condition of Baltic cod indicate that there has been likely a change in the food availability for cod. Although it could be difficult to implement these changes they could be quite important for reliable estimates of the model.

3. The inclusion of spatial dynamics in the food consumption of cod (either directly or via overlap coefficients) may improve the fit to data sources.

In 2019, the WGSAM reviewers made this recommendation, which has not been considered so far. As long as the spatial overlap between prey and predator is constant through time, incorporating an overlap coefficient should not have a strong effect on the model predictions as this is currently addressed via the vulnerability parameter. However, if there is a temporal trend in the spatial overlap between predator and the prey species, and/or if the prey age classes are unevenly distributed through space, an overlap coefficient (per prey age class) could make a difference for the model predictions.

4. Uncertainty and potential issues with the quality of the estimates for the age 0 prey classes before they could be used as input in a single species stock assessment. Discussion with the stocks experts from the assessment working group on possible alternative data sources on the weight-at-age of age0 clupeids would be beneficial.

5. Account for the uncertainty in cod number at age in the model, such that this uncertainty is reflected in the uncertainties of estimated values like M2 and SSB.

In 2019, the WGSAM reviewers made this recommendation, which has not been fully considered so far. Due to the current problems regarding the age-reading of Baltic cod,

only a length-based abundance estimate is available for this stock. Consequently, cod is considered as a predator with a constant density in the model. The effect of the uncertainties of cod stock densities and size distribution on the M2 predictions of SMS have not been tested.

6. Provide a multispecies F_{MSY} estimation that can be used as a comparison with the F_{MSY} values from the stock-by-stock (SS) frameworks currently used for advice.

The current model version of SMS of the Baltic Sea is not suitable for predictions of multispecies F_{MSY} values. Unfortunately, due to the non-dynamic nature of cod in the 2022 key run, it is not straightforward to predict how fisheries mortality on cod may affect the other species in the food web through indirect effects. It is advisable to calculate this metric in the future if possible. The consideration of optimal yield trade-offs amongst the commercially exploited species in the complex is an important facet in improving fisheries management advice and an ecosystem-based approach to fisheries management.

WGSAM 2025 recommendation

Most of the recommendations for the previous key run have not been followed up but are still valid.

Recommendations from the WGSAM (2025) report

The following recommendations are copied from the WGSAM (2025) report.

<u>For further work on the Baltic Sea SMS key run, WGSAM recommends the following:</u>

1. Additional analyses of the cod diet to get a better process understanding of what is driving the systematic changes in relative stomach contents. Inclusion of the newer stomach data from after 2014.

2. A split of Other Food in parts where the time dynamic can be considered (e.g., flounder and *Saduria entomon*) directly or via a proxy and a part that still needs to be assumed constant in time may be beneficial. The availability of the different food groups for cod in the Baltic Sea have changed strongly over time (e.g. Haase et al. 2020).

In 2019, the WGSAM reviewers made these two recommendations, on which no work has been done since then. The overestimations of the model in the amount of herring eaten by cod and the underestimations of other food and sprat (all in recent years, Figure 8), indicate that it is still a good idea to consider these recommendations. The recent changes in the condition of Baltic cod indicate that there has been likely a change in the food availability for cod. Although it could be difficult to implement these changes they could be quite important for reliable estimates of the model.

3. The inclusion of spatial dynamics in the food consumption of cod (either directly or via overlap coefficients) may improve the fit to data sources.

In 2019, the WGSAM reviewers made this recommendation, which has not been considered so far. As long as the spatial overlap between prey and predator is constant through time, incorporating an overlap coefficient should not have a strong effect on the model predictions as this is currently addressed via the vulnerability parameter. However, if there is a temporal trend in the spatial overlap between predator and the prey species, and/or if the prey age classes are unevenly distributed through space, an overlap coefficient (per prey age class) could make a difference for the model predictions.

4. A run with age 1 as recruits could be tried because input for the 0 group is highly uncertain.

In 2019, the WGSAM reviewers made this recommendation, which has not been considered so in the present key run. The idea is that the weight at age data for the age 0 classes of the prey species are so uncertain that the model cannot estimate the consumption correctly based on the data that is currently used. However, this may be a difficult change for the model as age 0 herring and sprat form an important part of the cod diet. It is relevant to highlight the uncertainty and potential issues with the quality of the estimates for the age 0 prey classes before they could be used as input in a single species stock assessment. Discussion with the stocks experts from the assessment working group on possible alternative data sources on the weight-at-age of age0 clupeids would be beneficial.

5. Account for the uncertainty in cod number at size in the model.

In 2019, the WGSAM reviewers made this recommendation, which has not been considered so far. Due to the current problems regarding the age-reading of Baltic cod, an age-length-based model is used for the stock assessment. Cod number-at-age in the stock are retrieved from the stock assessment and used to represent the cod stock in the SMS key run without uncertainty. In 2025 this issue is compounded by the key run using the forecasted cod data for 2024 and parts of 2023.

6. Provide a multispecies F_{MSY} estimation that can be used as a comparison with the F_{MSY} values from the stock-by-stock (SS) frameworks currently used for advice.

The current model version of SMS of the Baltic Sea is not suitable for predictions of multispecies F_{MSY} values. Unfortunately, due to the non-dynamic nature of cod in the 2025 key run, it is not possible to predict how fisheries mortality may affect the other species in the food web through indirect effects. It is advisable to calculate this metric in the future if possible. The consideration of optimal yield trade-offs amongst the commercially exploited species in the complex is an important facet in improving fisheries management advice and an ecosystem-based approach to fisheries management.

7. Consider inclusion of other predators than cod, such as seals or birds.

8. The value of M1 changes the estimate of M2 and F. A likelihood profile analysis could reveal which values of M1 for sprat and herring could lead to the best fit to data, and additionally test of the sensitivity of the current assumptions.

APPENDIX 1: SMS, a stochastic age–length structured multispecies model applied to North Sea and Baltic Sea stocks

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Overview

SMS (Stochastic Multi Species model) is a fish stock assessment model which includes estimations of predation mortalities from observation of catches, survey indices and stomach contents. Estimation of predation mortality is based on the theory for predation mortality as defined by Andersen and Ursin (1977) and Gislason and Helgason (1985). SMS is a "forward running" model that operates with a chosen number of time steps (e.g. quarters of the year). The default SMS is a one-area model, but the model has options for spatial explicit predation mortality given a known stock distribution.

The model parameters are estimated using a maximum likelihood (ML) technique. Uncertainties of the model parameters are estimated from the Hessian matrix and confidence limits of derived quantities like historical fishing mortalities and stock abundances are estimated from the parameter estimates and the delta-method. SMS can be used to forecast scenarios and Management Strategy Evaluations, where fishing mortalities are estimated dynamically from Harvest Control Rules.

This document describes the model structure and the statistical models used for parameter estimation.

Model Structure

Survival of the stocks

The survival of the stocks is described by the standard exponential decay equation of stock numbers (N).

$$N_{s,a,y,q+1} = N_{s,a,y,q}\, e^{-Z_{s,a,y,q}} \qquad \text{Eq. 1}$$

or

$$N_{s,a+1,y,+1,q=1} = N_{s,a,y,q=last\ season}\, e^{-Z_{s,a,y,q=last\ season}} \qquad \text{Eq. 2}$$

The instantaneous rate of total mortality, $Z_{s,a,y,q}$ by species s, age group a, year y and season q, is divided into three components; predation mortality (M2), fixed residual natural mortality (M1) and fishing mortality (F):

$$Z_{s,a,y,,q} = M1_{s,a,q} + M2_{s,a,y,q} + F_{s,a,y,q}$$

For non-assessment species which act as predators (e.g. grey seal and horse mackerel) stock numbers are assumed known and must be given as input.

Fishing mortality

Fishing mortality, $F_{s,a,y,q}$ is modelled from an extended separable model including age, year and season effects. However, as these effects may change over time a more flexible structure is assumed, allowing for such changes for specified periods. For convenience, the species index is left out in the following:

$$F_{a,y,q} = F^1_{Y,A1}\, F^2_y\, F^3_{Y,A2,q} \qquad \text{Eq. 3}$$

where indices $A1$ and $A2$ are grouping of ages, (e.g. ages 1–3, 4–7 and 8–9) and Y is grouping of years (e.g. 1975–1989, 1990–2011).

Eq. 3 defines that the years included in the model can be grouped into a number of period clusters (Y), in which the age selection (F^1) and seasonal selection (F^3) are assumed constant. $F^2$is the year effect, specifying the overall level of F for a particular year. The grouping of ages for age selection, $A1$, and season selection, $A2$, can be defined independently.

2.2.1 Options for year effect

Given a good relationship between F and effort the fishing mortality can be calculated from the observed effort.

$$F_{a,y,q} = F^1_{Y,A1}\, EFFORT_y\, F^1_{Y,A2,q}$$

Natural Mortality

Natural mortality is divided into two components, predation mortality (M2) caused by the predators included in the model and a residual natural mortality (M1), which is assumed to be known and is given as input.

M2 of a prey species, $prey$, with size group l_{prey} due to a predator species, $pred$, with size group l_{pred} is calculated as suggested by Andersen and Ursin (1977) and Gislason and Helgason (1985).

$$M2_{prey,l_{prey},y,q} = \sum_{pred}\sum_{l_{pred}} \frac{\bar{N}_{pred,l_{pred},y,a}\; RA_{pred,l_{pred},y,q}\; S_{prey,pred,q}(l_{prey}, l_{pred})}{AB_{pred,l_{pred},y,a}} \qquad \text{Eq. 4}$$

where RA denotes the total food ration (weight) of one individual predator per time unit, where S denotes the food suitability defined in section 0 and where AB is the total available (suitable) biomass. AB is defined as the sum of the biomass of preys weighted by their suitability. This total prey biomass includes also the so-called "other food" (OF) which includes all prey items not explicitly modelled, e.g. species of invertebrates and non-commercial fish species. Other food species are combined into one group, such that the total available prey biomass becomes:

$$AB_{pred,l_{pred},y,q} = \sum_{prey}\sum_{l_{prey}} \left(\bar{N}_{prey,l_{prey},y,q}\ W_{prey,l_{prey},y,q}\ S_{prey,pred,q}(l_{prey},l_{pred})\right) + OF_{pred,}\ S_{OF,pred,q}(l_{pred}) \quad \text{Eq. 5}$$

M2 cannot directly be calculated from Eq. 4 because M2 also is included in the right hand term of Eq. 6 to calculate $\bar{N}$.

$$\bar{N} = \frac{N\ (1 - e^{-(M1+M2+F)})}{M1 + M2 + F} \quad \text{Eq. 6}$$

As no analytical solution for $M2$ exists, $M2$ has to be solved numerically. If the time step considered is sufficiently small, for instance a quarter, $M2$ becomes small and can optionally be approximated by replacing the average number during the season, $\bar{N}$, on the right hand side of Eq. 4 by the stock at the beginning of the season, N. As the right hand side of equation now is independent of M2 this quantity can be calculated directly from Eq. 4 where AB (Eq. 5) is modified correspondingly.

Use of size distribution by age

The equations outlined in the section above provide M2 at-size groups. However, predation mortality by age is needed as well because F and catches are age-structured. If just one size group per age group of predators and preys is assumed Eq. 4 can be used directly where the age index substitutes the size group index in stock numbers ($\bar{N}_{prey,a,y,q} = \bar{N}_{prey,l_{prey},y,q}$)

Given more size groups per age, the calculation of M2 at-age requires age–length-keys to split N at age to N at size group.

$$N_{s,l_s,y,q} = \sum_{a} N_{s,a,y,q}\ ALK_{s,a,l_s,y,q} \quad \text{Eq. 7}$$

where $ALK_{s,l_s,a,y,q}$ denotes the observed proportion of size group ls for a given species and age group, i.e. $\sum_{l_s} ALK_{s,l_s,a,y,q} = 1$

Assuming that F and M1 depend only on age and that M2 only depends on length, M2 at-age is estimated by: (leaving out the species, year and quarter indices).

$$M2_a = Z_a \frac{\sum_l \bar{N}_{a,l}\, M2_{a,l}}{D_a} = \log\left(\frac{N_a}{N_a - D_a}\right) \frac{\sum_l \bar{N}_{a,l}\ M2_l}{D_a}$$

where

$$\bar{N}_{a,l} = N_{a,l}\ \frac{1 - e^{-(F_{a,l}+M1_{a,l}+M2_{a,l})}}{F_{a,l} + M1_{a,l} + M2_{a,l}} = N_{a,l}\ \frac{1 - e^{-(F_a+M1_a+M2_l)}}{F_a + M1_a + M2_l}$$

and where

$$D_a = \sum_l \bar{N}_{a,l}\ (F_a + M1_a + M2_l)$$

denotes the number of individuals at-age that died within a season.

Food suitability

As suggested by Andersen and Ursin (1977) and Gislason and Helgason (1985) the size-dependent food suitability of prey entity j for predator entity i is defined as the product of a species dependent vulnerability coefficient, $\rho_{i,j}$, a size preference coefficient $\varrho_{i,j}(l_i, l_j)$, and an overlap index $o_{i,j,q}$. Suitability is then defined as:

$$S_{pred,prey,q}(l_{pred}, l_{prey}) = \rho_{pred,prey} \ \varrho_{pred,prey}(l_{pred}, l_{prey}) \ o_{pred,prey,q} \quad \text{Eq. 8}$$

For the "other food" part suitability is defined as:

$$S_{OF,pred,q}(l_{pred}) = \rho_{OF,pred} \ o_{OF,pred,q} \exp\left(\upsilon_{pred} \log\left(W_{pred,l_{pred},q}/\overline{W}_{pred}\right)\right) \quad \text{Eq. 9}$$

Where $\overline{W}_{pred}$ is the average size of the predator species. Eq. 9 extends the original equation, to allow predator size dependent suitability for other food, for values of υ_{pred} different from zero. The overlap index may change between seasons, but is assumed independent of year and sizes.

Log-normal distributed size selection

Several functions can be used for the size preference for a prey. Andersen and Ursin (1977) assumed that a predator has a preferred prey size ratio and that a prey twice as big as the preferred size is as attractive as another half the prey size. This was formulated as a log-normal distribution:

$$\varrho_{pred,prey}(l_{pred}, l_{prey}) = \exp\left(-\frac{\left(\log\left(\frac{W_{l_{pred}}}{W_{l_{prey}}}\right) - \eta_{PREF\ pred}\right)^2}{2\,\sigma^2_{PREF\ pred}}\right); 0 < \varrho \leq 1 \quad \text{Eq. 10}$$

Where η_{PREF} is the natural logarithm of the preferred size ratio, $\sigma^2_{PREF}$is the "variance" of relative preferred size ratio, expressing how selective a predator is with respect to the size of a prey and where $W_{l_s}$is the mean weight for a species size group.

The basic size selection equation (Eq. 10) has been extended by modifying the preferred size ratio parameter.

$$\varrho_{pred,prey}(l_{pred}, l_{prey}) = \exp\left(-\frac{\left(\log\left(\frac{W_{l_{pred}}}{W_{l_{prey}}}\right) - \left(\eta_{PREF\ pred} + \xi_{prey} + \varpi_{pred} \log\left(W_{l_{pred}}\right)\right)\right)^2}{2\,\sigma^2_{PREF\ pred}}\right) \quad \text{Eq. 11}$$

Where ξ_{prey} specify a prey-specific adjustment term for the preferred size ratio, and where ϖ_{pred} specifies how the preferred size range can change by predator size.

Uniform size selection

Alternatively, a uniform size preference can be assumed within the range of the observed size ratio and zero size selection outside that ratio:

$$\varrho_{pred,prey}(l_{prep}, l_{prey}) = \begin{cases} 1 & \text{for } \eta_{MIN_{pred,prey}} \leq \dfrac{W_{l_{pred}}}{W_{l_{prey}}} \leq \eta_{MIN_{pred,prey}} \\ 0 & \text{for values outside observed range} \end{cases} \quad \text{Eq. 12}$$

where η_{MIN} and η_{MAX} are the observed minimum and maximum predator/prey size ratios.

7.2.3.2.2.1. Constraint uniform size selection

The uniform size preference does not take into account that the preferred predator/prey size ratio might change by size, such that larger individuals select relatively smaller preys (Floeter and Temming, 2005; Sharft *et al*., 2000). A way to account for that is to assume that the fixed minimum and maximum constants, η_{MIN} and η_{MAX}, depend on the predator size:

$$_{pred,prey}(l_{pred}, l_{prey}) = \begin{cases} 1 \text{ for } U1_{pred,prey} + U2_{pred,prey}\ \log(W_{l_{pred}}) \leq \log\left(\dfrac{W_{l_{pred}}}{W_{l_{prey}}}\right) \leq U3_{pred,prey} + U4_{pred,prey}\ \log(W_{l_{pred}}) \\ 0 \quad \text{for values outside regression range} \end{cases} \quad \text{Eq. 13}$$

The regression parameters are estimated externally by quantile regression (e.g. Koenker and Bassett, 1978) using e.g. the 2.5% and 97.5% percentiles of stomach content data. Figure 7.1 shows an example of such regression.

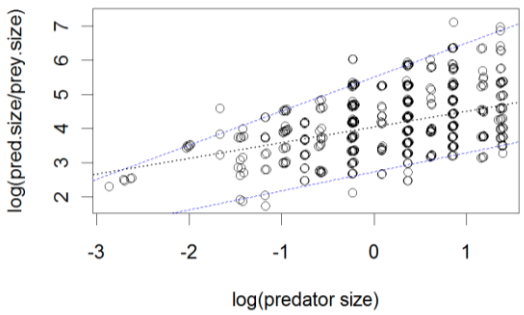


Figure 7.1. Quantile regression of stomach contents observations (Baltic cod eating cod), with 2.5%, 50% and 97.5% lines shown. Predator and prey size in weight.

Adjustment of age–size keys

For the North Sea configuration, age length keys were obtained from the IBTS surveys where the same gear (i.e. the GOV trawl) has been used in the period considered. This allows an adjustment of the observed ALK's to account for mesh size selection. Using a logistic length-dependent selection function, selection is defined as:

$$SL_s(l) = 1/\left(1 + e^{(S1_s - S2_s * l)}\right)$$

Where $S1_s$ and $S2_s$ are species-specific gear selection parameters.

The adjusted ALK can then be derived from the observed ALK by:

$$ALK_{s,l_s,a,y,a} = \text{ObservedALK}_{s,l_s,a,y,q} \; / \; SL_{s,l_s}$$

which finally has to be standardised to 1 for each age before used in Eq. 7.

Growth

Not implemented yet!

Food ration

Food ration, RA, pr. time step is given as input or estimated from mean weight by size group assuming an exponential relationship between ration and body weight W.

$$RA_{pred,l_{pred},q} = \gamma_{pred,q} \; W_{pred,l_{pred}}^{\varsigma_{pred}} \qquad \text{Eq. 2}$$

where the coefficient γ and ς are assumed to be known.

Body weight at-size group $lpred$ is estimated from mean length within the size group and a length–weight relationship.

Area-based SMS

SMS has three area explicit options:

1) Default one area model. Both F and M2 are calculated for the entire stock area;
2) M2 by area. M2 is calculated by subareas, but F is assumed global;
3) M2 and F by area. Both M2 and F are calculated by area (forecast only).

Stock distribution

For the area-based models, the stock is assumed redistributed between areas between each seasonal time step.

$$N_{s,a,y,q}^{area} = N_{s,a,y,q} \; DIST_{s,a,y,q,area}$$

Where DIST is a stock distribution key that sums up to 1

$$\sum_{area} DIST_{s,a,y,q,area} = 1$$

The calculation of M2 for Option 1) is provided in the previous section.

The method for option 3) is very similar, but the calculations must be done by each subarea separately.

$$Z_a^{area} = F_a^{area} + M1_a^{area} + M2_a^{area}$$

where $M2^{area}$ is calculated as given in Eq. 4.

Option 2) is the hybrid, where F is global but M is calculated by area.

$$Z_a^{area} = F_a + M1_a^{area} + M2_a^{area}$$

$\bar{N}$ in an area is calculated in the usual way

$$\bar{N}_a^{area} = N_a^{area} \frac{1 - e^{-Z_a^{area}}}{Z_a^{area}}$$

The total number of individuals died due to predation mortality (DM2) then becomes:

$$DM2_a = \sum_{area} M2_a^{area} \bar{N}_a^{area} \qquad \text{Eq. 3}$$

M2 for the whole stock can be estimated from:

$$M2_a = \log\left(\frac{N_a}{N_a - D_a}\right) \frac{DM2_a}{D_a}$$

where

$$D_a = \sum_{area} DF_a^{area} + DM1_a^{area} + DM2_a^{area}$$

and DF and $DM1$ represent the number that died due to fishery and residual mortality (M1) and are calculated in similar ways as specified for DM2 (Eq. 3).

Area based suitability parameters

For the "one area" SMS suitability is defined by Eq. 8.

The area-based version of suitability uses an area-specific vulnerability and overlap index, while the size preference (ϱ) is assumed independent of area.

$$S_{pred,prey,q}^{area}(l_{pred}, l_{prey}) = \rho_{pred,prey}^{area} \varrho_{pred,prey}(l_{pred}, l_{prey}) o_{pred,prey,q}^{area}$$

Statistical models

Three types of observations are considered: Total international catch-at-age; survey abundance indices and relative stomach content. For each type, a stochastic model is formulated and the likelihood function is calculated. As the three types of observations are independent, the total log likelihood is the sum of the contributions from three types of observations. A stock–recruitment (penalty) function is added as a fourth contribution.

Catch-at-age

Catch-at-age observations are considered stochastic variables subject to sampling and process variation. The probability model for these observations is modelled along the lines described by Lewy and Nielsen (2003):

Catch-at-age is assumed to be lognormally distributed with log mean equal to log of the standard catch equation. The variance is assumed to depend on age and season and to be constant over years. To reduce the number of parameters, ages and seasons can

be grouped, e.g. assuming the same variance for age 3 and age 4 in one or all seasons. Thus, the likelihood function, LCATCH, associated with the catches is:

$$L_{CATCH} = \prod_{s,a,y,q} \frac{1}{\sigma_{CATCH\ s,a,q}\sqrt{2\pi}} \; exp\left(-\frac{\left(\log\left(C_{s,a,y,q}\right) - E\left(\log\left(C_{s,a,y,q}\right)\right)\right)^2}{2\,\sigma^2_{CATCH\ s,a,q}}\right) \qquad \text{Eq. 4}$$

Where

$$E\left(\log\left(C_{s,a,y,q}\right)\right) = \log\left(F_{s,a,y,q}\,\bar{N}_{s,a,y,q}\right)$$

Leaving out the constant term, the negative log-likelihood of catches then becomes:

$$\begin{aligned} l_{CATCH} &= -\log(L_{CATCH}) \\ &\propto \text{NOY} \sum_{s,a,q} \log\left(\sigma_{CATCH\ s,a,q}\right) \\ &+ \sum_{s,a,y,q} \left(\log\left(C_{s,a,y,q}\right) - E\left(\log\left(C_{s,a,y,q}\right)\right)\right)^2 / 2\sigma^2_{CATCH\ s,a,q} \end{aligned} \qquad \text{Eq. 5}$$

Where NOY is the number of years in the time-series.

Annual catches

Catch-at-age numbers by quarter have not been available for some of the demersal North Sea stocks in recent years. For use in the default SMS configuration of the North Sea, where quarterly time step is used, it is assumed that the seasonal distribution (the F^3 parameter in Eq. 3) is known and given as input. The likelihood function is modified to make use of the observed annual catches.

$$E\left(\log\left(C_{s,a,y}\right)\right) = \log\left(\sum_{q} F_{s,a,y,q}\;\; \bar{N}_{s,a,y,q}\right)$$

$$L_{CATCH} = \prod_{s,a,y} \frac{1}{\sigma_{CATCH\ s,a}\sqrt{2\pi}} \; exp\left(-\frac{\left(\log\left(C_{s,a,y}\right) - E\left(\log\left(C_{s,a,y}\right)\right)\right)^2}{2\,\sigma^2_{CATCH\ s,a}}\right) \qquad \text{Eq. 6}$$

Survey indices

Similarly to the catch observations, survey indices, $CPUE_{survey,s,a,y,q}$ are assumed to be log-normally distributed with mean:

$$E\left(\log\left(CPUE_{survey,s,a,y,q}\right)\right) = \log\left(Q_{survey,a}\;\bar{N}_{SURVEY\ s,a,y,q}\right) \qquad \text{Eq. 7}$$

where Q denotes catchability by survey and $\bar{N}_{SURVEY}$ is mean stock number during the survey period. Catchability may depend on a single age or groups of ages. Similarly, the variance of log cpue, σ^2_{SURVEY} may be estimated individually by age or by clusters of age groups. The negative log-likelihood is on the same form as Eq. 4.

$$
\begin{aligned}
& l_{SURVEY} \\
& = -\log(L_{SURVEY}) \\
& \propto NOY_{survey,s} \sum_{survey,s,a} \log\left(\sigma_{SURVEY\ survey,s,a}\right) \\
& + \sum_{survey,s,a,y} \left(\log\left(CPUE_{survey,s,a,y}\right) - E\left(\log\left(CPUE_{survey,s,a,y}\right)\right)\right)^2 / 2\sigma^2_{SURVEY\ s,a}
\end{aligned}
\qquad \text{Eq. 8}
$$

Stomach contents

The stomach contents observations, which are the basis for modelling predator food preference, consist of the average proportions by weight of the stomach content averaged over the stomach samples in the North Sea. The model observations, $STOM_{pred,l_{pred},prey,l_{prey},y,q}$, are given for combinations of prey and predator species and size classes. In the following, we use entity i for a combination of predator species and predator size class (e.g. saithe 50–60 cm) and entity j for the combination of prey species and prey size class eaten by entity i. Model observations are therefore defined as $STOM_{i,j,y,q}$.

$STOM_{i,j,y,q}$ is assumed to be a set of stochastic variables subject to sampling and process variations. For a given predator entity the observations across prey entities i are continuous variables which sum to one. Thus, the probability distribution of the stomach observations for a given predator including all prey/length groups needs to be a multivariate distribution defined on the simplex. As far as the authors know, the Dirichlet distribution is the only distribution fulfilling this requirement. Leaving out the year and season index, the Dirichlet density function for a predator entity i with k observed diet proportions $STOM_{i,1}, \ldots STOM_{i,k-1} > 0$ and the parameters $p_1, \ldots, p_k > 0$ follows the probability density given by:

$$
\begin{aligned}
f_i = f\left(STOM_{i,1}, \ldots, STOM_{i,k-1} \mid p_{i,1}, \ldots, p_{i,k}\right) \\
= \frac{\Gamma(p_i)}{\prod_{j=1}^{k} \Gamma\left(p_{i,j}\right)} \prod_{j=1}^{k} STOM_{i,j}^{p_{i,j}-1}
\end{aligned}
\qquad \text{Eq. 9}
$$

Where

$$
STOM_{i,k} = 1 - \sum_{j=1}^{k-1} STOM_{i,j}
$$

and

$$
p_i = \sum_{j=1}^{k} p_{i,j}
$$

The mean and variance of the observations in the Dirichlet distribution are equal to:

$$
E\left(STOM_{i,j}\right) = \frac{p_{i,j}}{p_i}
$$

$$
Var\left(STOM_{i,j}\right) = \frac{E\left(STOM_{i,j}\right)\left(1 - E\left(STOM_{i,j}\right)\right)}{p_i + 1}
\qquad \text{Eq. 10}
$$

The expected value of the stomach contents observations is modelled using the theory developed by Andersen and Ursin (1977):

$$E\left(STOM_{i,j}\right) = \frac{\bar{N}_j \; W_j \; S_{i,j}(l_i, l_j)}{\sum_j \left(\bar{N}_j \; W_j \; S_{i,j}(l_i, l_j)\right) \; + \; OF_i \; S_{OF,i}(l_i)} = \frac{p_{i,j}}{p_i} \qquad \text{Eq. 11}$$

where the food suitability function, S, is defined by Eq. 8 and Eq. 9. We use the same assumptions as for the calculation of M2 (Eq. 4), the small time steps used in the model allow for an approximation of $\bar{N}_j$ by N_j in Eq. 11.

Regarding the variance of stomach content observations in unpublished analyses of the present authors of data from the North Sea stomach-sampling project 1991 (ICES, 1997), the authors indicate that the relationship between the variance and the mean of the stomach contents may be formulated in the following way:

$$Var\left(STOM_{i,j,y,q}\right) = \frac{E\left(STOM_{i,j,y,q}\right)\left(1 - E\left(STOM_{i,j,y,q}\right)\right)}{V_{pred} \; U_{i,y,q}} \qquad \text{Eq. 12}$$

where $U_{i,y,q}$ is a known quantity reflecting the sampling level of a predator entity, e.g. the number of hauls containing stomach samples of a given predator and size class. V_{pred} is a predator species-dependent parameter linking the sampling level and variance. Equating Eq. 10 and Eq. 12 implies that:

$$P_{i,y,q} = V_{pred} \; U_{i,y,q} - 1. \qquad \text{Eq. 13}$$

Insertion of Eq. 13 into Eq. 11 results in:

$$P_{i,j,y,q} = \left(V_{pred} \; U_{i,y,q} - 1\right) \frac{\bar{N}_j \; W_j \; S_{i,j}(l_i, l_j)}{\sum_j \left(\bar{N}_j \; W_j \; S_{i,j}(l_i, l_j)\right) \; + \; OF_i \; S_{OF,i}(l_i)}.$$

The parameters, $p_{i,j,y,q}$ are uniquely determined through stock numbers, total mortality, suitability parameters and V_{pred}.

Assuming that the diet observations for the predator length groups are independent, the negative log likelihood function including all predators and length groups is derived from Eq. 9:

$$l_{STOM} = -\log(L_{STOM}) = -\sum_{i,j,y,q} \log\left(f_{i,j,y,q}\right) \qquad \text{Eq. 14}$$

Modification of the stomach contents model

The stomach contents observations, $STOM_{prey,l_{prey},pred,l_{pred},y,q}$ are given for combinations of prey and predator species and size classes. For a diet consisting of a large proportion "other food" and several species and prey size classes, the proportion of the individual combination of species and size becomes small (less than 0.1%) for several prey entities. Very small proportions, in combination with a modest sampling size per stratum, make the estimation of parameters impossible in some cases. To overcome the problem, SMS has an option to let the likelihood use proportion summed over all size classes for a given prey species such that the prey entity equals the species.

The same grouping of all sizes from a prey is applied when the uniform size selection option (Eq. 12 and Eq. 1) is used. The likelihood function is the same as used for stomach observations that include prey size.

Stock–recruitment

In order to enable estimation of recruitment in the last year for cases where survey indices catch from the recruitment age is missing (e.g. saithe), and to estimate parameters for forecast use, a stock–recruitment relationship $R_{s,y} = R\left(SSB_{s,y} | \, \alpha_s, \beta_s\right)$ penalty function is included in the likelihood function.

Recruitment to the model takes place in the same season ($recq$) and at the same age (fa) for all species. It is estimated from the Spawning–Stock Biomass (SSB) in the first season (fq) of the year, and a stock–recruitment relation. SSB is calculated from stock numbers, proportion mature (PM) and mean weight in the sea.

$$SSB_{s,y} = \sum_{a} N_{s,y,a,q=recq} \; PM_{s,y,a,q=recq} \; W_{s,y,a,q=recq} \qquad \text{Eq. 15}$$

At present, the Ricker (Eq. 16), the Beverton and Holt (Eq. 17), segmented regression (Eq. 18) and geometric mean are implemented.

$$R_{s,y} = \alpha_s \; SSB_{s,y-fa,fq} \; e^{\left(\beta_s \; SSB_{s,y-fa,fq}\right)} \qquad \text{Eq. 16}$$

$$R_{s,y} = \frac{\alpha_s \; SSB_{s,y-fa,fq}}{1 + \beta_s \; SSB_{s,y-fa,q}} \qquad \text{Eq. 17}$$

$$R_{s,y} = \begin{cases} \alpha_s \; SSB_{s,y-fa,fq} & \text{for } SSB_{s,y-fa,fq} < \beta_s \\ \alpha_s \; \beta_s & \text{for } SSB_{s,y-fa,fq} < \beta_s \end{cases} \qquad \text{Eq. 18}$$

Assuming that recruitment is lognormally distributed, the negative log likelihood, l_{SR}, equals:

$$\begin{aligned} & l_{SR} \\ & = -\log(L_{SR}) \\ & \propto NOY \sum_{s} \log(\sigma_{SR\,a}) \\ & + \sum_{s,a,y} \left(\log\left(N_{ss,a=fa,y,q=recq}\right) - E\left(\log\left(R_{s,y}\right)\right)\right)^2 / 2\sigma_{SR\,s}^2 \end{aligned} \qquad \text{Eq. 19}$$

Where NOY gives the number of years selected and where Eq. 20 gives the expected recruitment for the Ricker case.

$$E(\log(R_s)) = \log\left(\alpha_s\, SSB_{s,y-fa,fq}\, e^{\left(\beta_s\, SSB_{s,y-fa,fq}\right)}\right) \qquad \text{Eq. 20}$$

Total likelihood function and parameterisation

The total negative log likelihood function, l_{TOTAL}, is found as the sum of the four terms:

$$l_{TOTAL} = l_{CATCH} + l_{SURVEY} + l_{STOM} + l_{SR}$$

To ensure uniquely determined parameters it is necessary to fix part of them. For the F-at-age model (Eq. 3) the year selection in the beginning of each year range (Y) has been fixed to one ($F^2_{y=\text{first year in each group of years}} = 1$). The season effect in the last season of all years and ages is also fixed ($F^3_{y,a,q=last\ season} = 1/\text{number of seasons}$).

Eq. 4 and Eq. 8 indicate that it is only possible to determine relative vulnerability parameters, $\rho_{pred,prey}$. We have chosen to fix the vulnerability of other food for all predators to 1.0. Similarly, the biomass of other food OFpred has arbitrarily been set (e.g. at 1 million tonnes) for each predator. The actual value by predator was chosen to obtain estimates of vulnerability parameters for the fish prey at around 1. Other parameters than suitability are practically unaffected of the actual choice of biomass of other food.

In the food suitability function (Eq. 8 and Eq. 9), vulnerability and overlap effects cannot be distinguished. Hence the overlap parameters were fixed for at least one season. In practice, however, several combinations of overlap have to be fixed (at e.g. 1).

Initial stock size, i.e. the stock numbers in the first year and recruitment over years are used as parameters in the model while the remaining stock sizes are considered as functions of the parameters determined by Eq. 1 and Eq. 2.

The year effect ($F^2_{y,s}$) in the separable model for fishery mortality (Eq. 3) takes one parameter per species for each year in the time-series which sum up to a considerable number of parameters. To reduce this high number of parameters, the year effect can optionally be modelled from a cubic spline function which requires fewer parameters. The number of knots must be specified if this option is used.

Another way to reduce the number of parameters is to substitute the parameters $\sigma_{CATCH}, \sigma_{SURVEY}$ and σ_{SR} used in the likelihood functions by their empirical estimates. This optional substitution has practically no effect on the model output and the associated uncertainty.

Appendage 1 gives an overview of parameters and variables in the model.

The parameters are estimated using maximum likelihood (ML) i.e. by minimizing the negative log likelihood, l_{TOTAL}. The variance/covariance matrix is approximated by the inverse Hessian matrix. Uncertainties of functions of the estimated parameters (such as biomass and mean fishing mortality) are calculated using the delta method.

SMS forecast

SMS is a forward-running model and can as such easily be used for forecast scenarios and Management Strategy Evaluation (MSE). SMS used the estimated parameters to calculate the initial stock numbers and exploitation pattern used in the forecast. Exploitation pattern is assumed constant in the forecast period, but is scaled to a specified average F, derived dynamically from Harvest Control Rules (HCR). Recruits are produced from the stock–recruitment relation, input parameters and a noise term.

Recruitment

Recruitment is estimated from the available stock–recruitment relationships, f(SSB), (see Section 0) and optionally a lognormal distributed noise term with standard deviation std.

$$R = f(SSB)\ e^{(std\ NORM(0,1))} \qquad \text{Eq. 21}$$

Where NORM(0,1) is a random number drawn from a normal distribution with mean=0 and standard deviation 1. A default value for std can be obtained from the estimated variance of stock–recruitment relationship, $\sigma^2_{SR_S}$ (Eq. 19)

Application of the noise function for the lognormally distributed recruitment gives on average a median recruitment as specified by f(SSB). Optionally, recruitment can be adjusted with half of the variance, to obtain, on average, a mean recruitment given by f(SSB).

$$R = f(SSB)\ e^{(std\ NORM(0,1))}\ e^{\left(-(std^2/2)\right)} \qquad \text{Eq. 22}$$

Harvest Control Rules

Several HCRs have been implemented, e.g. constant F and the ICES interpretation of management according to MSY for both short- and long-lived species. Selected, more complex management plans in force for the North Sea and Baltic Sea species have also been implemented.

Model validation

Model validation (in the years 2004–2009), was focused on the performance of the model using simulated data from an independent model and simulated data produced by the SMS model itself. The independent model was implemented using R (R Development Core Team. 2011) and include a medium complex North Sea configuration (nine species, of which four are predators and eight species preys). The simulation model follows the SMS model specification with an addition of von Bertallanfy growth curves to model mean length-at-age. Variance around mean length-at-age was assumed to increase by increasing age. This combined age–length approach made it possible to simulate all the data needed for model verification. Test

dataset from the simulation model included 20 years of catch data, one survey time-series per species covering all years and ages, and four quarterly stomach samples in year ten including stomach observations for all predator length groups. Data from the independent simulation model was used to verify that the SMS model actually works as intended and to investigate model sensitivity with respect to observation errors on catch, survey cpue and stomach data.

To test if model parameters were identifiable when uncertainties estimated from real data were applied, the SMS model was modified to produce observations with the estimated observation noise of catch, survey and stomach data. The experiment consists of the following steps:

1) Estimate model parameters using the SMS model and available North Sea data.

2) Generate 100 set of input data from SMS output (expected catch numbers, survey indices and stomach observations) and their associated variance of these values).

3) Let SMS estimate 100 sets of parameters from the 100 sets of input data.

This procedure results in one set of "true parameters", $\theta = (\theta_1, \dots, \theta_k)$ and 100 sets of estimated parameters, $\hat{\theta}_j = \left(\hat{\theta}_{1,j}, \dots, \hat{\theta}_{k,j}\right),\ j = 1, \dots, k$. Based on the 100 repetitions, and for each of the k parameters, the mean and the standard deviation of the mean $\bar{\hat{\theta}}_i$ and σ_i and hence the 95% confidence limits was calculated. Finally, the proportion of the parameters was calculated for which θ_i lies in the 95% confidence interval of $\bar{\hat{\theta}}_i$.

The test showed that parameters are identifiable for most "real" North Sea configurations. For some species with relatively few diet observations, size selection parameters (Eq. 11) and the variance parameter (V) linking the stomach sampling level to the variance of Dirichlet distribution (Eq. 12 and Eq. 13), were outside the 95% confidence interval of $\bar{\hat{\theta}}_i$.

A more informal testing of the model has been done by simply using the model. SMS has been applied to produce the so called key run for both the species rich North Sea system (ten species with stock number estimation including seven prey species, and 16 species of "other predators") (ICES, WGSAM 2011) and the species poor Baltic Sea (cod, herring and sprat, one predator and three prey species) (WGSAM 2008, WKMAMPEL 2009). In addition, the model has been used in single-species mode for the ICES advice of blue whiting in the North East Atlantic (WGWIDE 2011) since 2005 and several sandeel stocks in the North Sea since 2009 (WGNSSK 2011). For MSE purposes, the model has been applied for sandeel and Norway pout in the North Sea (AGSANNOP 2007), blue whiting and pelagic stocks in the Baltic (WKMAMPEL 2009) in both single and multispecies mode.

SMS is essentially an extension of the statistical models normally used for single-species stock assessment. This allows the use the long list of available diagnostics tools, e.g. residuals plots, and retrospective analysis, developed for model testing of submodels for catch-at-age and survey indices. For stomach observations, however, fewer established methods are available. To apply reliable residual plots for stomach observations residuals need to be independent, which are not the case for the stomach contents model as the observations with respect to prey entity sum to one. Instead, we do the following: Let the predator entity, year and quarter be given and consider the stomach contents observations following the Dirichlet distribution:

$$STOM_r = (STOM_{r,1}, \dots, STOM_{r,k-1}) \sim Dir(p_{r,1}, \dots, p_{r,k})$$

Where r equals the combined entity of predator entity, year and quarter and where $p_{r,j}, j = 1, \dots, k$ represent the Dirichlet parameters estimated. Instead of considering the weight proportions, $STOM$, we consider absolute weights in the stomachs, $W_{r,j}$, $j = 1, \dots, k$, where

$$STOM_{r,j} = \frac{W_{r,j}}{\sum_j W_{r,j}}$$

If we assume that $W_{r,j}$, $j = 1, \dots, k$ are independent and follow gamma distributions with the same scale parameter, θ_r, i.e.

$$W_{r,j} \sim \Gamma(p_{r,j}, \theta_r) \;\; j = 1, \dots, k$$

it is well known that $STOM_r$ follows the Dirichlet distribution. We now assume that the opposite is the case (we have to prove that!) and hence assume that the absolute weights, $W_{r,j}$ are independent gamma distributed variables. We then transform these observations to obtain normal distributed residuals: Leaving out the indices, we get that $U = pgamma(W, p, \theta)$, where $pgamma$ is the distribution function of the gamma distribution, is uniformly distributed. To obtain normally distributed variables, U is finally transformed to $V = qnorm(U)$, where $qnorm$ equals the inverse of the distribution function of the standardized normal distribution. This means that V is our new residuals for stomach contents observations.

To obtain the absolute weight of the prey entities form the relative stomach content, $STOM$, we have to know the total stomach weight for the predator entity. We have not extracted those from the basic observations, but simply assumed that the total weight in the stomach is proportional to the number of stomachs sampled for a given predator entity.

Implementation

The SMS has been implemented using the AD Model Builder (Fournier *et al.*, 2011), which is freely available from ADMB Foundation (www.admb-project.org). ADMB is an efficient tool including automatic differentiation for Maximum likelihood estimation of many parameters in nonlinear models.

SMS configurations may contain more than 1000 parameters of which less than 5% are related to predation mortality. It is not possible to estimate all parameters simultaneously without sensible initial parameter values. Such values are obtained in three phases:

1) Estimate "single-species" stock numbers, fishing mortality and survey catchability parameters assuming that natural mortality (M1+M2) are fixed and known (i.e. as used by the ICES single-species assessments).

2) Fix all the "single-species" parameters estimated in step 1 and use the fixed stock numbers to estimate initial parameter values for the predation parameters.

3) Use the parameter values from step 1 and 2 as initial parameter values and re-estimate all parameters simultaneously in the full model including estimation of predation mortality M2.

Optimisation might potentially be dependent on the initial parameter values, however the same final result was obtained using the three steps above or using a configuration where step two is omitted. Using step two however, in general makes the estimation process more robust as extreme values and system crash are avoided.

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Appendage 1. Notation, parameters and variables

Indices

a	age
area	area with specific predation mortality
A1, A2	group of ages
Fa	first age group in the model
i	prey entity, combination of prey species and prey size group
j	predator entity, combination of predator group and predator size group
l	species size class
lpred	predator size class
lprey	prey size class
other	other food "species"
pred	predator species
prey	prey species
q	season of the year, e.g. quarter
recq	recruitment season
s	species
survey	survey identifer
y	year
Y	group of years

Parameters and variables

AB	available (suitable) prey biomass for a predator
ALK	proportion at-size for a given age group. Input
C	catch in numbers. Observations
Cpue	catch in numbers per unit of effort. Observations
D	number died
DM1	number died due to M1
DM2	number died due to M2
DF	number died due to F
F	instantaneous rate of fishing mortality
F^1	age effect in separable model for fishing mortality. Estimated parameter
F^2	year effect in separable model for fishing mortality. Estimated parameter
F^3	season effect in separable model for fishing mortality. Estimated parameter
M1	instantaneous rate of residual natural mortality. Input
M2	instantaneous rate of predation mortality estimated in the model
N	stock number
Ns,a,y=first year,q=1	Stock number in the first year of the model. Estimated parameters
Ns,a=fa,q=recq	Stock numbers at youngest age (recruitment). Estimated parameter
OF	Biomass of other food for a predator. Input
Q	catchability, proportion of the population caught by one effort unit. Estimated
Rs,y	recruitment calculated from stock–recruitment model
RA	food ration, biomass consumed by a predator. Input
S	suitability of a prey entity as food for a predator entity
S1, S2	mesh selection parameters. Estimated
SSB	spawning–stock biomass
STOM	weight proportion of prey i found in the stomach of predator j. Observations
U	sampling intensity of stomachs. Observation

V variance of diet observations in relation to sampling intensity. Estimated Parameter

W body weight. Input

Z instantaneous rate of total mortality

α stock–recruitment parameter. Estimated

β stock–recruitment parameter. Estimated

ϱ prey size preference of a predator. Estimated parameter

γ food ration coefficients. Input

ς food ration exponent. Input

υ parameter for size dependent preference for other food. Estimated parameter

$\eta PREF$ natural logarithm of the preferred predator prey size ratio. Estimated parameter

ηMIN observed minimum relative prey size for a predator species. Input

ηMAX observed maximum relative prey size for a predator species. Input

o spatial overlap between predator and prey species. Estimated parameter

ρ coefficient of species vulnerability. Estimated parameter

$\sigma CATCH$ standard deviation of catch observations. Estimated parameter

$\sigma PREF$ parameter expressing how particular a predator is about the size of its prey. Parameter

σSR standard deviation of stock–recruitment estimate. Estimated parameter

$\sigma STOM$ standard deviation of stomach content observations (used with lognormal distribution)

$\sigma SURVEY$ standard deviation of survey cpue observations. Estimated parameter

APPENDIX 2: Option file for SMS-key runs

Key run 2025

```
# sms.dat option file
# the character "#" is used as comment character, such that all text and numbers
# after # are skipped by the SMS program
#
######################################
# Produce test output (option test.output)
#  0 no test output
#  1 output file sms.dat and  file fleet.info.dat as read in
#  2 output all single species input files as read in
#  3 output all multi species input files as read in
#  4 output option overview
#
# 11 output between phases output
# 12 output iteration (obj function) output
# 13 output stomach parameters
# 19 Both 11, 12 and 13
#
# Forecast options
# 51 output hcr_option.dat file as read in
# 52 output prediction output summary
# 53 output prediction output detailed
1
######################################
# Produce output for SMS-OP program. 0=no, 1=yes
0
######################################
# Single/Multispecies mode (option VPA.mode)
# 0=single species mode
# 1=multi species mode, but Z=F+M (used for initial food suitability parm. est.)
# 2=multi species mode, Z=F+M1+M2
2
######################################
# Number of areas for multispecies run (default=1)
1
#
#&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&
#
# single species parameters
#
#&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&
#
## first year of input data (option first.year)
1974
######################################
## first year used in the model (option first.year.model)
1974
######################################
## last year of input data (option last.year)
2024
######################################
## last year used in the model (option last.year.model)
2024
######################################
##  number of seasons (option last.season). Use 1 for annual data
4
######################################
## last season last year (option last.season.last.year). Use 1 for annual data
4
######################################
## number of species (option no.species)
3
######################################
# Species names, for information only. See file species_names.in
#  Cod Herring Sprat
######################################
## first age all species (option first.age)
0
######################################
## recruitment season (option rec.season). Use 1 for annual data
3
######################################
## maximum age for any species(max.age.all)
11
######################################
## various information by species
# 1. last age
# 2. first age where catch data are used (else F=0 assumed), First age if species is other
predator
# 3. last age with age dependent fishing selection
# 4. Esimate F year effect from effort data. 0=no, 1=yes
# 5. Last age included in the catch at age likelihood (normally last age)
```

```
# 6. plus group, 0=no plus group, 1=plus group
# 7. predator species, 0=no, 1=VPA predator, 2=Other predator
# 8. prey species, 0=no, 1=yes
# 9. Stock Recruit relation
#      1=Ricker, 2=Beverton & Holt, 3=Geom mean,
#      4= Hockey stick, 5=hockey stick with smoother,
#      51=Ricker with estimated temp effect,
#      52=Ricker with known temp effect,
#      >100= hockey stick with known breakpoint (given as input)
# 10. Spawning season (not used yet, but set to 1)
# 11. Additional data for Stock Recruit relation
##
11 0 0 0 0 0 2 0 0 0 0  # 1 Cod
8 1 5 0 8 1 0 1 3 0 0  # 2 Herring
7 1 4 0 7 0 0 1 3 0 0  # 3 Sprat
########################################
## use input recruitment estimate (option use.known.rec)
#   0=estimate all recruitments
#   1=yes use input recruitment from file known_recruitment.in
0
########################################
## adjustment factor to bring the beta parameter close to one (option beta.cor)
#      Herring       Sprat
        1e+06       1e+06
########################################
## year range for data included to fit the R-SSB relation (option SSB.R.year.range)
# first (option SSB.R.year.first) and last (option SSB.R.year.last) year to consider.
# the value -1 indicates the use of the first (and last) available year in time series
# first year by species
#      Herring       Sprat
            -1        1990
# last year by species
#      Herring       Sprat
              -1          -1
########################################
## Objective function weighting by species (option objective.function.weight)
# first=catch observations,
# second=CPUE observations,
# third=SSB/R relations
# fourth=stomach observations, weight proportions
# fifth=stomach observations, number at length
##
0 0 0 1 0  # 1 Cod
1 1 0.05 0 0  # 2 Herring
1 1 0.05 0 0  # 3 Sprat
########################################
## parameter estimation phases for single species parameters
# phase.rec (stock numbers, first age) (default=1)
1
# phase.rec.older (stock numbers, first year and all ages) (default=1)
1
# phase.F.y (year effect in F model) (default=1)
1
# phase.F.y.spline (year effect in F model, implemented as spline function)
-1
# phase.F.q (season effect in F model) (default=1)
1
# phase.F.a (age effect in F model) (default=1)
1
# phase.catchability (survey catchability) (default=1)
1
# phase.SSB.R.alfa (alfa parameter in SSB-recruitment relation) (default=1)
1
# phase.SSB.R.beta (beta parameter in SSB-recruitment relation) (default=1)
1
########################################
## minimum CV of catch observation used in ML-estimation (option min.catch.CV)
0.1
########################################
## minimum CV of catch SSB-recruitment relation used in ML-estimation (option min.SR.CV)
0.1
########################################
## Use proportion landed information in calculation of yield (option calc.discard)
#    0=all catches are included in yield
#    1=yield is calculated from proportion landed (file proportion_landed.in)
#      Herring       Sprat
              0           0
########################################
## use seasonal or annual catches in the objective function (option combined.catches)
# do not change this options from default=0, without looking in the manual
#    0=annual catches with annual time steps or seasonal catches with seasonal time steps
#   1=annual catches with seasonal time steps, read seasonal relative F from file F_q_ini.in
(default=0)
#      Herring       Sprat
              0           0
########################################
## use seasonal or common combined variances for catch observation
# seasonal=0, common=1 (use 1 for annual data)
```

```
#      Herring       Sprat
             1           0
#######################################
##
# catch observations: number of separate catch variance groups by species
#     Herring       Sprat
            2           4
#  first age group in each catch variance group
1 2      #  Herring
1 2 3 6  #  Sprat
#######################################
##
# catch observations: number of separate catch seasonal component groups by species
#     Herring       Sprat
            3           2
#  first ages in each seasonal component group by species
1 2 3    #  Herring
1 2      #  Sprat
#######################################
## first and last age in calculation of average F by species (option avg.F.ages)
3 6  # Herring
3 5  # Sprat
#######################################
## minimum 'observed' catch, (option min.catch). You cannot log zero catch at age!
#
# 0 ignore observation in likelihood
#
# negative value gives percentage (e.g. -10 ~ 10%) of average catch in age-group for input
catch=0
# negative value less than -100 substitute all catches by the option/100 /100 *average catch
in the age group for catches less than (average catch*-option/10000
#
# if option>0 then will zero catches be replaced by catch=option
#
#  else  if  option<0  and  option  >-100  and  catch=0  then  catches  will  be  replaced  by
catch=average(catch at age)*(-option)/100
# else if option<-100  and catch < average(catch at age)*(-option)/10000 then catches will
be replaced by catch=average(catch at age)*(-option)/10000
#      Herring       Sprat
             0           0
#######################################
##
# catch observations: number of year groups with the same age and seasonal selection
#     Herring       Sprat
            3           2
#  first year in each group (please note #1 will always be changed to first model year)
1974 1989 2005   #  Herring
1974 2000        #  Sprat
#######################################
##
# number of nodes for year effect Fishing mortality spline
# 1=no spline (use one Fy for each year), >1 number of nodes
#     Herring       Sprat
            1           1
#  first year in each group
1974     #  Herring
1974     #  Sprat
#######################################
## year season combinations with zero catch (F=0) (option zero.catch.year.season)
# 0=no, all year-seasons have catches,
# 1=yes there are year-season combinations with no catch.
#   Read from file zero_catch_seasons_ages.in
# default=0
0
#######################################
## season age combinations with zero catch (F=0) (option zero.catch.season.ages)
# 0=no, all seasons have catches,
# 1=yes there are seasons with no catch. Read from file zero_catch_season_ages.in
# default=0
0
#######################################
## Factor for fixing last season effect in F-model (default=1) (fix.F.factor))
#      Herring       Sprat
             1           1
#######################################
## Uncertainties for catch, CPUE and SSB-R observations (option calc.est.sigma)
#  values: 0=estimate sigma as a parameter (the right way of doing it)
#          1=Calculate sigma and truncate if lower limit is reached
#          2=Calculate sigma and use a penalty function to avoid lower limit
#  catch-observation, CPUE-obs, Stock/recruit
            0           0           0
#######################################
# Read HCR_option file (option=read.HCR) default=0
#  0=no  1=yes
0
#######################################
#
#&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&&
```

```
#
# multispecies parameters
#
#$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$
#
# Exclude year, season and predator combinations where stomach data are not incl.(option
incl.stom.all)
#   0=no, all stomach data are used in likelihood
#   1=yes there are combinations for which data are not included in the likelihood.
#      Read from file: incl_stom.in
#   default(0)
1
########################################
##  N in the beginning of the period or N bar for calculation of M2 (option use.Nbar)
#  0=use N in the beginning of the time step (default)
#  1=use N bar
0
########################################
## Maximum M2 iterations (option M2.iterations) in case of use.Nbar=1
5
########################################
## convergence criteria (option max.M2.sum2) in case of use.Nbar=1
#  use max.M2.sum2=0.0 and M2.iterations=7 (or another high number) to make Hessian
0
########################################
## likelihood model for stomach content observations (option stom.likelihood)
#  1 =likelihood from prey weight proportions only (see option below)
#   2 =likelihood from prey weight proportions and from prey numbers to estimate size
selection
#  3 =Gamma distribution for prey absolute weight and size selection from prey numbers
1
########################################
# Variance used in likelihood model for stomach contents as prey weight proportion (option
stomach.variance)
#  0 = not relevant,
#  1 = log normal distribution,
#  2 = normal distribution,
#  3 = Dirichlet distribution (with estimation of 'concentration' parameter)
#  4 = Dirichlet distribution (with input of 'concentration' parameter)
#  5 = Dirichlet distribution (with input of maximum 'concentration' parameter)
3
########################################
## Usage of age-length-keys for calc of M2 (option simple.ALK))
#  0=Use only one size group per age (file lsea.in or west.in)
#  1=Use size distribution per age (file ALK_all.in)
0
########################################
## Usage of food-rations from input values or from size and regression parameters (option
consum)
#  0=Use input values by age (file consum.in)
#  1=use weight at age (file west.in) and regression parameters (file consum_ab.in)
#   2=use length at age (file lsea.in), l-w relation and regression parameters (file
consum_ab.in)
0
########################################
## Size selection model based on (option size.select.model)
#  1=length:
#      M2 calculation:
#         Size preference:
#           Predator length at age from file: lsea.in
#           Prey     length at age from file: lsea.in
#         Prey mean weight is weight in the sea from file: west.in
#      Likelihood:
#         Size preference:
#           Predator mean length per length group (file: stom_pred_length_at_sizecl.in)
#           Prey mean length per ength group (file stomlen_at_length.in
#                 Prey mean weight from mean weight per prey length group (file:
stomweight_at_length.in
#  2=weight:
#      M2 calculation:
#         Size preference:
#           Predator weight at age from file: west.in
#           Prey     weight at age from file: west.in
#         Prey mean weight is weight in the sea from file: west.in
#      Likelihood:
#         Size preference
#            Predator mean weight is based on mean length per predator length group (file:
stom_pred_length_at_sizecl.in)
#              and l-w relation (file: length_weight_relations.in),
#           Prey mean weight per prey length group (file: stomweight_at_length.in)
#                 Prey mean weight from mean weight per prey length group (file:
stomweight_at_length.in
#  3=weight:
#       M2 calculation: Same as option 2
#       Likelihood:
#         Size preference:
#            Predator mean weight is based on mean length per predator length group (file:
stom_pred_length_at_sizecl.in)
```

```
#              and l-w relation (file: length_weight_relations.in),
#             Prey mean weight per prey length group (file: stomlen_at_length.in) and l-w
relation (file:length_weight_relations.in)
#               Prey mean weight from prey mean length per prey length group (file:
stomlen_at_length.in) and l-w relation (file: length_weight_relations.in)
#  4=weight:
#       M2 calculation:
#         Size preference:
#             Predator mean weight from file lsea.in (length in the sea) and l-w relation
(file: length_weight_relations.in)
#            Prey mean weight from file lsea.in (length in the sea) and l-w relation (file:
length_weight_relations.in)
#       Likelihood:  Same as option 3
#  5=weight in combination with simple.ALK=1:
#       M2 calculation:
#         Size preference:
#             Predator weight based on length from file ALK_all.in (length distribution at
age) and l-w relation (file: length_weight_relations.in)
#             Prey     weight based on length from file ALK_all.in (length distribution at
age) and l-w relation (file: length_weight_relations.in)
#         Prey mean weight based on length from file ALK_all.in (length distribution at age)
and l-w relation (file: length_weight_relations.in)
#       Likelihood: Same as for option 2
#  6=weight in combination with simple.ALK=1:
#       M2 calculation: Same as option 5
#       Likelihood: Same as option 3
2
########################################
# Adjust Length at Age distribution by a mesh selection function (option L50.mesh)
#  Please note that options simple.ALK shoud be 1 and option size.select.model should be 5
# L50 (mm) is optional given as input. Selection Range is estimated by the model
# L50= -1 do not adjust
# L50=0, estimate L50 and selection range
# L50>0, input L50 (mm) and estimate selection range
# by VPA species
#      Herring       Sprat
            -1          -1
########################################
## spread of size selection (option size.selection)
#   0=no size selection, predator/preys size range defined from observations
#   1=normal distribution size selection
#   3=Gamma distribution size distribution
#   4=no size selection, but range defined by input min and max regression parameters (file
pred_prey_size_range_param.in)
#   5=Beta distributed size distribution, within observed size range
#   6=log-Beta size distributed, within observed size range
#
# by predator
#          Cod
            1
########################################
## sum stomach contents over prey size for use in likelihood for prey weight proportions
(option sum.stom.like)
#   0=no, use observations as they are; 1=yes, sum observed and predicted stomach contents
before used in likelihood for prey weight proportions
#
# by predator
#          Cod
            1
########################################
## # Use estimated scaling factor to link number of observation to variance for stomach
observation likelihood (option stom_obs_var)
#    0=no, do not estimate factor (assumed=1);  1=yes, estimate the factor;  2=equal weight
(1) for all samples
#
# by predator
#          Cod
            1
########################################
## # Upper limit for Dirichlet sumP. A low value (e.g. 10) limits the risk of overfitting.
A high value (e.g. 100) allows a full fit. (option stom_max_sumP)
# by predator
#          Cod
          1000
########################################
## Scaling factor (to bring parameters close to one) for relation between no of stomachs
sampling and variance
#  value=0: use default values i.e. 1.00 for no size selection and otherwise 0.1 (option
var.scale.stom)
#          Cod
            0
########################################
## other food suitability size dependency  (option size.other.food.suit)
#  0=no size dependency
#  1=yes, other food suitability is different for different size classes
#          Cod
            1
########################################
```

```
## Minimum observed relative stomach contents weight for inclusion in ML estimation (option
min.stom.cont)
#          Cod
         0.001
########################################
## Upper limit for no of samples used for calculation of stomach observation variance (option
max.stom.sampl)
#          Cod
           500
########################################
## Max prey size/ pred size factor for inclusion in M2 calc (option max.prey.pred.size.fac)
#          Cod
           0.3
########################################
## inclusion of individual stomach contents observations in ML for weight proportions (option
stom.type.include)
# 1=Observed data
# 2= + (not observed) data within the observed size range (=fill in)
# 3= + (not observed) data outside an observed size range. One obs below and one above
(=tails)
# 4= + (not observed) data for the full size range of a prey species irrespective of predator
size (=expansion)
#          Cod
             1
########################################
## use overlap input values by year and season (use.overlap)
#   0: overlap assumed constant or estimated within the model
#   1: overlap index from file overlap.in (assessment only, use overlap from last year in
forecast)
#   2: overlap index from file overlap.in (assessment and forecast)
0
########################################
## parameter estimation phases for predation parameters
#  the number gives the phase, -1 means no estimation
#
#  vulnerability (default=2) (phase phase.vulnera)
2
# other food suitability slope (default=-1) (option phase.other.suit.slope)
2
# preferred size ratio (default=2) (option phase.pref.size.ratio)
2
#     predator     size     ratio     adjustment     factor     (default=-1)     (option
phase.pref.size.ratio.correction))
-1
# prey species size adjustment factor (default=-1) (option phase.prey.size.adjustment)
-1
# variance of preferred size ratio (default=2) (option phase.var.size.ratio)
2
# season overlap (default=-1) (option phase.season.overlap)
3
# Stomach variance parameter (default=2) (option phase.Stom.var)
2
# Mesh size selection of stomach age length key (default=-1) (option phase.mesh.adjust)
-1
########################################
```